



# **LONWORKS® Router User's Guide**



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## Welcome

A LONWORKS® router connects two communications channels within a LONWORKS network, and routes LonTalk® messages between them. Using a LONWORKS router supports the installation of small or large networks, with dozens to thousands of nodes.

This document describes how to design and develop a LONWORKS router based on the Echelon Router 5000 chip, the Echelon FT Router 5000 chip, or the Echelon RTR-10 Router Core Module.

This document does not describe any of the following Echelon prepackaged router products: MPR-50 Multi-Port Router, i.LON® 600 LONWORKS/IP Server, CRD 3000 Power Line/RF Bridge, or LonPoint Router.

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## Audience

This guide provides user instructions for users of LONWORKS routers. It also provides information for developers who plan to develop a LONWORKS router based on either the Router 5000 chip or the FT Router 5000 chip, or who plan to integrate the RTR-10 router into embedded or standalone routers.

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## Related Documentation

The following manuals are available from the Echelon Web site ([www.echelon.com](http://www.echelon.com)) and provide additional information that can help you develop applications for Neuron® Chip or Smart Transceiver devices:

- *Connecting a Neuron 5000 Processor to an External Transceiver* Engineering Bulletin (005-0202-01D). This bulletin describes how to connect a Neuron 5000 Processor's communications port to external transceivers for TP/XF-1250 channels or for EIA-485 networks, using an external transceiver circuit. It also describes how to connect a Neuron 5000 Processor to a link-power TP/FT-10 channel using a LONWORKS LPT-11 Link Power Transceiver.
- *FT 3120 / FT 3150 Smart Transceiver Data Book* (005-0139-01D). This manual provides detailed technical specifications on the electrical interfaces, mechanical interfaces, and operating environment characteristics for the FT 3120® and FT 3150® Smart Transceivers.
- *Introduction to the LONWORKS Platform* (078-0391-01B). This manual provides an introduction to the ISO/IEC 14908 (ANSI/CEA-709.1 and EN14908) Control Network Protocol, and provides a high-level introduction to LONWORKS networks and the tools and components that are used for developing, installing, operating, and maintaining them.
- *Junction Box and Wiring Guidelines for Twisted Pair LonWorks Networks* (005-0023-01P). This bulletin identifies the different types of junction boxes and interconnections that can be used in twisted pair LONWORKS networks in building and industrial control applications.

- *LonMark® Application Layer Interoperability Guidelines*. This manual describes design guidelines for developing applications for open interoperable LONWORKS devices, and is available from the LonMark Web site, [www.lonmark.org](http://www.lonmark.org).
- *LonWorks FTT-10A Free Topology Transceiver User's Guide* (078-0156-01G). This manual provides specifications and user instructions for the FTT-10A Free Topology Transceiver.
- *LonWorks LPT-11 Link Power Transceiver User's Guide* (078-0198-01A). This manual provides technical specifications on the electrical and mechanical interfaces and operating characteristics for the LPT-11 Link Power Transceiver.
- *LonWorks TPT Twisted Pair Transceiver Module User's Guide* (078-0025-01C). This manual provides detailed specifications on the electrical and mechanical interfaces and operating environment characteristics for the TPT/XF-78 and TPT/XF-1250 transceiver modules.
- *NodeBuilder® FX User's Guide* (078-0405-01A). This manual describes how to develop a LONWORKS device using the NodeBuilder tool.
- *PL 3120 / PL 3150 / PL 3170 Power Line Smart Transceiver Data Book* (005-0193-01C). This manual provides detailed technical specifications on the electrical interfaces, mechanical interfaces, and operating environment characteristics for the PL 3120, PL 3150, and PL 3170™ Smart Transceivers.
- *Series 5000 Chip Data Book* (005-0199-01C). This manual provides detailed specifications on the electrical interfaces, mechanical interfaces, and operating environment characteristics for the FT 5000 Smart Transceiver and Neuron 5000 Processor.

All of the Echelon documentation is available in Adobe® PDF format. To view the PDF files, you must have a current version of the Adobe Reader®, which you can download from Adobe at: [get.adobe.com/reader](http://get.adobe.com/reader).

For information about previous generation Neuron Chips, see one of the Neuron Chip Data Books: Motorola® *LONWORKS Technology Device Data*, Toshiba *Neuron Chip TMPN3150/3120*, or Cypress™ *Neuron Chip Technical Reference Manual*.

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## Getting Support

You can get technical support for any of Echelon's current product offerings by contacting Echelon Support: [www.echelon.com/support](http://www.echelon.com/support).

You can also search the Echelon Knowledge Base for known product issues: [www.echelon.com/support/kb/search.asp](http://www.echelon.com/support/kb/search.asp). The Knowledge Base contains a wealth of information about Echelon products and technologies, including technical articles that range from "How to" articles that describe how to complete a specific task to "Bug" articles that document known issues with Echelon products.

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## FCC Notice

The RTR-10 Router Core Module is designed to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. The Router 5000 chip is designed to comply with FCC Part 15 Subpart B and EN 55022 Level B.

These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation.

**Caution: Changes or modifications not covered in this manual must be approved in writing by the manufacturer's Regulatory Engineering department. Changes or modifications made without written approval may void the user's authority to operate this equipment.**

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## VDE Notice

The RTR-10 Router Core Module product is designed to comply with VDE 0871 Level B as a peripheral device. To ensure continued compliance, this product should only be used in conjunction with other compliant devices.

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## Canadian DoC Notice

The RTR-10 Router Core Module digital apparatus does not exceed the Class A limits for radio noise emissions from digital apparatus set out in the Radio Interference Regulations of the Canadian Department of Communications.

L' appareil RTR-10 Router Core Module numérique n'émet pas de bruits radioélectriques dépassant les limites applicables aux appareils numériques de la classe A prescrites dans le règlement sur la brouillage radioélectrique édicté par le Ministère des Communications du Canada.



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# 1

## Introduction to LONWORKS Routers

This chapter describes the router theory of operation, including router types, LonTalk protocol support for routers, and router use of message buffers.

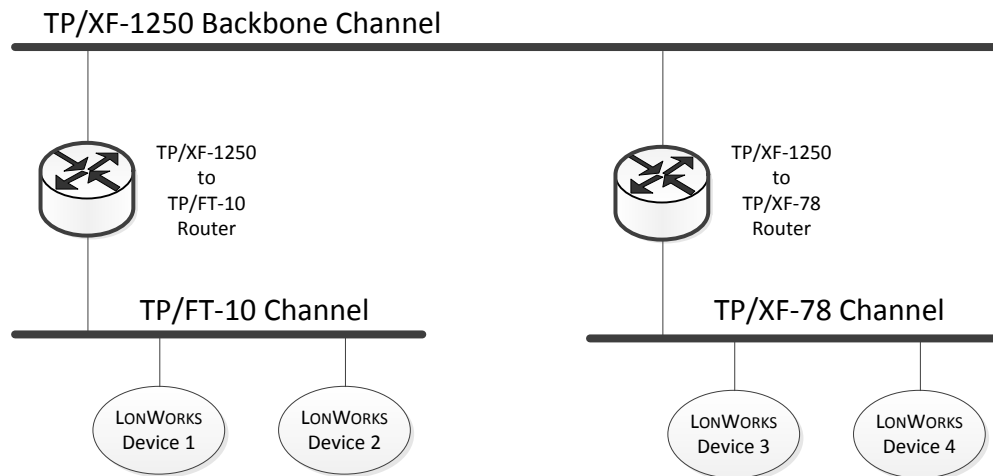
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## Introduction

In general terms, a router is a device that forwards data packets between communications networks. The router connects to the data lines from each network, and reads address information in each data packet to determine the packet's destination.

A LONWORKS router connects two communications channels within a LONWORKS network, and routes LonTalk messages between them. Using a LONWORKS router supports the installation of small or large networks, with dozens to thousands of nodes.

**Figure 1** shows a typical router installation, with a free topology channel and a 78 kbps bus topology channel connected to a 1.25 Mbps backbone twisted pair channel. Because the network includes the routers, applications on each of the LONWORKS devices can communicate with each other transparently, as if they were installed on a common channel.



**Figure 1.** Sample Router Installation

A single router can connect two channels, or multiple routers (called *redundant routers*) can connect the same pair of channels. Redundant routers provide fault tolerance by providing more than one routing path from one channel to another. They are also required when not all devices on a given channel are able to hear one another (referred to as an “ear shot problem”), for example on a radio frequency channel. For a router to function as a redundant router, the router must be configured to be a Configured router (see *Router Types*).

LONWORKS routers are used to:

- *Extend the limits of a single channel.* You can use a router to add a channel to a LONWORKS network to support additional devices or to extend the maximum channel length. You can add multiple routers, depending on the capacity or distance needed.
- *Interface different communications media, or bit rates, in a LONWORKS network.* For example, you might want to trade data rate for distance on portions of the network, or to use a 1.25 Mbps backbone twisted pair channel to connect several 78 kbps free topology and link power channels. Alternatively, you might want to use power line for a portion of the

network where the devices are subject to frequent physical relocation, or if cable installation is difficult. For each of these cases, you use a router to connect the dissimilar LONWORKS channels.

- *Enhance the reliability of the LONWORKS network.* The two channels that connect to a router are logically isolated, so a failure on one channel does not affect the other channel. For example, in an industrial control network, isolation among connected cells might be desirable to prevent a failure in a single cell from bringing down multiple cells. You can achieve this goal by dedicating channels to individual cells and isolating them from one another with routers.
- *Improve overall network performance.* You can use routers to isolate traffic within subsystems. For example, in a cluster of industrial cells, most of the communications might be between devices within cells rather than across cells. Using intelligent routers across cells avoids forwarding messages addressed to devices within specific cells, thus increasing the capacity and decreasing the response time of the overall network.

The use of routers across channels is transparent to the application programs within devices. Thus, you can develop applications without needing to know the workings of the routers or even if the device's channel will use a router. You only need to consider routers when determining the network image of a device. When you move a device from one channel to another, you need only change the network image. Use a network management tool, such as OpenLNS CT, to manage network images.

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## LONWORKS Router Products

Echelon provides the following router products:

- MPR-50 Multi-Port Router (Model: 42150)

Five-channel (one TP/XF-1250 channel and four TP/FT-10 channels) LONWORKS router. The MPR-50 can be used to connect two, three, or four TP/FT-10 channels together, or it can be used to connect these TP/FT-10 channels to a high-speed TP/XF-1250 backbone.

- i.LON 600 LONWORKS/IP Server (Model: 7260x)

An EIA-852 compliant LonTalk-to-IP router. The i.LON 600 provides secure Internet access to LONWORKS devices and transforms the Internet (or other IP-based network) into a pathway for LONWORKS control information.

- CRD 3000 Power Line/RF Bridge (Model: 76520R)

A Power Line (PL) to RF communications device, designed primarily for intelligent LONWORKS street lighting networks.

- LonPoint Router (Model: 4210x)

A two-channel router for TP/FT-10, TP/XF-78, or TP/XF-1250 LONWORKS channels. Three models are available for various network connection combinations.

- RTR-10 Router Core Module (Model: 61000R)

A compact module used by OEMs to build LONWORKS routers. The RTR-10 consists of the core electronics and firmware needed to implement a router.

- Router 5000 (Model: 14315R)

A semiconductor product used by OEMs to build half-routers or full routers for various LONWORKS channel types. The Router 5000 includes the firmware required to implement a half-router.

- FT Router 5000 (Model: 14285R)

A semiconductor product used by OEMs to build half-routers or full routers for LONWORKS Free Topology (FT) channels. The FT Router 5000 includes the firmware required to implement a half-router.

Packaged routers eliminate the need to build hardware and obtain the necessary electrical interference and safety certifications. Thus, they allow direct, off-the-shelf integration into the user's LONWORKS network. This manual does not describe how to use the Echelon prepackaged router products. See the Echelon router Web page ([www.echelon.com/products/routers](http://www.echelon.com/products/routers)) for information about the pre-packaged Echelon router products.

This manual describes those Echelon router products that allow OEMs to design and build their own custom routers for LONWORKS channels: the RTR-10 Router Core Module, the Router 5000 chips, and the FT Router 5000 chips.

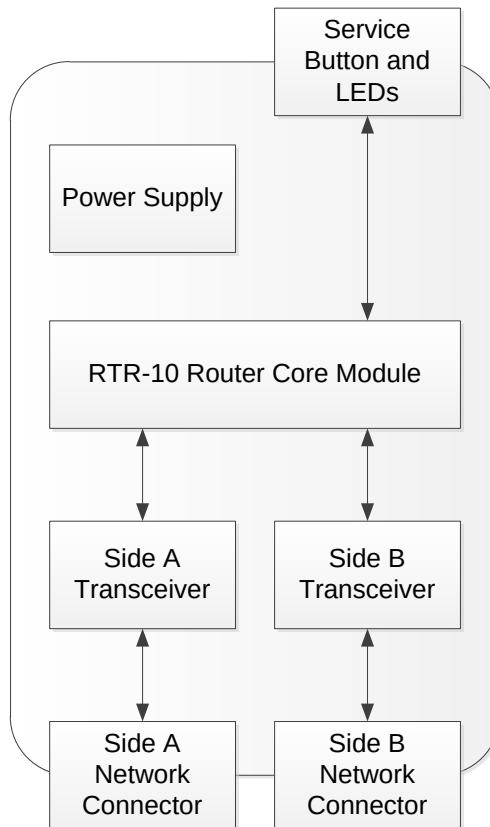
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## *RTR-10 Overview*

The RTR-10 Router Core Module is a compact module used by OEMs to design and build LONWORKS routers. A LONWORKS router connects two communications channels and route LonTalk messages between them. They support installation of networks with dozens to thousands of devices.

The RTR-10 Module consists of the core electronics and firmware required to implement a router. Its compact single inline module (SIM) form factor minimizes the board space required to implement a router. Vertical SIM sockets are available to minimize board space; right-angle SIM sockets are also available to minimize component height.

A complete router, using an RTR-10 module, consists of the module, two transceivers, and a motherboard to connect the RTR-10 router to the two transceivers, as shown in **Figure 2**.



**Figure 2.** Block Diagram of a LonWorks Router Based on the RTR-10

As the figure shows, an RTR-10 router and two transceiver modules (one to handle each of two channels connected by the router) can be mounted on a motherboard, along with a single power supply and two network connectors. This sub-assembly constitutes a LONWORKS router. It can be packaged in an enclosure to meet unique form factor and environmental requirements. Depending on the application, the package could contain a single router sub-assembly, or could include other application-specific hardware. Multiple routers can be packaged together for some applications, such as a backbone connecting multiple channels.

The RTR-10 router comes preconfigured with many common LONWORKS transceiver parameters. Two sets of five transceiver identification (XID) pins on the RTR-10 router select the appropriate transceiver type for each side. The transceiver ID inputs eliminate a manufacturing step by automatically configuring the RTR-10 router for most transceivers. A special transceiver ID is reserved for programming any custom type.

One side of the RTR-10 router has a fixed input clock rate of 10 MHz. This side can be used with transceivers running at interface bit rates from 9.8 kbps to 1.25 Mbps. The second side of the RTR-10 router can be tied to the 10 MHz output of the first side, requiring no external components for interface bit rates from 9.8 kbps to 1.25 Mbps. Alternatively, the 10 MHz output can be divided to a lower frequency with external hardware and used as the input clock for the second side to support transceivers running at bit rates as low as 610 bps.

Any pair of channel types can be connected by a router by selecting the appropriate pair of transceivers. The RTR-10 router is compatible with all

LONWORKS transceivers, including standard transceivers for free topology, link power, twisted pair, and power line. Using multiple communications media can minimize installation costs and increase system performance by allowing easily installed media, such as power line or link power, to be combined with high-performance media such as TP/XF-1250 twisted pair.

## Router 5000 and FT Router 5000 Overview

The Router 5000 chip and the FT Router 5000 chip (generically referred to as Series 5000 router chips) are Echelon semiconductor products, based on the Echelon Neuron 5000 Core, that are used to build half-routers and full routers for LONWORKS channels. A LONWORKS router connects two communications channels and route LonTalk messages between them. They support installation of networks with dozens to thousands of devices.

Both the Router 5000 and FT Router 5000 include the Router firmware required to implement a half-router. Each chip's compact form factor minimizes the space required to develop a half-router. You can implement two half-routers to develop a full router for the same, or different, external transceiver types.

The FT Router 5000 integrates the high performance Neuron 5000 Core with a free topology (FT) twisted pair transceiver. Thus, the FT Router 5000 supports the LONWORKS TP/FT-10 channel type.

**Table 1** lists commonly used channel and transceiver types for Router 5000-based router halves; see *Developing a Router with the Router 5000 Chip* for additional information about connecting a Router 5000 chip to these external transceiver types. These external transceivers can run at interface bit rates from 9.8 kbps to 1.25 Mbps.

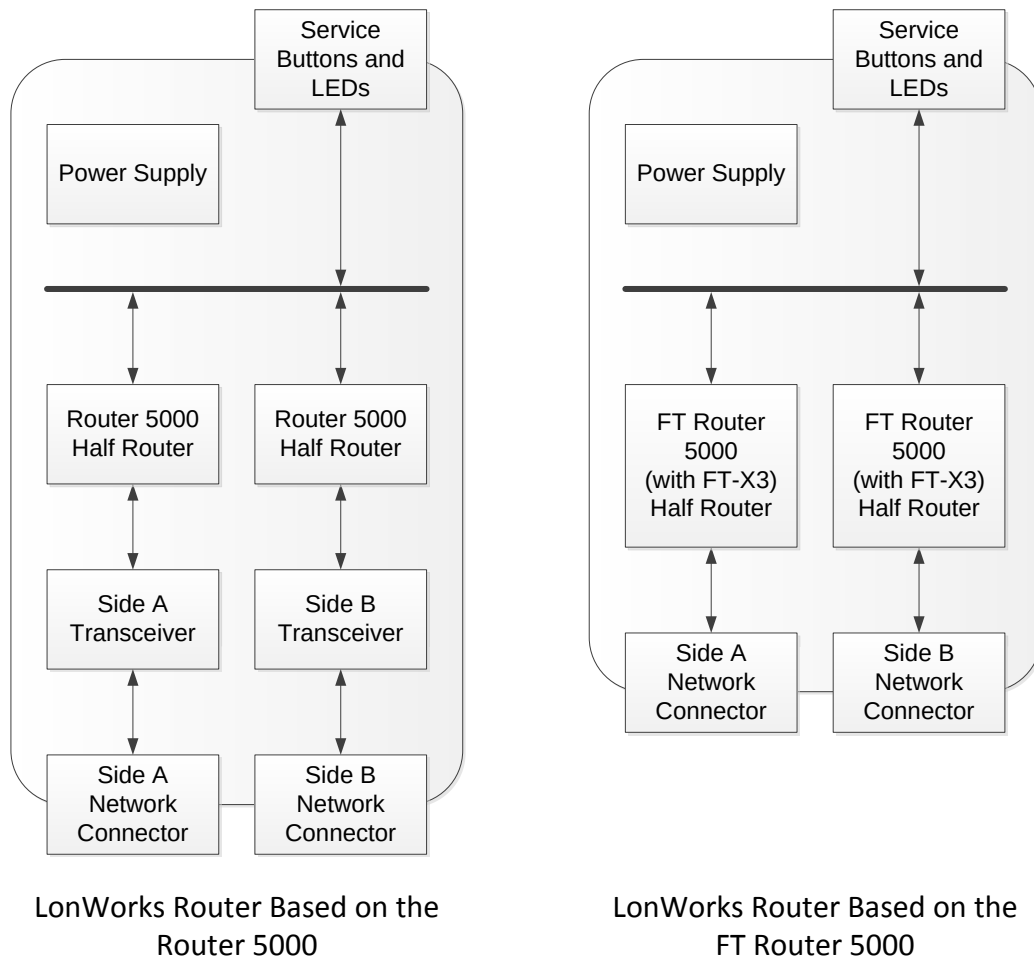
**Table 1.** Common Channel and Transceiver Types for a Router 5000

| Channel Type | Transceiver for Half Router                               | Notes                                                                                                                                                                                                                          |
|--------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TP/FT-10     | Echelon FTT-10A Free Topology Transceiver (Model 50051)   | Connection to the Router 5000 is similar to the connection to a Neuron Chip.<br><br>Use an FT Router 5000 for a standard (non-link-powered) TP/FT-10 channel or for a locally powered device on a link-power TP/FT-10 channel. |
| TP-RS485     | Any EIA-485 transceiver                                   | Can use any standard 3.3V or 5V EIA-485 transceiver.                                                                                                                                                                           |
| TP/XF-78     | Echelon TPT Twisted Pair Transceiver Module (Model 50010) | Add differential driver and differential comparator circuits (contact Echelon Support).                                                                                                                                        |
| TP/XF-1250   | Echelon TPT Twisted Pair Transceiver Module (Model 50020) | Add differential driver and differential comparator circuits.                                                                                                                                                                  |

| Channel Type | Transceiver for Half Router                         | Notes                                       |
|--------------|-----------------------------------------------------|---------------------------------------------|
| Link-power   | Echelon LPT-11 Link Power Transceiver (Model 50040) | Add linear regulator and TX buffer circuit. |

Echelon provides special licensing for other transceiver types, such as a Power Line Smart Transceiver; contact Echelon Support for additional information.

A complete router using the Router 5000 consists of two Router 5000 half routers, two transceivers, and a motherboard to connect the two half routers. A complete router using the FT Router 5000 consists of two FT Router 5000 half routers (each with its own FT-X3 Communications Transformer), and a motherboard to connect the two half routers. **Figure 3** shows block diagrams of both types of Series 5000 based routers.



**Figure 3.** Block Diagrams of LONWORKS Routers Based on a Series 5000 Router Chip

In the left side of the figure, two Router 5000 half routers and two transceiver modules, one to handle each of two channels connected by the router, can be mounted on a motherboard, along with a single power supply and two network connectors. This sub-assembly constitutes a LONWORKS router. In the right side of the figure, two FT Router 5000 half routers, one to handle each of two channels connected by the router, can be mounted on a motherboard, along with a single power supply and two network connectors. This sub-assembly constitutes a

LONWORKS router. Although not shown in the figure, you can place a Router 5000 half router (with its transceiver module) and FT Router 5000 half router on the same motherboard to create a LONWORKS router.

A complete LONWORKS router can be packaged in an enclosure to meet unique form factor and environmental requirements. Depending on the application, the package could contain a single router sub-assembly, or could include other application-specific hardware. Multiple routers can be packaged together for some applications, such as a backbone connecting multiple channels.

Unlike an RTR-10 router, you store the LONWORKS transceiver parameters for each Router 5000 or FT Router 5000 half router in external EEPROM, thus allowing you to define the appropriate transceiver type for each side.

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## Comparing the RTR-10, Router 5000, and FT Router 5000

**Table 2** lists some of the major characteristics of the three router products described in this manual.

**Table 2.** Comparing the RTR-10, Router 5000, and FT Router 5000

| Parameter      | RTR-10                                        | Router 5000                                                                                            | FT Router 5000                                             |
|----------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Package        | Integrated module with both A side and B side | 7mm x 7mm QFN chip                                                                                     | 7mm x 7mm QFN chip                                         |
| Voltage        | 5 V                                           | 3.3 V                                                                                                  | 3.3 V                                                      |
| Clock rate     | 10 MHz A side<br>≤10 MHz B side               | 5, 10, 20, or 40 MHz system clock (10 MHz crystal)                                                     | 5, 10, 20, or 40 MHz system clock (10 MHz crystal)         |
| Router Buffers | 1408 bytes                                    | 25 KB                                                                                                  | 25 KB                                                      |
| Transceivers   | Any                                           | TP/FT-10<br>TP-RS485<br>TP/XF-78<br>TP/XF-1250<br>LPT-11 Link-power<br>Other (contact Echelon Support) | TP/FT-10<br>(requires an FT-X3 Communications Transformer) |

The main difference between the Router 5000 and the FT Router 5000 is their transceiver support. The FT Router 5000 supports only the TP/FT-10 channel, but because the transceiver is integrated into the chip, you do not need to use an FTT-10A Free Topology Transceiver (and associated circuitry to accommodate the different voltage standards between the half router and transceiver).

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## Router Types

A LONWORKS router can use one of four routing algorithms: configured router, learning router, bridge, and repeater. This selection allows you to trade system performance for ease of installation. The configured router and learning router algorithms create intelligent routers that selectively forward messages based on network topology. Both sides of a router must use the same routing algorithm.

The following general rules apply to all four routing algorithms:

- For a message to be forwarded, it must fit into the router's input and output message buffers. A free input message buffer must be available.
- For a message to be forwarded, it must have a valid cyclic redundancy check (CRC) code.
- Priority messages are forwarded as priority messages, but with the priority level of the transmitting side rather than the priority level of the originator of the message. If the transmitting side has not been installed with a priority value, then priority messages are not forwarded in a priority slot. The priority message is still flagged as a priority message, so that if it passes through a second router that is installed with a priority level, the second router transmits the message in a priority slot.

---

### Repeater

A *Repeater* is a router that forwards all messages in both directions, regardless of the message's destination or domain. That is, a repeater forwards all valid messages (that is, messages with a valid CRC code) to the other channel.

A *Permanent Repeater* behaves similarly, but its type cannot be changed after creation.

---

### Bridge

A *Bridge* is a router that forwards all messages received on either of the router's domains, regardless of the message's destination. That is, a bridge forwards packets received on one channel to the other channel, if the packet is sent on a domain to which the bridge belongs. Use a bridge to span domains. In a single domain network, a bridge functions essentially the same as a repeater.

A *Permanent Bridge* behaves similarly, but its type cannot be changed after creation.

---

### Configured Router

A *Configured Router* determines which packets to forward based on internal routing tables. A configured router forwards only those messages which are received on either of the router's domains and which meet the forwarding rules shown in **Figure 4** and **Figure 5**. Configured routers maintain their routing tables in non-volatile memory, and thus retain them after a reset. These tables control forwarding of subnet and group-addressed messages, and are managed by a network management tool.

A forwarding table is used for each domain on each side of the router. Each forwarding table contains a forwarding flag for each of the 255 subnets and 255 groups in a domain. As shown in **Figure 4** and **Figure 5**, these flags determine whether or not a message should be forwarded or dropped based on the destination subnet or group address of the message.

A network management tool initializes the forwarding tables using the network management messages described in Chapter 7, *Network Management Messages*. By configuring the routing tables based on network topology, a network management tool can optimize network performance and make the most efficient use of available bandwidth. Configured routers should be used for looping topologies; see *Loop Topology*.

For a LONWORKS router, there are two sets of forwarding tables, one in non-volatile memory (typically EEPROM) and one in RAM. The non-volatile table is copied to the RAM table when the router is initially powered-up, after a reset, and when the router receives the **Set Router Mode** command with the *Initialize Routing Table* option. The RAM table is used for all forwarding decisions.

Several of the operations in shown in **Figure 4** and **Figure 5** help prevent message loops for service-pin messages. Service-pin messages require special handling because they are broadcast to all nodes on the zero-length domain, and have a source subnet ID of zero. When a router receives a service-pin message with a source subnet ID of zero, the router modifies the source subnet field of the message to be the router's subnet on the receiving side. If the receiving side is installed in two domains, two service-pin messages are forwarded, one for each domain. Thus, the router can drop the service-pin message if a loop causes the message to be received again on the same side.

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## Learning Router

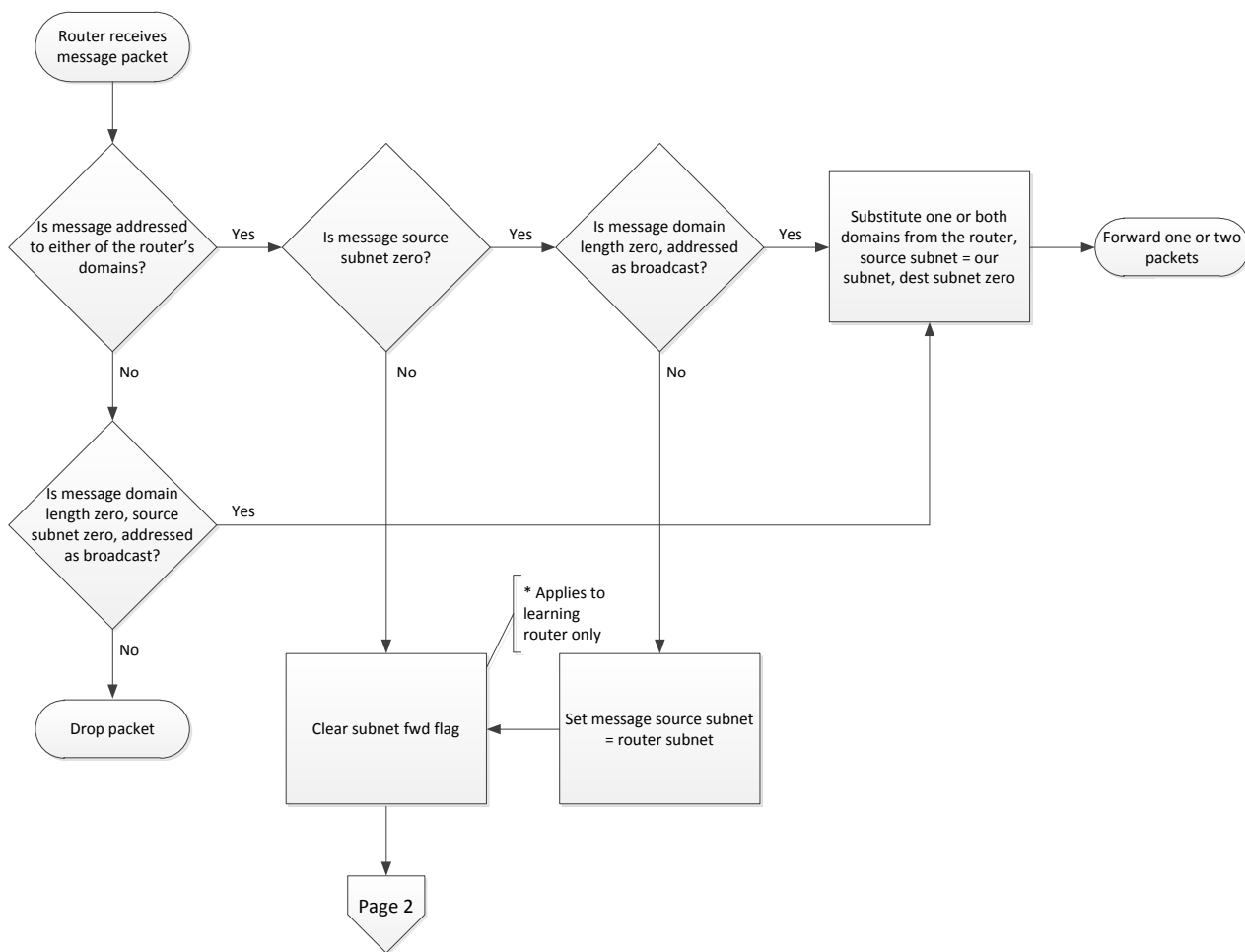
A *Learning Router*, like a configured router, determines which packets to forward based on internal routing tables. A learning router forwards only those messages which are received on either of the router's domains and which meet the forwarding rules shown in **Figure 4** and **Figure 5**. A learning router always forwards all group-addressed messages. Learning routers maintain their routing tables in non-volatile memory, and thus retain them after a reset. These tables control forwarding of subnet and group-addressed messages, and are updated automatically by the router firmware, rather than their being configured by a network management tool. The group forwarding tables are configured to always forward (flood) all messages with group destination addresses.

When a router receives a packet with a destination address using a subnet ID, it uses the subnet ID to determine whether to forward the packet. Learning routers learn network topology by examining the source subnet of all messages received by the router. Whenever a learning router receives a packet from one of its channels, it uses the source subnet ID to learn the network topology. It sets the corresponding routing table entries to indicate that the subnet in question is to be found in the direction from which the packet was received. Because subnets cannot span two channels connected to an intelligent router, the router can learn which side a subnet is on whenever that subnet ID appears in the source address.

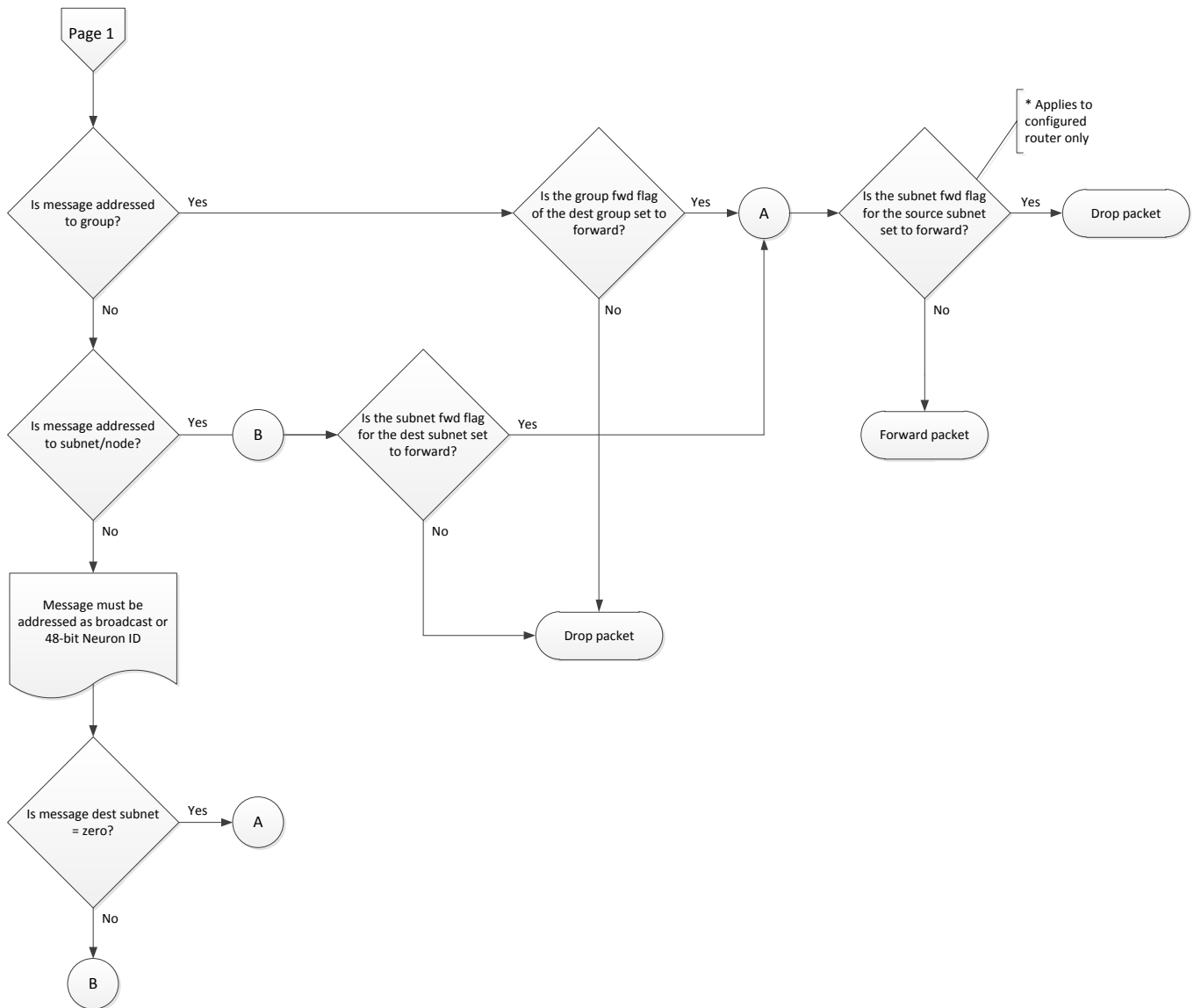
The subnet forwarding tables are initially configured to forward all messages with subnet destination addresses. Each time a new subnet ID is observed in the source address field of a message, its corresponding flag is cleared (that is, forwarding is disabled) in the subnet forwarding table. The forwarding flag for

the destination address is then checked to determine whether the message should be forwarded or dropped. The forwarding flags are all cleared whenever the router is reset, so the learning process restarts after a reset.

The forwarding flag for a given subnet should never be cleared on both sides of a router. However, the flag can be cleared on both sides if a device is moved from one side of a router to the other side. For example, if subnet 1 is located on side A of a router, the router will learn subnet 1's location as soon as it receives a message generated by any device in subnet 1. If any subnet 1 device is moved to side B without reinstalling it, the router will learn that subnet 1 is also on side B, and will stop forwarding subnet 1 messages to side A. The router detects this error and logs it, as described in Chapter 7, *Network Management Messages*.



**Figure 4.** Configured and Learning Router Forwarding Rules, Part 1



**Figure 5.** Configured and Learning Router Forwarding Rules, Part 2

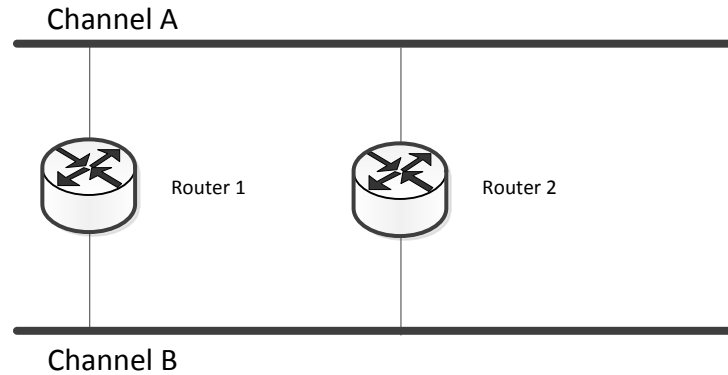
As with configured routers, learning routers sometimes modify source addresses for service-pin messages to help prevent message loops.

Learning routers, in general, are less efficient in using channel bandwidth because they always forward all messages with group destination addresses. Their advantage is simplified installation because the installation tool does not need to know the network topology to configure the router.

## Loop Topology

A *looping topology* is a network topology that has the potential for message *loops*. A loop is a path through two or more routers that forwards a message from a channel to itself. For example, **Figure 6** shows a looping topology with two channels and two routers. A message on channel A could be forwarded by router

1 to channel B, then the same message could be forwarded by router 2 back to channel A, starting an endless loop of forwarded messages.



**Figure 6.** Looping Topology

The LonTalk protocol does not support topologies where loops can occur. However, looping topologies can be desirable for the following reasons:

- *Increased Reliability.* Redundant routers can increase system reliability by providing multiple paths between two channels.
- *Support for Open Media.* Open media (such as radio frequency [RF] communications) might require redundant routers with overlapping coverage to ensure complete coverage of an area.

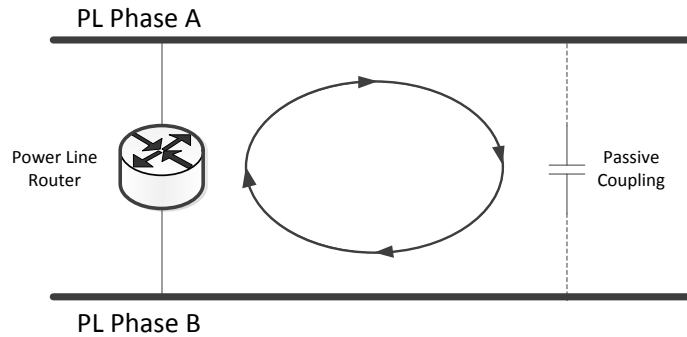
You can use configured routers (see *Configured Router*) to support looping topologies by configuring the routers to prevent message loops. For example, the topology in **Figure 6** can be supported if both routers are configured to forward all messages addressed to subnets on channel B from channel A; and all messages addressed to subnets on channel A from channel B. Any groups with members on both channels can only be forwarded by one of the two routers.

Network management tools, such as OpenLNS CT, can automatically set up the forwarding tables for configured redundant routers.

---

## Power Line Routers

A looping topology can be inadvertently created when using power line (PL) media. Passive coupling between different phases of a power line system can cause packets transmitted on one phase to be received by devices installed on another phase. A loop can be formed when active coupling provided by a router is combined with passive coupling. **Figure 7** shows an example looping topology with a power line router.



**Figure 7.** A Looping Topology with One Router

Routers can be used between power line channels only if the two channels are fully isolated. Such isolation is generally not the case between two phases on the same circuit, but can be the case between phases on different distribution transformers. Use an Echelon PLCA-22 Power Line Communication Analyzer to confirm isolation between power line channels before installing power-line-to-power-line routers.

---

## LonTalk Protocol Support for Routers

The LonTalk protocol<sup>1</sup> is designed to provide transparent routing of messages between devices that communicate through routers. To increase the efficiency of routers, the LonTalk protocol defines a hierarchical form of addressing using domain, subnet, and device (node) addresses. An intelligent router operates at the subnet level. The router determines which subnets lie on each of its two sides, and forwards packets accordingly.

Subnets do not span intelligent routers, which allow intelligent routers to make routing decisions based on the subnet component of a device's logical address. To further facilitate the addressing of multiple dispersed devices, the LonTalk protocol defines another class of addresses using domain and group addresses. Intelligent routers also can be configured to make routing decisions based on the group addressing component of a message.

In general, a network management tool, such as OpenLNS CT, is responsible for domain, subnet, node, and group address assignments.

See the ISO/IEC 14908 Control Network Protocol specification for detailed information about the LonTalk protocol.

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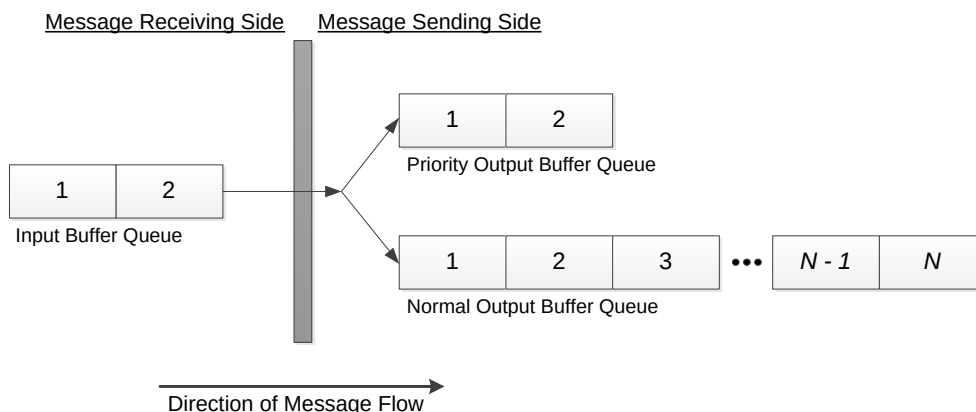
## Message Buffers

As messages are received by a router, they are placed in an input buffer queue. By default, this queue is limited to two message buffers to ensure that priority messages are never enqueued behind more than one non-priority message. When forwarded to the transmitting side of the router, priority messages have their own

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<sup>1</sup> Echelon's implementation of the ISO/IEC 14908 Control Network Protocol is called the LonTalk protocol. Echelon has implementations of the LonTalk protocol in several product offerings, including the Neuron firmware, OpenLNS<sup>®</sup> Server, SmartServers, and various network interfaces. This document refers to the ISO/IEC 14908-1 Control Network Protocol as the "LonTalk protocol," although other interoperable implementations exist.

outgoing buffer queue. Thus, priority processing of these outgoing messages is assured because the transmitting side will send messages from the priority output buffer queue before sending messages from the non-priority output buffer queue. **Figure 8** shows the message flow through the input and output buffer queues. This message flow is duplicated for messages moving in the opposite direction, that is, another set of input and output buffer queues exist for messages flowing in the opposite direction.



**Figure 8.** Buffering Scheme for a LONWORKS Router

The size and count of the message buffers is limited by the amount of RAM on the router.

---

## *RTR-10 Message Buffers and Transaction Records*

There are three different versions of the router firmware for the RTR-10 router. To determine the router firmware version of your router, see Appendix B, *Determining RTR-10 Firmware Version*.

Some versions of the RTR-10 routers were shipped with a reduced buffer and transaction record capacity. This does not impact system performance. However, a catastrophic failure of the router can occur if the buffers or transaction records on a router with reduced buffer and transaction record capacity are reconfigured and assume that a larger buffer and transaction record space is available. This could occur if you are using a manufacturing test station to reconfigure routers based on the higher router buffer and transaction record capacity. It can also occur if you use a network installation tool that reconfigures router buffers or transaction records when replacing a router using a non-standard configuration. The failure condition that occurs is that the router will go into the application-less state and will no longer function as a router. This failure mode cannot be recovered in the field. If you are making the change with a tool based on the OpenLNS Server or the OpenLNS Commissioning Tool (or LNS and the LonMaker Turbo Integration Tool), in some cases, the tool will prevent an invalid configuration; however, in other cases the tool may allow an invalid configuration and the router will fail. Use the tables, below, to understand the correct buffer and transaction record configurations for your RTR-10 router.

Depending on your firmware, buffers and transaction records are allocated according to the following tables. For versions A and C, **Table 3** shows the transaction record configurations for receive and transmit transactions, and the

buffer configurations for input buffers, output buffers, priority buffers, and non-priority buffers.

**Table 3.** RTR-10 Default Buffer and Transaction Record Configuration, Firmware A and C

| Type                                                                                                            | Count | Size (Bytes) | Total Bytes |
|-----------------------------------------------------------------------------------------------------------------|-------|--------------|-------------|
| Receive Transaction Record                                                                                      | 3     | 13           | 39          |
| Transmit Transaction Record                                                                                     | 2     | 28           | 56          |
| Application Buffer In                                                                                           | 1     | 42           | 42          |
| Application Buffer Out                                                                                          | 1     | 42           | 42          |
| Network Buffer In                                                                                               | 2     | 66           | 132         |
| Network Buffer Out                                                                                              | 15    | 66           | 990         |
| Application Buffer Out Priority                                                                                 | 1     | 42           | 42          |
| Network Buffer Out Priority                                                                                     | 2     | 66           | 132         |
| <b>Total Allocated Bytes</b>                                                                                    |       |              | <b>1475</b> |
| Unused                                                                                                          | 1     | 25           | 25          |
| <b>Total Available Bytes for Transaction Records and Buffers</b>                                                |       |              | <b>1500</b> |
| <b>Total Available Bytes for Buffers, including Default Transaction Records shown in first two lines, above</b> |       |              | <b>1405</b> |

The default buffer sizes allow the router to handle packets with maximum address overhead and data size for any network variable message and explicit messages with up to 40 bytes of data; this is large enough for any network management or network diagnostic message.

For RTR-10 routers with version B firmware, **Table 4** shows the transaction record configurations for receive and transmit transactions, and the buffer configurations for input buffers, output buffers, priority buffers, and non-priority buffers.

**Table 4.** RTR-10 Default Buffer and Transaction Record Configuration, Firmware B

| Type                        | Count | Size (Bytes) | Total Bytes |
|-----------------------------|-------|--------------|-------------|
| Receive Transaction Record  | 3     | 13           | 39          |
| Transmit Transaction Record | 2     | 28           | 56          |
| Application Buffer In       | 1     | 42           | 42          |
| Application Buffer Out      | 1     | 42           | 42          |
| Network Buffer In           | 3     | 66           | 198         |
| Network Buffer Out          | 11    | 66           | 726         |

|                                                                                                                 |   |    |             |
|-----------------------------------------------------------------------------------------------------------------|---|----|-------------|
| Application Buffer Out Priority                                                                                 | 1 | 42 | 42          |
| Network Buffer Out Priority                                                                                     | 3 | 66 | 198         |
| <b>Total Allocated Bytes</b>                                                                                    |   |    | <b>1343</b> |
| Unused                                                                                                          | 1 | 65 | 65          |
| <b>Total Available Bytes for Transaction Records and Buffers</b>                                                |   |    | <b>1408</b> |
| <b>Total Available Bytes for Buffers, including Default Transaction Records shown in first two lines, above</b> |   |    | <b>1313</b> |

You will not have a problem interchanging routers with Router Firmware Versions A, B, or C if you are not changing the router buffer or transaction record configuration. If you need to change the configuration, make sure that the total number of bytes required for the buffers and transaction records does not exceed the capacity for the version of the router firmware that you are using. For example, to use the OpenLNS Commissioning Tool or the LonMaker Turbo Integration Tool to safely change the buffer configuration for a router, right-click the router shape in the OpenLNS CT or LonMaker drawing and then click **Properties** on the shortcut menu. Click the **Buffers** tab to display and change the buffer configuration. As you change the buffer configuration, the required memory for each side of the router for your buffer configuration is displayed under the **Memory** heading. Verify that the memory required is less than the buffer capacity listed in Table 3 or Table 4 (as appropriate for your router firmware version) before clicking **OK** or **Apply**. When you click **OK** or **Apply**, the buffer configuration you selected is written to the router if you are attached to the network and you are OnNet. If the memory required is larger than the buffer and transaction record capacity, the router will fail to operate.

In applications that must route large explicit messages with more than 40 bytes of data, the buffer size must be increased, and the count of nonpriority buffers decreased. See the *Neuron C Programmer's Guide* to understand how the network buffer sizes are calculated. See *Network Management Messages* in this guide for a description of how to change the size and count of buffers. You can also use the NodeUtil Node Utility, which you can download from the Echelon Web site. However you allocate the transaction record counts and the buffer sizes and counts, the total memory required by the transaction records and buffer must not exceed the total available memory size in Table 3 or Table 4 (as appropriate for your router firmware version).

The default buffer configuration places the bulk of the buffers on the output queues of the router. The reasoning behind this configuration is to keep buffered packets on the output queues, after they have been processed for forwarding. This processing includes checking for priority packets. Priority packets are sensed and forwarded through the router's priority output buffers, so that priority packets are processed as quickly as possible, rather than allowing them to be delayed behind non-priority packages in a large input queue.

There are applications, however, where the network traffic can be "bursty", where many packets appear on the network almost at the same time. In these cases, the traffic bursts could cause the input queue to become full and lose excess packets.

In this case, reduce the number of lost messages by moving more of the packet buffering from the output queue to the input queue by increasing the size of the input queue and decreasing the size of the output queue. A router with a larger input queue can handle larger bursts of traffic, at the risk of priority messages being queued behind a number of non-priority messages.

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## Router 5000 and FT Router 5000 Message Buffers

Each router side has maximum 26 623 bytes of buffer space available. Because both the Router 5000 and FT Router 5000 have sufficient RAM available for any router configuration, you can allocate this space with any combination of buffers, for example, seven input buffers, two priority output buffers, and seven non-priority buffers. You can specify any valid buffer size (see the *Neuron C Programmer's Guide* for information about valid buffer sizes), but, in general, there is no reason not to specify the maximum size of 255 bytes. **Table 5** shows a general buffer configuration.

**Table 5.** General Series 5000 Router Buffer Configuration

| Queue                            | Count | Size (Bytes) | Total Bytes |
|----------------------------------|-------|--------------|-------------|
| Input Buffer Queue               | 7     | 255          | 1785        |
| Priority Output Buffer Queue     | 2     | 255          | 510         |
| Non-Priority Output Buffer Queue | 7     | 255          | 1785        |
| <i>Total</i>                     |       |              | <i>4080</i> |

The buffer size of 255 bytes allows the router to handle packets with maximum address overhead and data size for any network variable message or explicit message. See *Configuring a Series 5000 Half-Router* for a description of how to change the size and count of buffers. However you allocate the buffer sizes and counts, the total memory required by the three buffer queues must not exceed 25 K bytes.

The general buffer configuration shown in **Table 5** balances the buffers between the input and output queues of the router. For systems with large bursts of traffic, you could specify additional non-priority output buffers. Priority packets are sensed and forwarded through the router's priority output buffers, so that priority packets are processed as quickly as possible, rather than allowing them to be delayed behind non-priority packages in a large input queue.

---

## Router Performance

A major criterion of router performance is network throughput. An optimal router would be able to forward traffic at the wire-rate, with zero packet loss and minimal delay. Thus, an optimal router would forward traffic from 9.8 kbps to 1.25 Mbps, depending on the router's transceiver type.

A real router typically does not perform at the wire-rate because of latency within the router, including the time to receive and buffer the incoming packet at the near side, the time to forward the packet between the halves, and the time to buffer and transmit the packet at the far side. You should measure your router device's latency to determine if its design meets your system's needs.

**Example:** For a 20 MHz Router 5000 device (where both halves use the Router 5000 chip), a measured data transfer rate for sending a service-pin message between the router halves was approximately 1.2  $\mu$ s per byte (or 830 kbytes/sec). Some additional latency was seen for the time between the beginning of the original packet transmission and the beginning of the forwarded packet transmission.

For slower channel types, this router latency is not significant, but could become significant for faster channel types.

The latency between router halves is relatively invariant, with respect to router configuration, whereas overall router latency depends on the router type and configuration. For an RTR-10 device, the maximum data transfer rate between router halves is approximately 2.4  $\mu$ s per byte (or 416 kbytes/sec). For a Router 5000 device (where both halves use the Router 5000 chip), the maximum data transfer rate between router halves is approximately 600 ns per byte (or 1.6 Mbytes/sec, assuming a 40 MHz system clock for both halves; this rate scales with the system clock setting). An FT Router 5000 performs similarly to the Router 5000.



# 2

## **LONWORKS Router Electrical Interfaces**

This chapter provides an overview of the electrical interfaces for the RTR-10 Router Core Module, the Router 5000 chip, and the FT Router 5000 chip.

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## Overview

This chapter describes the electrical interface and power requirements for a LONWORKS router.

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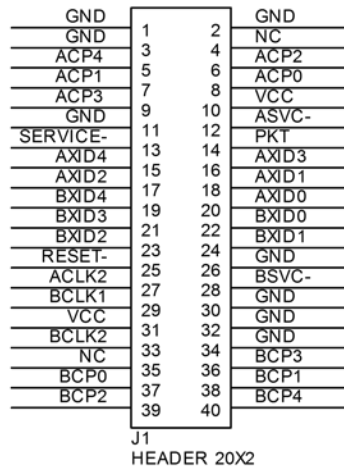
## Electrical Interface

The following sections describe the electrical interface for a LONWORKS router, including detailed descriptions of each of the RTR-10, Router 5000, and FT Router 5000 pins.

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### RTR-10 Electrical Interface

**Figure 9** shows a schematic view of a connector for the RTR-10 Router Core Module, and **Table 6** shows the pinout of the RTR-10 Router Core Module. See the *Neuron Chip Data Book* for more information about the use of the Neuron Chip communications port pins.



**Figure 9.** RTR-10 Header Pinout

**Table 6.** RTR-10 Pinout

| Pin Name | Pin Description                     | Pin Number |
|----------|-------------------------------------|------------|
| ACLK2    | A-side output clock                 | 27         |
| ACP0     | A-side network communication port 0 | 8          |
| ACP1     | A-side network communication port 1 | 7          |
| ACP2     | A-side network communication port 2 | 6          |
| ACP3     | A-side network communication port 3 | 9          |
| ACP4     | A-side network communication port 4 | 5          |

| Pin Name | Pin Description                     | Pin Number                  |
|----------|-------------------------------------|-----------------------------|
| ASVC~    | A-side Service output               | 12                          |
| AXID0    | A-side transceiver ID 0 (LSB)       | 20                          |
| AXID1    | A-side transceiver ID 1             | 18                          |
| AXID2    | A-side transceiver ID 2             | 17                          |
| AXID3    | A-side transceiver ID 3             | 16                          |
| AXID4    | A-side transceiver ID 4 (MSB)       | 15                          |
| BCLK1    | B-side input clock                  | 29                          |
| BCLK2    | B-side output clock                 | 33                          |
| BCP0     | B-side network communication port 0 | 37                          |
| BCP1     | B-side network communication port 1 | 38                          |
| BCP2     | B-side network communication port 2 | 39                          |
| BCP3     | B-side network communication port 3 | 36                          |
| BCP4     | B-side network communication port 4 | 40                          |
| BXID0    | B-side transceiver ID 0 (LSB)       | 22                          |
| BXID1    | B-side transceiver ID 1             | 24                          |
| BXID2    | B-side transceiver ID 2             | 23                          |
| BXID3    | B-side transceiver ID 3             | 21                          |
| BXID4    | B-side transceiver ID 4 (MSB)       | 19                          |
| BSVC~    | B-side Service output               | 28                          |
| GND      | Ground                              | 1, 2, 3, 11, 26, 30, 32, 34 |
| PKT      | Packet forward output               | 14                          |
| RESET~   | Reset input and output)             | 25                          |
| SERVICE~ | Combined Service input              | 13                          |
| VCC      | +5 VDC input                        | 10, 31                      |
| NC       | No Connect                          | 4, 35                       |

## ACLK2, BCLK1, and BCLK2

A 10 MHz crystal is provided for Side A of the RTR-10 router, which can run at only 10 MHz. This clock rate allows Side A to be used with transceivers running at interface bit rates from 9.8 kbps to 1.25 Mbps. The 10 MHz clock is output on the **ACLK2** pin, which allows Side B to be tied directly to the same clock through pin **BCLK1**. Thus, no external components are required to support the same range of bit rates on Side B.

The 10 MHz output can be divided to a lower frequency with external hardware, and used as the input clock for Side B to support transceivers running at interface bit rates as low as 610 bps.

**ACLK2** can drive five LS-TTL loads.

## ACP[4..0] and BCP[4..0]

The **ACP[4..0]** and **BCP[4..0]** signals are connected to the **CP[4..0]** pins of the core module Neuron Chips. The function of these pins is described in the *Neuron Chip Data Book*.

## ASVC~ and BSVC~

Each side of the RTR-10 router has an independent service-pin output: **ASVC~** for the A Side and **BSVC~** for the B Side. You can connect these output pins to service LEDs, as shown in **Figure 27** (in chapter 4). The function of the service pin is described in the *Neuron Chip Data Book*. The internal pullup resistor for the service pin on each side is enabled.

The service LEDs reflect the firmware status:

- Blinking means that the router side is unconfigured
- Off means that the side is configured
- On means that the side has failed

## AXID[4..0] and BXID[4..0]

The RTR-10 router comes preconfigured with many common LONWORKS transceiver parameters. Two sets of five transceiver identification (ID) pins on the RTR-10 router select the appropriate transceiver type for each side. The transceiver ID inputs eliminate a manufacturing step by automatically configuring the RTR-10 router for most transceivers. A special transceiver ID is reserved for programming any custom transceiver type; this value causes the communication port pins to be configured as inputs so that no line will be driven by both the transceiver and RTR-10 Neuron before the RTR-10 Neuron Chips can be properly configured.

The RTR-10 firmware reads the transceiver ID inputs on power up and reset. If the router is being powered-up for the first time, or if the transceiver ID is different from the last time it was powered-up, the parameters specified in the table on page 25 are loaded. If the router is being re-powered-up, and the transceiver ID is not 30 (0x1E), the RTR-10 firmware compares the network bit rate and input clock for the specified transceiver to the current transceiver parameters. If these parameters do not match, all transceiver parameters are

reinitialized. This reinitialization allows a network services tool to change parameters, such as the number of priority slots, without the new values' being overwritten by the RTR-10 firmware.

**Table 7. RTR-10 Router Transceiver IDs**

| ID                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Name         | Media                             | Bit Rate (bps) | Input Clock |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------------------------|----------------|-------------|
| 01 (0x01)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | TP/XF-78     | Transformer-isolated twisted pair | 78k            | 10 MHz      |
| 03 (0x03)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | TP/XF-1250   | Transformer-isolated twisted pair | 1.25M          | 10 MHz      |
| 04 (0x04)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | TP/FT-10     | Free Topology and Link Power      | 78k            | 10 MHz      |
| 05 (0x05)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | TP/RS485-39  | EIA-485 twisted pair              | 39k            | 10 MHz      |
| 07 (0x07)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | RF-10        | Radio Frequency (49 MHz)          | 4.9k           | 5 MHz       |
| 09 (0x09)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | PL-10        | Power Line spread-spectrum        | 10k            | 10 MHz      |
| 10 (0x0A)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | TP/RS485-625 | EIA-485 twisted pair              | 625k           | 10 MHz      |
| 11 (0x0B)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | TP/RS485-125 | EIA-485 twisted pair              | 1.25M          | 10 MHz      |
| 12 (0x0C)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | TP/RS485-78  | EIA-485 twisted pair              | 78k            | 10 MHz      |
| 16 (0x10)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | PL-20C       | Power Line C-Band                 | 5.4k           | 10 MHz      |
| 17 (0x11)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | PL-20N       | Power Line C-Band                 | 5.4k           | 10 MHz      |
| 18 (0x12)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | PL-30        | Power Line A-Band                 | 2.7k           | 10 MHz      |
| 24 (0x18)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | FO-10        | Direct Connect                    | 1.25M          | 10 MHz      |
| 27 (0x1B)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | DC-78        | Direct Connect                    | 78k            | 10 MHz      |
| 28 (0x1C)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | DC-625       | Direct Connect                    | 625k           | 10 MHz      |
| 29 (0x1D)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | DC-1250      | Direct Connect                    | 1.25M          | 10 MHz      |
| 30 (0x1E)                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Custom       | Custom                            | Custom         | Custom      |
| <b>Notes:</b> <ul style="list-style-type: none"> <li>• Type 07 (0x07) can be used for Side B only.</li> <li>• PL-20C channels use the CENELEC protocol; PL-20N channels do not use the CENELEC protocol.</li> <li>• Type 30 (0x1E) can be used for any transceiver type; the communications port is initially defined as all inputs to prevent circuit conflicts. The side using type 30 (0x1E) must be reprogrammed through the other router side.</li> </ul> |              |                                   |                |             |

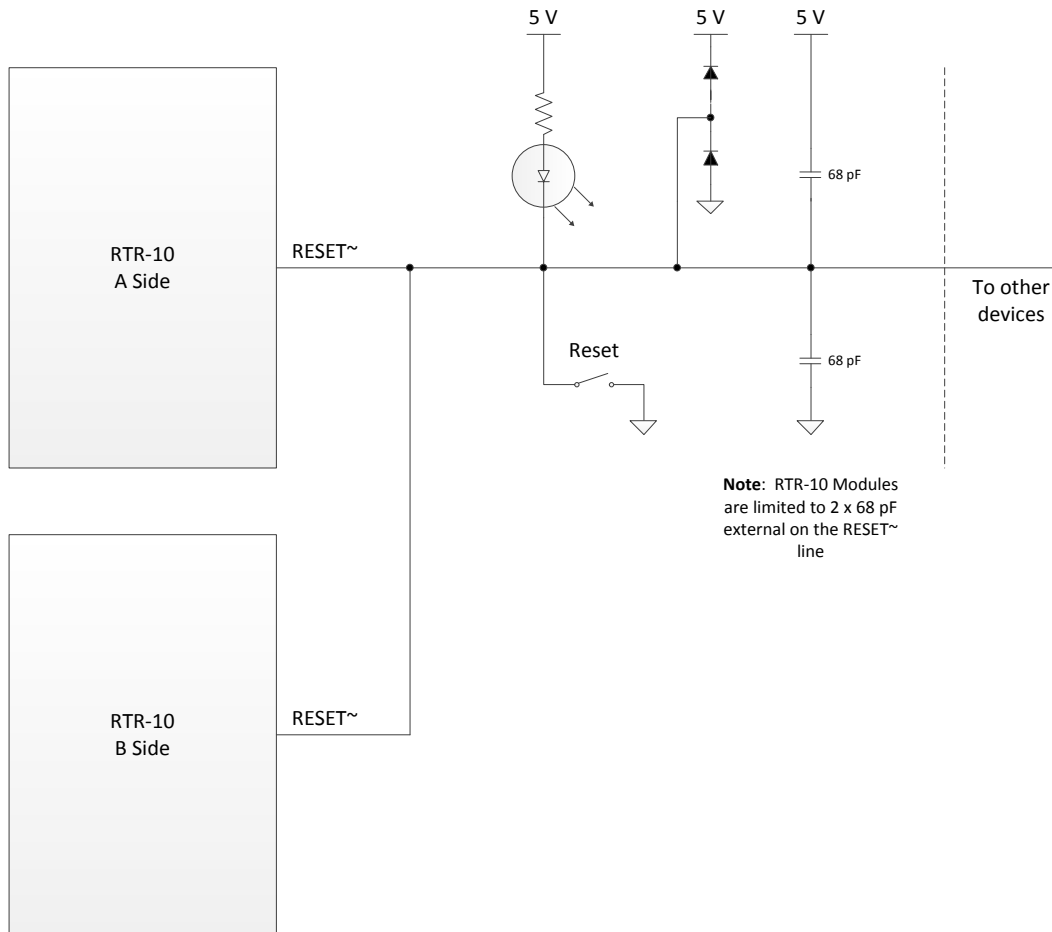
See Appendix A, *Communications Parameters for LONWORKS Routers*, for a listing of the communications parameters for each transceiver type.

## PKT

The **PKT** output can be used as a network activity indicator. When packets are passed between the router sides, **PKT** is active. This signal uses the unbuffered IO0 signal from the Neuron Chips. You can add a pulse stretcher circuit driven by **PKT** to make an activity LED flash, as in the example circuit shown in **Figure 27** in chapter 4.

## RESET~

The Neuron Chip reset pins are tied together and brought out on one pin. **Figure 10** shows the reset circuitry on the RTR-10 router.



**Figure 10.** RTR-10 Reset Circuit

Typical applications do not require debounce conditioning of a momentary pushbutton attached to the **RESET~** pin. The software response time associated with this input is long enough to effectively provide a software debounce for switches with a contact bounce settling time as long as 20 milliseconds. The **RESET~** signal must be driven low by a low voltage protection circuit on the router motherboard as described in *Low Voltage Protection*.

## SERVICE~

The **SERVICE~** input drives both sides of the RTR-10 router from a single input. You can connect a pushbutton to this pin broadcast each side's 48-bit Neuron ID on its channel (for example, during installation).

Typical applications do not require debounce conditioning of a momentary pushbutton attached to the **SERVICE~** pin. The software response time associated with this input is long enough to effectively provide a software debounce for switches with a contact bounce settling time as long as 20 milliseconds.

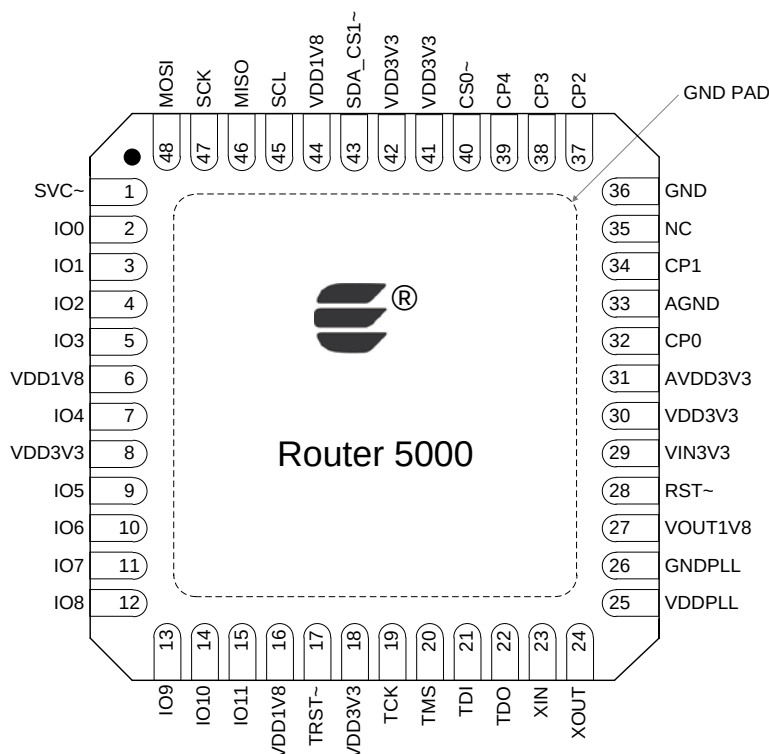
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## Series 5000 Router Electrical Interface

The electrical interfaces for the Router 5000 chip and FT Router 5000 chip are similar to the electrical interfaces of the Neuron 5000 Processor and the FT 5000 Smart Transceiver, as described in the *Series 5000 Chip Data Book*.

## Router 5000 Pinout

**Figure 11** shows the pinout for the Router 5000 chip. The central rectangle in the figure represents the bottom pad (pin 49), which must be connected to ground.



**Figure 11.** Router 5000 Chip Pinout

**Table 8** lists the pin assignments for the Router 5000 chip. All digital inputs are low-voltage transistor-transistor logic (LVTTL) compatible, 5 V tolerant, with low leakage. All digital outputs are slew-rate limited to reduce Electromagnetic Interference (EMI) concerns.

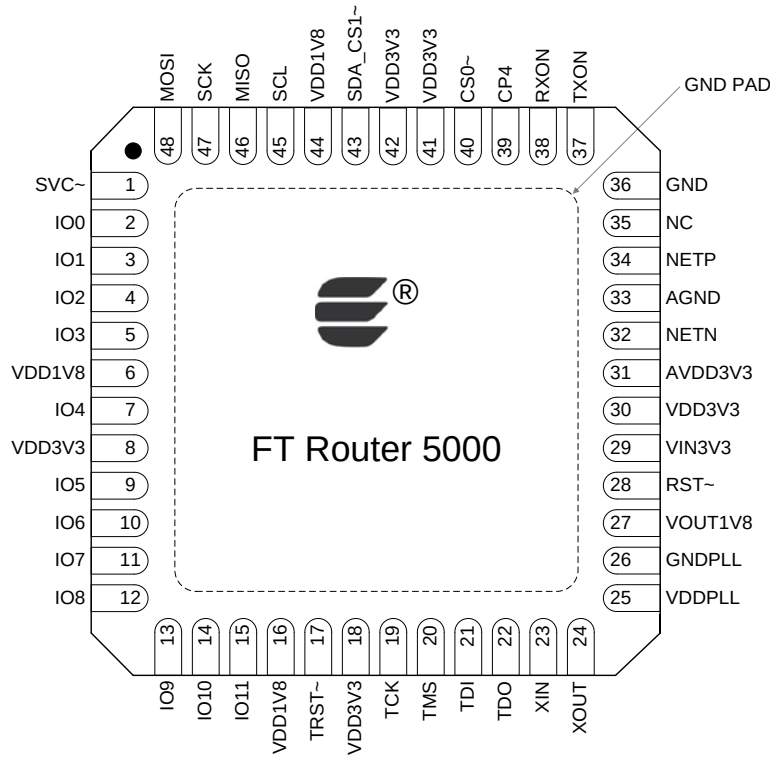
**Table 8. Router 5000 Chip Pin Assignments**

| Name   | Pin Number | Type           | Description                                            |
|--------|------------|----------------|--------------------------------------------------------|
| SVC~   | 1          | Digital I/O    | Service (active low)                                   |
| IO0    | 2          | Digital I/O    | IO0 (side A to side B)                                 |
| IO1    | 3          | Digital I/O    | IO1 (side A to side B)                                 |
| IO2    | 4          | Digital I/O    | IO2 (side A to side B)                                 |
| IO3    | 5          | Digital I/O    | IO3 (side A to side B)                                 |
| VDD1V8 | 6          | Power          | 1.8 V Power Input<br>(from internal voltage regulator) |
| IO4    | 7          | Digital I/O    | IO4 (side A to side B)                                 |
| VDD3V3 | 8          | Power          | 3.3 V Power                                            |
| IO5    | 9          | Digital I/O    | IO5 (side A to side B)                                 |
| IO6    | 10         | Digital I/O    | IO6 (side A to side B)                                 |
| IO7    | 11         | Digital I/O    | IO7 (side A to side B)                                 |
| IO8    | 12         | Digital I/O    | IO8 (side A to side B)                                 |
| IO9    | 13         | Digital I/O    | IO9 (side A to side B)                                 |
| IO10   | 14         | Digital I/O    | IO10 (side A to side B)                                |
| IO11   | 15         | Digital I/O    | IO11 (not used for routers)                            |
| VDD1V8 | 16         | Power          | 1.8 V Power Input<br>(from internal voltage regulator) |
| TRST~  | 17         | Digital Input  | JTAG Test Reset (active low)                           |
| VDD3V3 | 18         | Power          | 3.3 V Power                                            |
| TCK    | 19         | Digital Input  | JTAG Test Clock                                        |
| TMS    | 20         | Digital Input  | JTAG Test Mode Select                                  |
| TDI    | 21         | Digital Input  | JTAG Test Data In                                      |
| TDO    | 22         | Digital Output | JTAG Test Data Out                                     |
| XIN    | 23         | Oscillator In  | Crystal oscillator input                               |
| XOUT   | 24         | Oscillator Out | Crystal oscillator output                              |
| VDDPLL | 25         | Power          | 1.8 V Power Input<br>(from internal voltage regulator) |
| GNDPLL | 26         | Power          | Ground                                                 |
| VOU1V8 | 27         | Power          | 1.8 V Power Output<br>(of internal voltage regulator)  |

| Name     | Pin Number | Type                   | Description                                                       |
|----------|------------|------------------------|-------------------------------------------------------------------|
| RST~     | 28         | Digital I/O            | Reset (active low)                                                |
| VIN3V3   | 29         | Power                  | 3.3 V Power Input                                                 |
| VDD3V3   | 30         | Power                  | 3.3 V Power                                                       |
| AVDD3V3  | 31         | Power                  | 3.3 V Power                                                       |
| CP0      | 32         | Comm                   | CP0: Receive serial data                                          |
| AGND     | 33         | Ground                 | Ground                                                            |
| CP1      | 34         | Comm                   | CP1: Transmit serial data                                         |
| NC       | 35         | N/A                    | Do Not Connect                                                    |
| GND      | 36         | Ground                 | Ground                                                            |
| CP2      | 37         | Comm                   | CP2: External transceiver enable output                           |
| CP3      | 38         | Comm                   | CP3: Do Not Connect                                               |
| CP4      | 39         | Comm                   | CP4: Collision detect input                                       |
| CS0~     | 40         | Digital I/O for Memory | SPI slave select 0 (active low)                                   |
| VDD3V3   | 41         | Power                  | 3.3 V Power                                                       |
| VDD3V3   | 42         | Power                  | 3.3 V Power                                                       |
| SDA_CS1~ | 43         | Digital I/O for Memory | I <sup>2</sup> C: serial data<br>SPI: slave select 1 (active low) |
| VDD1V8   | 44         | Power                  | 1.8 V Power Input<br>(from internal voltage regulator)            |
| SCL      | 45         | Digital I/O for Memory | I <sup>2</sup> C serial clock                                     |
| MISO     | 46         | Digital I/O for Memory | SPI master input, slave output (MISO)                             |
| SCK      | 47         | Digital I/O for Memory | SPI serial clock                                                  |
| MOSI     | 48         | Digital I/O for Memory | SPI master output, slave input (MOSI)                             |
| PAD      | 49         | Ground Pad             | Ground                                                            |

## FT Router 5000 Pinout

**Figure 12** shows the pinout for the FT Router 5000 chip. The central rectangle in the figure represents the bottom pad (pin 49), which must be connected to ground.



**Figure 12.** FT Router 5000 Chip Pinout

**Table 9** lists the pin assignments for the FT Router 5000 chip. All digital inputs are low-voltage transistor-transistor logic (LVTTL) compatible, 5 V tolerant, with low leakage. All digital outputs are slew-rate limited to reduce Electromagnetic Interference (EMI) concerns.

**Table 9.** FT Router 5000 Chip Pin Assignments

| Name   | Pin Number | Type        | Description                                            |
|--------|------------|-------------|--------------------------------------------------------|
| SVC~   | 1          | Digital I/O | Service (active low)                                   |
| IO0    | 2          | Digital I/O | IO0 (side A to side B)                                 |
| IO1    | 3          | Digital I/O | IO1 (side A to side B)                                 |
| IO2    | 4          | Digital I/O | IO2 (side A to side B)                                 |
| IO3    | 5          | Digital I/O | IO3 (side A to side B)                                 |
| VDD1V8 | 6          | Power       | 1.8 V Power Input<br>(from internal voltage regulator) |
| IO4    | 7          | Digital I/O | IO4 (side A to side B)                                 |
| VDD3V3 | 8          | Power       | 3.3 V Power                                            |
| IO5    | 9          | Digital I/O | IO5 (side A to side B)                                 |
| IO6    | 10         | Digital I/O | IO6 (side A to side B)                                 |
| IO7    | 11         | Digital I/O | IO7 (side A to side B)                                 |

| Name    | Pin Number | Type                   | Description                                                |
|---------|------------|------------------------|------------------------------------------------------------|
| IO8     | 12         | Digital I/O            | IO8 (side A to side B)                                     |
| IO9     | 13         | Digital I/O            | IO9 (side A to side B)                                     |
| IO10    | 14         | Digital I/O            | IO10 (side A to side B)                                    |
| IO11    | 15         | Digital I/O            | IO11 (not used for routers)                                |
| VDD1V8  | 16         | Power                  | 1.8 V Power Input<br>(from internal voltage regulator)     |
| TRST~   | 17         | Digital Input          | JTAG Test Reset (active low)                               |
| VDD3V3  | 18         | Power                  | 3.3 V Power                                                |
| TCK     | 19         | Digital Input          | JTAG Test Clock                                            |
| TMS     | 20         | Digital Input          | JTAG Test Mode Select                                      |
| TDI     | 21         | Digital Input          | JTAG Test Data In                                          |
| TDO     | 22         | Digital Output         | JTAG Test Data Out                                         |
| XIN     | 23         | Oscillator In          | Crystal oscillator input                                   |
| XOUT    | 24         | Oscillator Out         | Crystal oscillator output                                  |
| VDDPLL  | 25         | Power                  | 1.8 V Power Input<br>(from internal voltage regulator)     |
| GNDPLL  | 26         | Power                  | Ground                                                     |
| VOUT1V8 | 27         | Power                  | 1.8 V Power Output<br>(of internal voltage regulator)      |
| RST~    | 28         | Digital I/O            | Reset (active low)                                         |
| VIN3V3  | 29         | Power                  | 3.3 V Power Input                                          |
| VDD3V3  | 30         | Power                  | 3.3 V Power                                                |
| AVDD3V3 | 31         | Power                  | 3.3 V Power                                                |
| NETN    | 32         | Comm                   | Network Port (polarity insensitive)                        |
| AGND    | 33         | Ground                 | Ground                                                     |
| NETP    | 34         | Comm                   | Network Port (polarity insensitive)                        |
| NC      | 35         | N/A                    | Do Not Connect                                             |
| GND     | 36         | Ground                 | Ground                                                     |
| TXON    | 37         | Comm                   | TxActive for optional network activity LED                 |
| RXON    | 38         | Comm                   | RxActive for optional network activity LED                 |
| CP4     | 39         | Comm                   | Connect to VDD33 through a 4.99 k $\Omega$ pullup resistor |
| CS0~    | 40         | Digital I/O for Memory | SPI slave select 0 (active low)                            |
| VDD3V3  | 41         | Power                  | 3.3 V Power                                                |

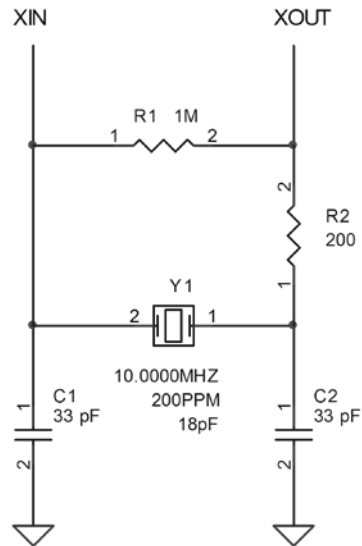
| Name     | Pin Number | Type                   | Description                                                       |
|----------|------------|------------------------|-------------------------------------------------------------------|
| VDD3V3   | 42         | Power                  | 3.3 V Power                                                       |
| SDA_CS1~ | 43         | Digital I/O for Memory | I <sup>2</sup> C: serial data<br>SPI: slave select 1 (active low) |
| VDD1V8   | 44         | Power                  | 1.8 V Power Input<br>(from internal voltage regulator)            |
| SCL      | 45         | Digital I/O for Memory | I <sup>2</sup> C serial clock                                     |
| MISO     | 46         | Digital I/O for Memory | SPI master input, slave output (MISO)                             |
| SCK      | 47         | Digital I/O for Memory | SPI serial clock                                                  |
| MOSI     | 48         | Digital I/O for Memory | SPI master output, slave input (MOSI)                             |
| PAD      | 49         | Ground Pad             | Ground                                                            |

## Clock Pins (XIN and XOUT)

Both the Router 5000 chip and FT Router 5000 chip require a 10 MHz external crystal or oscillator to provide its input clock signal. The chip then multiplies the input frequency by an amount specified in the device's hardware template (specified during device development using the NodeBuilder FX Development Tool; see *NodeBuilder Hardware Template*) to derive its internal system clock frequency. For multipliers greater than one, the chip uses a phase-locked loop (PLL) to drive and manage the internal on-chip system clock frequency.

A Series 5000 router chip requires a 10.0 MHz external clock signal for operation. An example part that meets the requirements for a Series 5000 router chip is the Abracon Corporation ABMM2-100000MHz-D1 Ceramic Surface Mount Low Profile Quartz Crystal.

The crystal must have a load capacitance rating of 18 pF. The internal capacitance for the **XIN** and **XOUT** pins is approximately 4.5 pF. To maintain the crystal's load capacitance, add a pair of 33 pF external capacitors, as shown in **Figure 13**. Note that **Figure 13** applies to a single Series 5000 half-router. Also, you must consider trace capacitance when calculating the values of the external capacitors. In the figure, the values for **R1** (feedback resistor) and **R2** (damping resistor) apply to any crystal used.



**Figure 13.** Series 5000 Chip Clock Generator Circuit

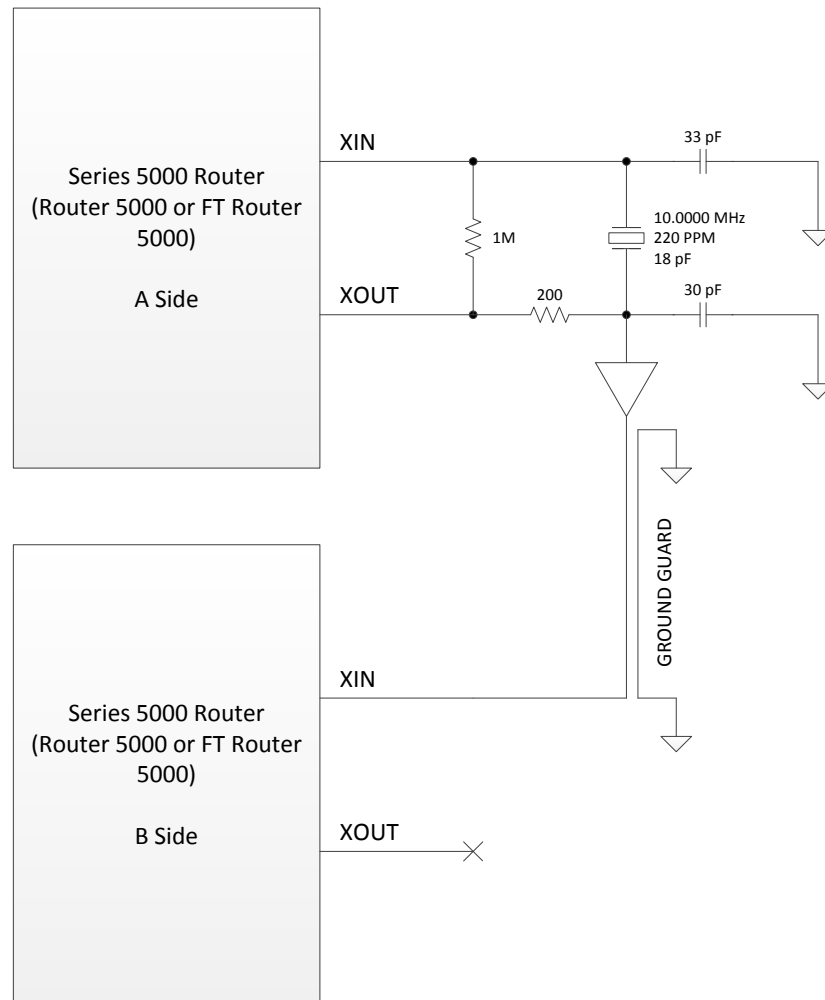
To ensure proper oscillator startup, the equivalent series resistance specification for the crystal should be  $\leq 50 \Omega$ , and the crystal shunt capacitance should be no greater than 7 pF.

Using a 33 pF capacitor for **C2** (in **Figure 13**), the Series 5000 router chip's **XOUT** pin cannot be used to drive an external CMOS load. However, if you maintain the required capacitance for the **XOUT** pin, you can drive an external clock, for example, for another Series 5000 half-router.

If your Series 5000 router device requires a common clock signal for both router halves, you can adjust the value for **C2** (in **Figure 13**), add a buffer, and leave the B Side **XOUT** unconnected, as shown in **Figure 14**. Clock traces should be kept short ( $\leq 2$  cm,  $\leq 0.8$  inch). Keep the crystal circuit close to the Router 5000 chips and isolated from communications lines. In addition, a logic ground guard must be added for the clock trace to minimize clock noise and to help keep EMI levels low. However, this ground guard should not be used as a ground source for digital circuitry.

In addition, the connection between A Side **XOUT** pin and the B Side **XIN** pin includes standard (inverting or non-inverting) bus buffer/line driver.

**Important:** Because the Series 5000 router A Side **XOUT** pin drives an input buffer, the values of the external capacitors are not equal. The value for A Side **XOUT** is specified as 30 pF based on an internal input capacitance of 4.5 pF of the **XIN/XOUT** pins and internal input capacitance for the buffer/line driver of 3 pF at 25 °C (so that the total capacitance for the A Side **XOUT** pin is 33 pF). For some bus buffer/line drivers, input capacitance can vary over temperature, up to 10 pF. If your device is likely to experience extreme temperatures, consider changing the value for the A Side **XOUT** capacitor to 27 pF to allow for the change in capacitance over temperature.



**Figure 14.** Common Clock Connections

See the *Series 5000 Chip Data Book* for more information about the clock requirements for a Series 5000 chip, including the Router 5000 and FT Router 5000.

If your router uses a Series 3100 half-router for one of its sides, and the Series 3100 Smart Transceiver is a Power Line transceiver or uses an input clock frequency other than 10 MHz, do not connect the Series 5000 router **XOUT** pin to the Series 3100 **CLK1** pin. Instead, use separate crystals for each router-half. If you do drive the Series 3100 **CLK1** pin from the Series 5000 router **XOUT** pin, you must connect them through a standard (inverting or non-inverting) bus buffer/line driver that supports TTL-compatible input and 5V CMOS output, such as an NXP® 74AHCT1G04 (or a 74AHCT1G126 with OE tied high).

See the *FT 3120 / FT 3150 Smart Transceiver Data Book* or the *PL 3120/PL 3150/PL 3170 Power Line Smart Transceiver Data Book* for more information about the clock pins for a Series 3100 chip.

## CP[4..0] – Router 5000 Only

The Router 5000 has a very versatile communications port, the **CP[4..0]** pins (39, 38, 37, 34, and 32). It consists of five pins that can be configured to interface to a

wide variety of media interfaces (network transceivers) and operates over a wide range of data rates.

The communications port for the Router 5000 is configured to operate in single-ended mode. **Table** lists the pin assignments for the communications port pins.

**Table 10.** Communications Port Pin Assignments

| Pin | Drive Current | Single-Ended Mode (3.3 V) | Connect To                           |
|-----|---------------|---------------------------|--------------------------------------|
| CP0 | N/A           | Data input                | Transceiver RXD                      |
| CP1 | 8 mA          | Data output               | Transceiver TXD                      |
| CP2 | 8 mA          | Transmit Enable output    | Transmit Enable (single ended mode)  |
| CP3 | N/A           | Do Not Connect            |                                      |
| CP4 | 8 mA          | Collision Detect input    | Collision Detect (single ended mode) |

Before programming, a Router 5000 uses its default communications parameters, which define a simplified single-ended mode 78 kbps channel. The default communications parameters allow you to load an application image over a 78 kbps network, for example during device manufacturing. Devices that use a 78 kbps transceiver (such as a 78 kbps EIA-485 transceiver or an LPT-11 Link Power Transceiver) can use the default communications parameters within development or manufacturing test networks. For production networks (networks with many devices), you should ensure that each device has communications parameters defined for the channel; see Appendix A, *Communications Parameters for LONWORKS Routers*.

Note that devices defined for a TP/XF-1250 channel cannot use the default communications parameters; each device's external serial non-volatile memory must be loaded with the correct communications parameters before connecting to the network.

See the *Series 5000 Chip Data Book* for more information about the communications port for the Neuron 5000 Processor, which is functionally equivalent to the Router 5000 communications port.

## NETP and NETN – FT Router 5000 Only

The FT Router 5000 has a simple communications port for connecting to TP/FT-10 network channels. It consists of two pins, **NETP** and **NETN**, which you connect to the corresponding pins of the FT-X3 Communications Transformer.

See the *Series 5000 Chip Data Book* for more information about connecting these pins to the transformer and connecting the transformer to the FT network.

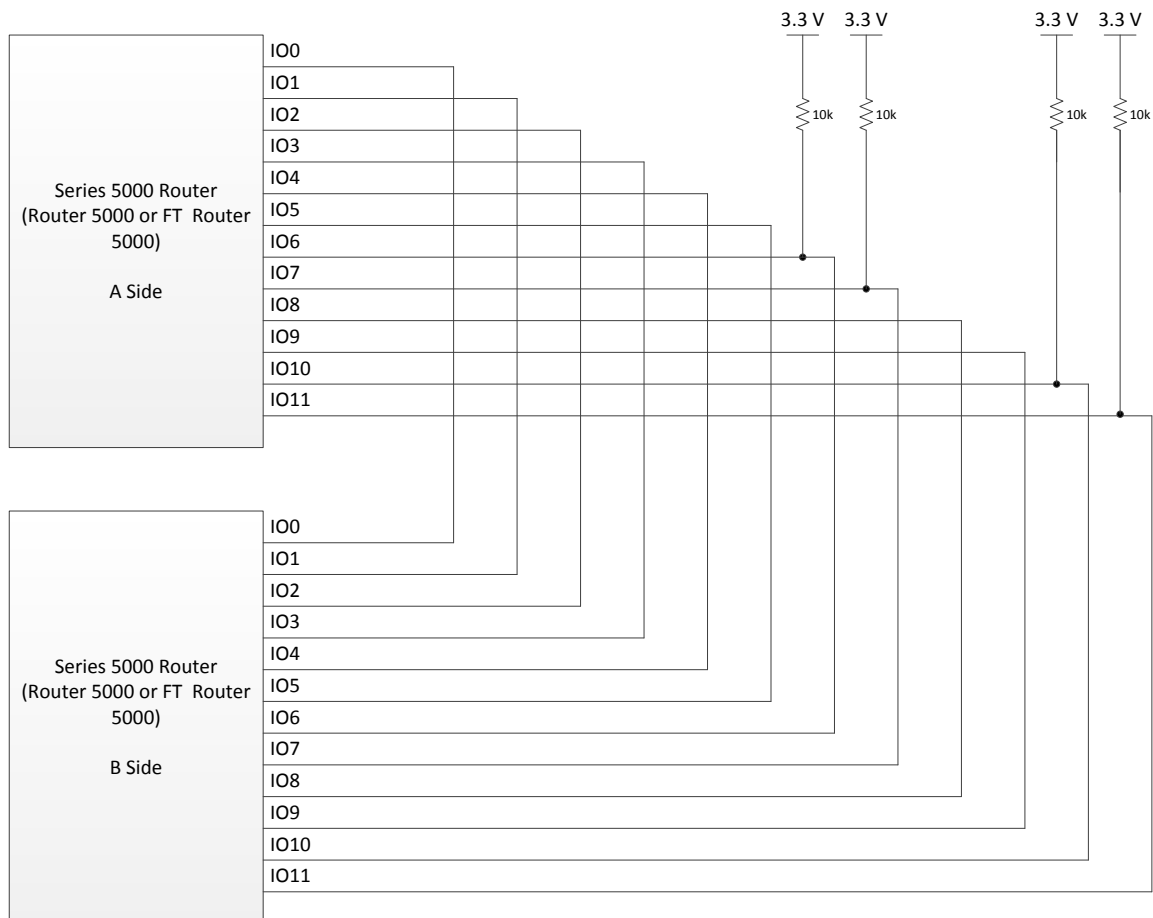
Like the Router 5000, an FT Router 5000's default communications parameters define a simplified single-ended mode 78 kbps channel – a TP/FT-10 channel.

## IO[11..0]

These digital I/O pins provide the communications between the A side and B side of a Series 5000 router device. Connect the IO pins for one router side to the corresponding IO pin on the other router side, as shown in **Figure 15**.

Note that you must provide 10 k $\Omega$  pull-up resistors for the **IO6**, **IO7**, and **IO10** pins. During power-up, the router half performs signal arbitration tests that require the pull-ups on **IO6** and **IO7**. The **IO10** pull-up is for the handshake signal between router halves. The **IO11** pin is not used for either router half, but it should be pulled up with a 10 k $\Omega$  pull-up resistor.

If your router uses a Series 3100 half-router for one of its sides, and the Series 3100 Smart Transceiver does not have an **IO11** pin, tie the Series 5000 half-router's **IO11** pin high with a 10 k $\Omega$  pull-up resistor.



**Figure 15.** Digital IO Pin Connections

**Important:** When routing the **IO[11..0]** signals between the two router halves of your Series 5000 router device, keep the traces as short as possible.

See the *Series 5000 Chip Data Book* for more information about the digital I/O pins for a Series 5000 chip, including the Router 5000 and FT Router 5000.

## JTAG Interface (TCK, TDI, TDO, TMS, and TRST~)

All Series 5000 chips (including the Router 5000 and FT Router 5000) provide an interface for the Institute of Electrical and Electronics Engineers (IEEE) Standard Test Access Port and Boundary-Scan Architecture (IEEE 1149.1-1990) of the Joint Test Action Group (JTAG) to allow a Series 5000 chip to be included in the boundary-scan chain for device production tests.

See the *Series 5000 Chip Data Book* for more information about the JTAG pins for a Series 5000 chip, including the Router 5000 and FT Router 5000.

## Memory Interface (CS0~, MISO, MOSI, SCK, SCL, and SDA\_CS1~)

The interface for accessing off-chip non-volatile memory (NVM) is a serial interface that follows either of the following protocols: serial Inter-Integrated Circuit (I<sup>2</sup>C) or serial peripheral interface (SPI). Although a Series 5000 chip supports both Electrically Erasable Programmable Read-Only Memory (EEPROM) devices and flash memory devices, a typical Router 5000 or FT Router 5000 device uses a single 2 KB EEPROM device (using either the I<sup>2</sup>C protocol or the SPI protocol). This EEPROM device contains configuration data for the router. If you supply an EEPROM device larger than 2 KB, the additional memory space is not used.

**Recommendation:** Your router design should allow for in-circuit programmability of the serial EEPROM device, unless the EEPROM devices must be programmed before device assembly.

See the *Series 5000 Chip Data Book* for more information about how to use the memory interface pins for a Series 5000 chip, including the Router 5000 and FT Router 5000.

## Power and Ground

Connect the **VDD3V3** pins (8, 18, 29, 30, 41, and 42) to V<sub>DD33</sub>. Also connect the **AVDD3V3** pin (31) to an analog V<sub>DD33</sub> source, if different from the digital V<sub>DD33</sub> source. In general, the **VDD3V3** pins and the **AVDD3V3** pin connect to the same V<sub>DD33</sub> source.

The **VOUT1V8** pin (27) is the output of the on-chip voltage regulator. Connect the **VDD1V8** pins (6, 16, and 44) to the **VOUT1V8** pin (27) to connect the 1.8 V input pins to the output of the internal voltage regulator.

**Important:** Do not connect an external 1.8 V source to any of the **VDD1V8** pins (6, 16, and 44). Connect these pins to the **VOUT1V8** pin (27) only. **Using an external 1.8 V source voids the warranty for the chip, and can cause unpredictable and possibly irreparable results.**

Connect the **VDDPLL** pin (25) to the **VOUT1V8** pin (27), with an associated chip ferrite bead. Connect the **GNDPLL** pin (26) to GND, with an associated chip ferrite bead.

Connect the **GND** pin (36) and the chip's pad (pin 49) to logic ground. Also connect the **AGND** pin (33) to logic ground.

See the *Series 5000 Chip Data Book* for more information about the power and ground requirements for a Series 5000 chip, including the Router 5000 and FT Router 5000.

If your router uses a Series 3100 half-router for one of its sides, you can use a low-dropout voltage regulator to provide power for the Series 5000 half-router (+5 V input from the Series 3100 router half power supply, and +3.3 V output for the Router 5000 chip. See *Connecting Half-Routers: Series 5000 and Series 3100* for more information.

## RST~

The **RST~** pin is both an input and an output. As an input, the **RST~** pin is internally pulled high by a resistor. The **RST~** pin becomes an output when any of the following events occur:

- Internal LVI detects a low voltage condition
- Software reset initialization
- Watchdog Timer event (times out)
- Traps

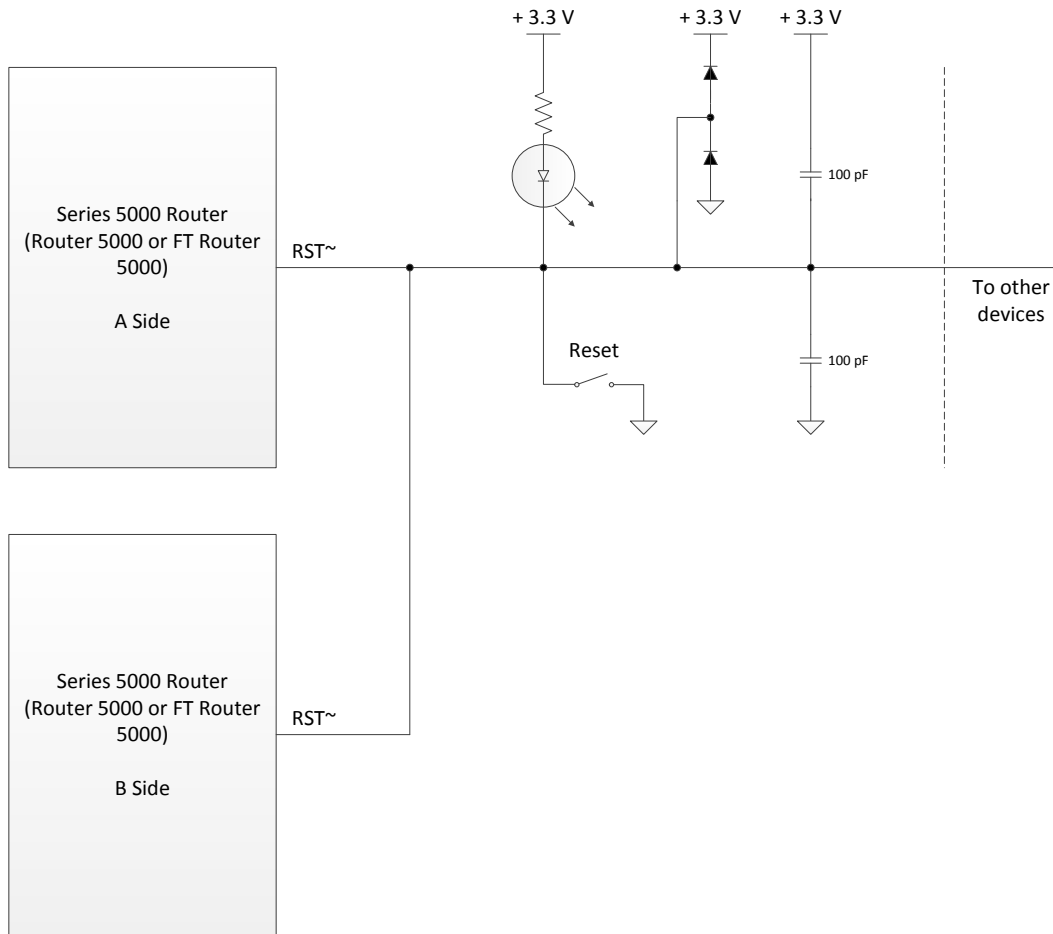
In some cases it is desirable to use the input capability of the **RST~** pin to allow other devices to reset the Series 5000 half-router. Examples of external devices that can be used for this purpose include push button switches, microcontrollers, and external low-voltage detectors.

**Important:** If the proper external reset circuitry is not used, the Series 5000 router can become applicationless or unconfigured. The applicationless or unconfigured state occurs when the checksum error verification routine detects corruption in memory which could have been falsely detected because of an improper reset sequence.

The following guidelines must be followed in order for the Series 5000 router's reset functions to operate reliably:

- Any device connected to the **RST~** pin must have an open-drain (or equivalent) output. If an external device were to actively drive the **RST~** pin high, contention between that device and the Series 5000 router's internal circuitry could result in anomalous behavior ranging from applicationless errors to device failure.
- A capacitor should be connected between **RST~** and ground to provide noise immunity. The value of this capacitor should be at least 100 pF, and must not exceed 1000 pF. For even greater noise immunity, two capacitors (totaling  $\leq 1000$  pF) can be used, with one connected from the **RST~** pin to ground and the other from **RST~** to  $V_{DD33}$ . These capacitors should be located within 5 mm of the Series 5000 router chip's **RST~** pin.
- During board level in-circuit testing (ICT), the **RST~** pin should be hard wired to ground through a "pogo pin".

**Figure 16** shows an example reset circuit, where the A side and B side reset pins are tied together.



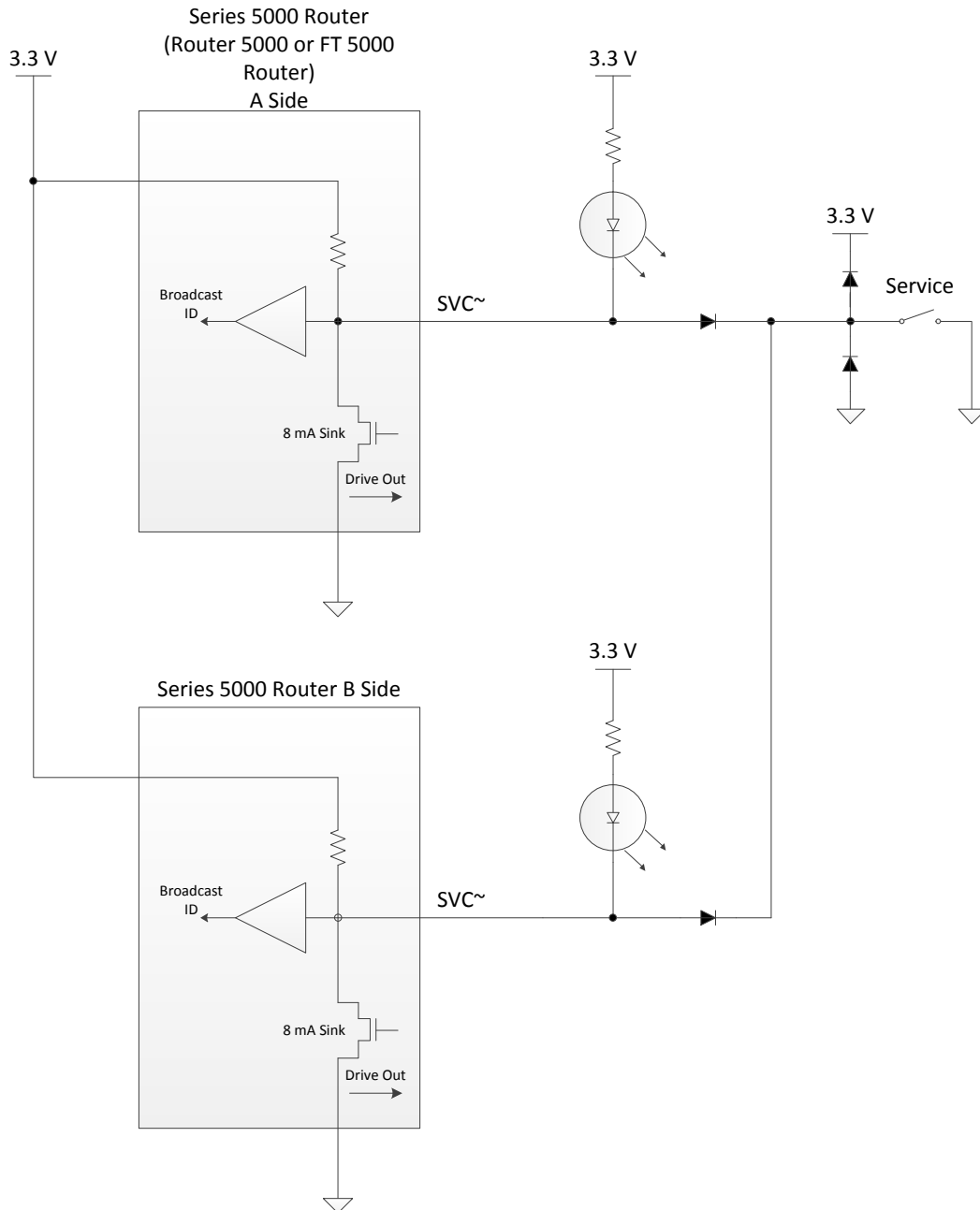
**Figure 16.** Reset Circuit – Series 5000 Router for Both Halves

If your router uses a Series 3100 half-router for one of its sides, you can connect the Router 5000's **RST~** pin to the Series 3100 **RESET~** pin, as shown in as shown in **Figure 17**, using the Series 3100 Smart Transceiver's +5 V power supply for the Reset switch, LED, diode clamps, and EMC capacitors.



its unique 48-bit Neuron ID and the application's program ID. This information can then be used by a network management tool to install and configure the router. **Table 11** lists the state of the Service LED for various device states. The Neuron firmware samples the **SVC~** pin whenever it is not actively driving the pin low.

A typical circuit for the **SVC~** pin, where the A side and B side service pins are tied together, but with separate Service LEDs, is shown in **Figure 18**. During reset, each **SVC~** pin is pulled high by its internal pull-up resistor. Alternatively, you could provide separate service pin buttons for each router side.



**Figure 18.** Service Circuit – Series 5000 Router for Both Halves

**Table 11.** Service LED Behavior during Different States

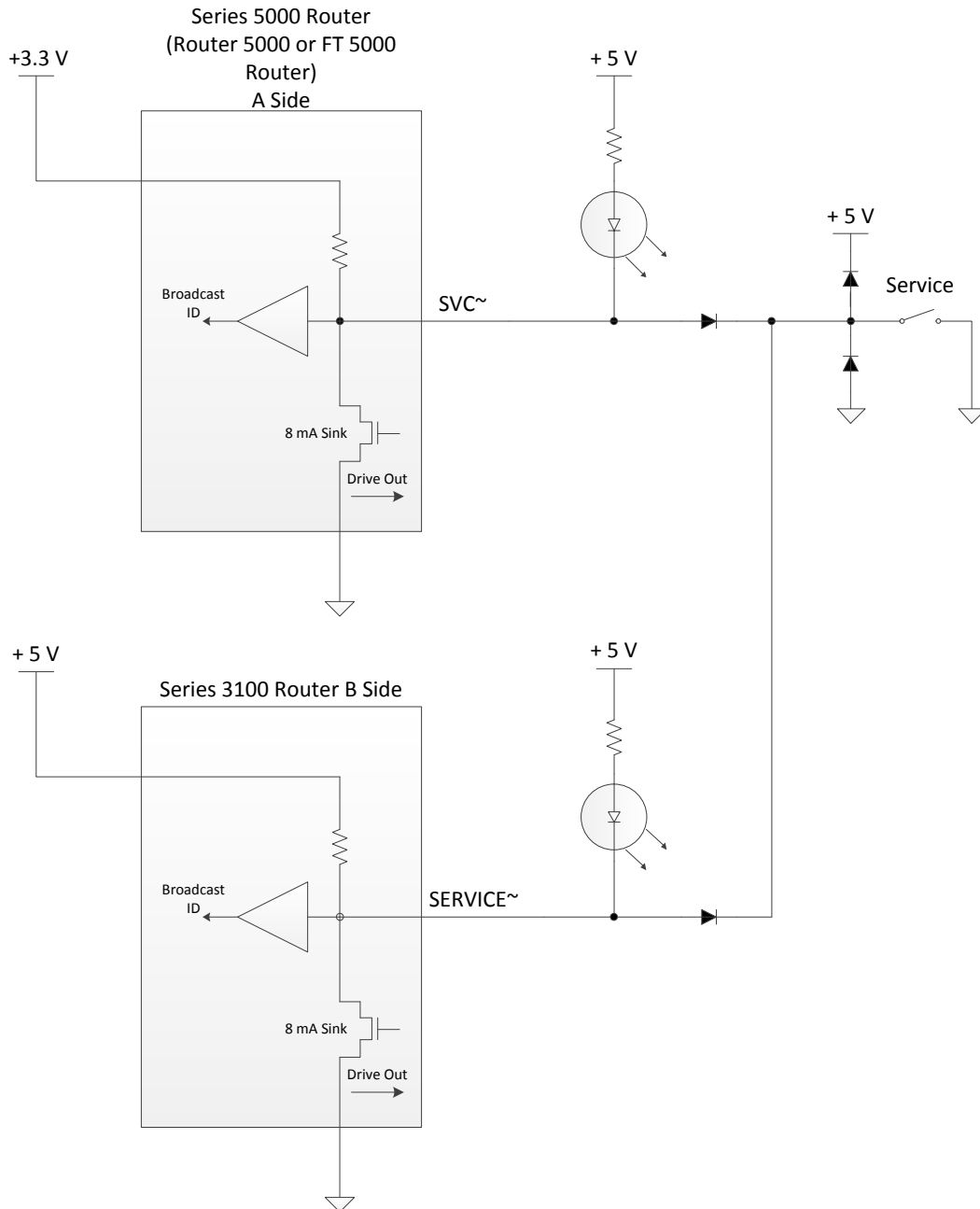
| Device State                           | State Code | Service LED |
|----------------------------------------|------------|-------------|
| Applicationless and Unconfigured       | 3          | On          |
| Unconfigured (but with an Application) | 2          | Flashing    |
| Configured, Hard Offline               | 6          | Off         |
| Configured                             | 4          | Off         |
| Defective External Memory              | —          | On          |

The **SVC~** pin is active low, and the service pin message is sent once per **SVC~** pin transition. The service pin message goes into the next available non-priority output network buffer.

Typical applications do not require debounce conditioning of a momentary pushbutton attached to the **SVC~** pin. The software response time associated with this input is long enough to effectively provide a software debounce for switches with a contact bounce settling time as long as 20 milliseconds.

If your router uses a Series 3100 half-router for one of its sides, you can connect the Series 5000 router's **SVC~** pin to the Series 3100 **SERVICE~** pin, as shown in **Figure 19**, using the Series 3100 Smart Transceiver's +5 V power supply for the Service switch, LEDs and diode clamps.

See the *Series 5000 Chip Data Book* for more information about the **SVC~** pin for a Series 5000 chip, including the Router 5000 and FT Router 5000.



**Figure 19.** Service Circuit – Series 5000 Router with Series 3100 Router

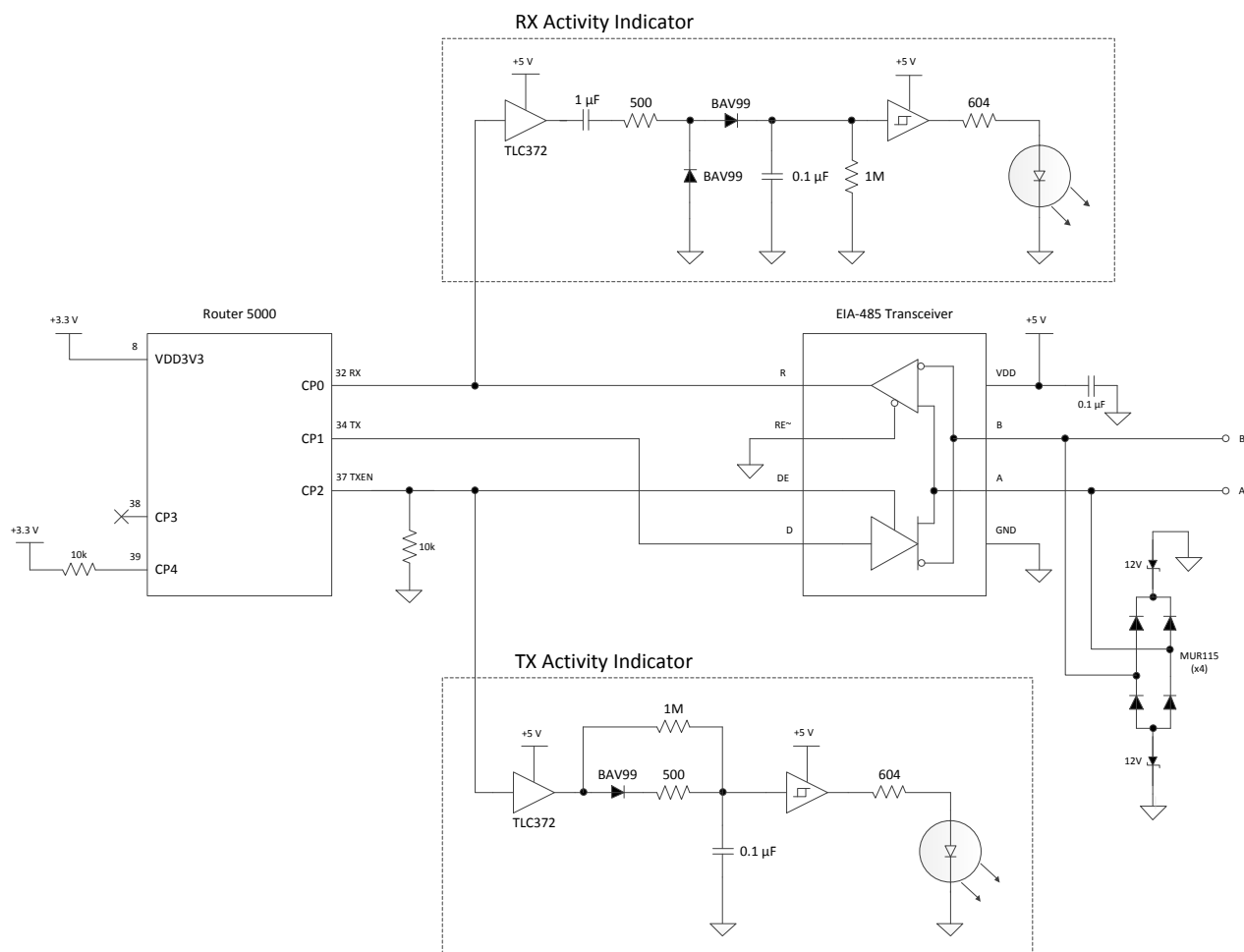
## Network Activity Indicator – Router 5000

Although the Router 5000 does not provide separate network indicator pins, you can use the **CP[4..0]** pins with appropriate indicator circuits to provide this functionality. For an RX network activity indicator, connect an appropriate LED circuit with pulse stretching to the Router 5000 **CP0** pin. For a TX network activity indicator, connect an appropriate LED circuit with pulse stretching to the Router 5000 **CP2** pin. If you add LEDs to the **CP0** or **CP2** pins, you must add a buffer to ensure that the LED's operation does not interfere with network communications.

**Figure 20** shows example RX and TX network activity indicator circuits for a Router 5000 half-router connected to an EIA-485 transceiver. You can use the same network activity circuits for other transceiver types, although other transceiver types have different connections to the Router 5000. See *Developing a Router with the Router 5000 Chip* for more information about connecting external transceivers to a Router 5000.

Both network indicator circuits use non-inverting bus buffer/line drivers that support TTL-compatible input and 5V CMOS output (assuming a transceiver that requires 5 V supply voltage). Both circuits also use standard rectifying diodes; if your transceiver uses 3.3 V supply voltage, consider replacing these diodes with Schottkey diodes.

For the TX network indicator, the Router 5000 **CP2** pin (TX Enable) is low when idle. However, for the RX network indicator, the Router 5000 **CP0** pin (RX) retains its state from the end of the previous received bit, and thus can be high or low when idle.



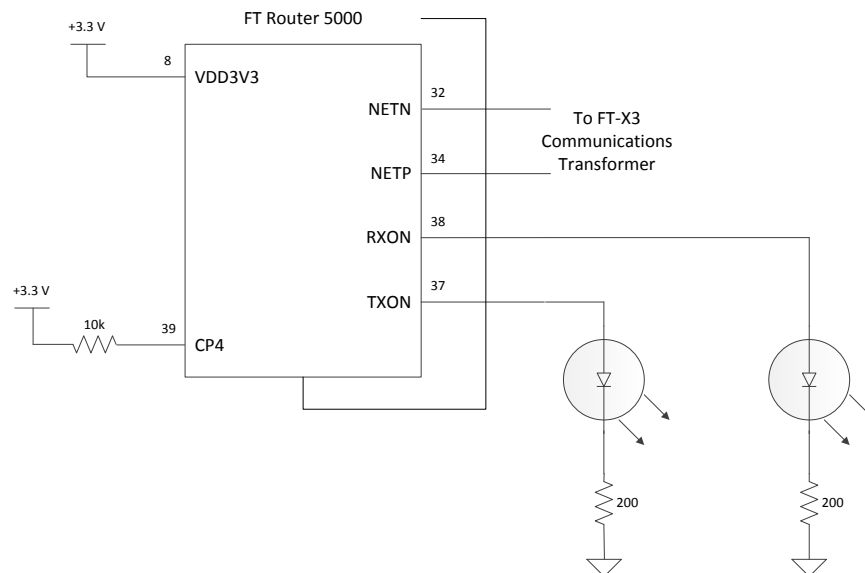
**Figure 20.** RX and TX Network Activity Indicator Circuits – Router 5000

When packets are transmitted, the TX network activity LED is active for the duration of the entire data transmission. When packets are received, the RX network activity LED is active for each bit received, and inactive between bytes. For both circuits, the approximate time constant for LED visibility is 100 ms.

## Network Activity Indicator – FT Router 5000

The FT Router 5000 provides two network indicator pins, **RXON** and **TXON**, that you can connect to LEDs to show network activity, as shown in **Figure 21**.

You can optionally add pulse-stretching circuits to increase the visibility of the LEDs, although they are likely visible enough without pulse stretching. Also, because the **RXON** and **TXON** pins are not directly involved in communications, you do not need to add buffers.



**Figure 21.** RX and TX Network Activity Indicators – FT Router 5000

---

## Power Requirements

The following sections describe the power requirements for a LONWORKS router.

---

### *RTR-10 Power Requirements*

An RTR-10 router requires a +5 VDC  $\pm 10\%$  at 200 mA.

---

### *Series 5000 Router Power Requirements*

A Series 5000 router chip requires a +3.3 VDC power source with sufficient current to power the chip in all modes of operation.

The supply current requirements for the Series 5000 router chip are outlined in **Table 12**, including typical requirements for the different operating states of the Router 5000 at various system clock rates.

**Important:** Although general Series 5000 chips support 80 MHz operations, neither the Router 5000 chip nor the FT Router 5000 chip supports this system clock setting.

**Table 12.** Series 5000 Router Current Requirements

| Active           | SysClk     | Typical                 | Maximum                 |
|------------------|------------|-------------------------|-------------------------|
| Receive Current  | 5 MHz      | 9 mA                    | 15 mA                   |
|                  | 10 MHz     | 9 mA                    | 15 mA                   |
|                  | 20 MHz     | 15 mA                   | 23 mA                   |
|                  | 40 MHz     | 23 mA                   | 33 mA                   |
| Transmit Current | 5 – 40 MHz | Receive Current + 15 mA | Receive Current + 18 mA |

The Series 5000 router chip requires a 3.3 V nominal power supply (3.0 V to 3.6 V range). The current requirements assume no load on digital I/O pins, and that the I/O lines are not switching. In addition, the current consumption in transmit mode represents a peak value rather than a continuous usage value because a Series 5000 router does not typically transmit data continuously.

Note that the stated current requirements do not include the requirements for performing reads or writes to the external memory (the two-wire serial EEPROM), which typically add 1 to 2 mA. When not in use, the EEPROM typically requires only 2  $\mu$ A.

---

## Power Supply Decoupling and Filtering

The design for a LONWORKS router power supply must consider filtering and decoupling requirements of the router. The power supply filter must prevent noise generated by the router from conducting onto external wires, and in the case of DC-DC switching power supplies, must prevent noise generated by the supply from interfering with router operation. Switching power supply designs must also consider the effects of radiated EMI.

An RTR-10 router or a Series 5000 half-router each requires a clean power supply to prevent RF noise from conducting onto the network through active drive circuits. Power supply noise near the network transmission frequency could degrade network performance.

The RTR-10 router includes 2.2  $\mu$ F and 0.1  $\mu$ F power supply bypass capacitors close to pins 10 and 31. In general, high-frequency decoupling capacitors valued at 0.1  $\mu$ F or 0.01  $\mu$ F placed near pins 10 and 31 on the motherboard are necessary to reduce EMI.

See the *Series 5000 Chip Data Book* for information about power-supply decoupling and filtering for Series 5000 chips, including the Router 5000 and FT Router 5000.

---

## Low Voltage Protection

For a RTR-10 design, it is necessary to include a low voltage protection circuit on the router motherboard to drive the **RESET~** line of the RTR-10 router. See Section 9.4 of the *Neuron Chip Data Book*. Failure to include such protection may cause data corruption to configuration data maintained in EEPROM on the

RTR-10 Neuron Chips. In the sample circuit of **Figure 27**, protection is provided by a Motorola MC33164.

See the *Series 5000 Chip Data Book* for information about internal low-voltage indications for Series 5000 chips, including the Router 5000 and FT Router 5000.



# 3

## **LONWORKS Router Mechanical Interfaces**

This chapter provides an overview of the mechanical interfaces for the RTR-10 Router Core Module, the Router 5000 chip, and the FT Router 5000 chip.

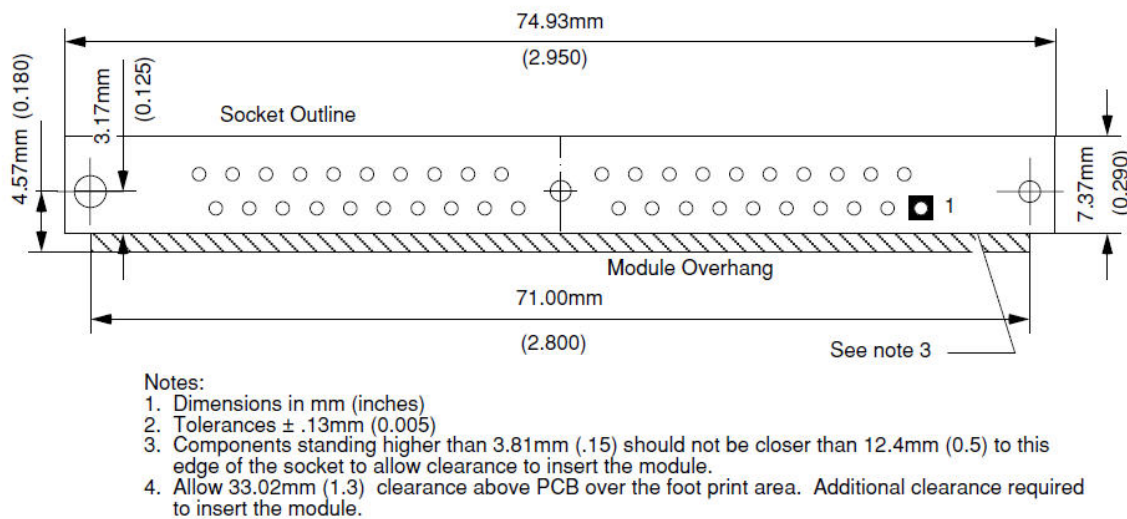
## RTR-10 Mechanical Description

The RTR-10 Router Core Module consists of a 67 mm by 23 mm by 7 mm (2.65 in by 0.9 in by 0.3 in) module with the core electronics and firmware required to implement a router. The RTR-10 is attached to a motherboard, using a 40-position 0.050-inch spacing SIMM socket, such as a Molex® Incorporated 1.27mm (.050") Pitch SIMM Socket:

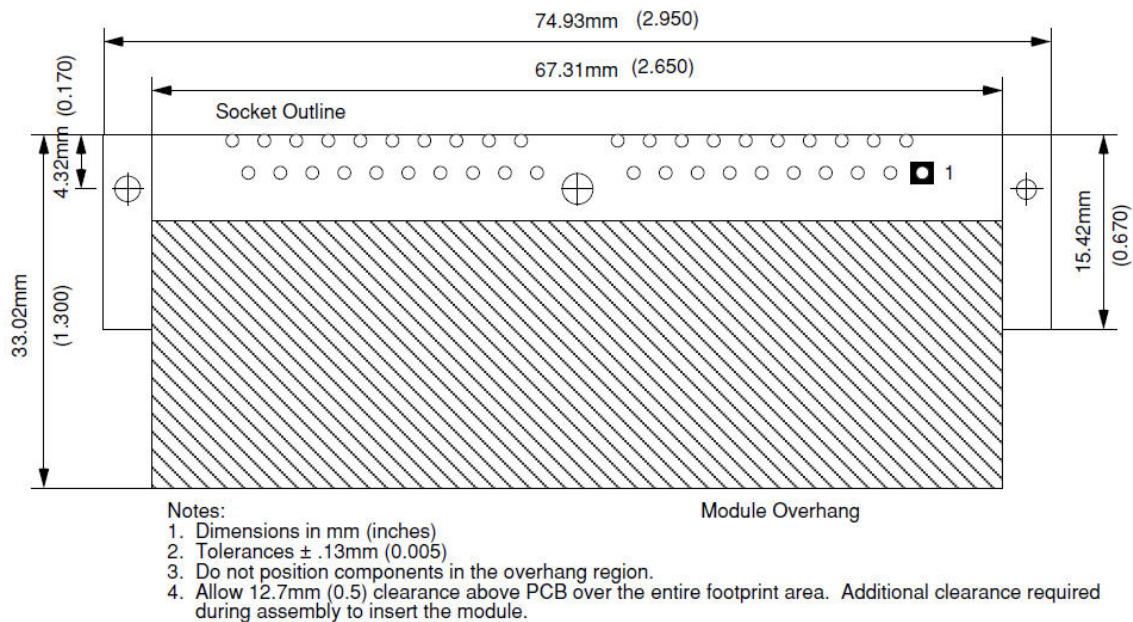
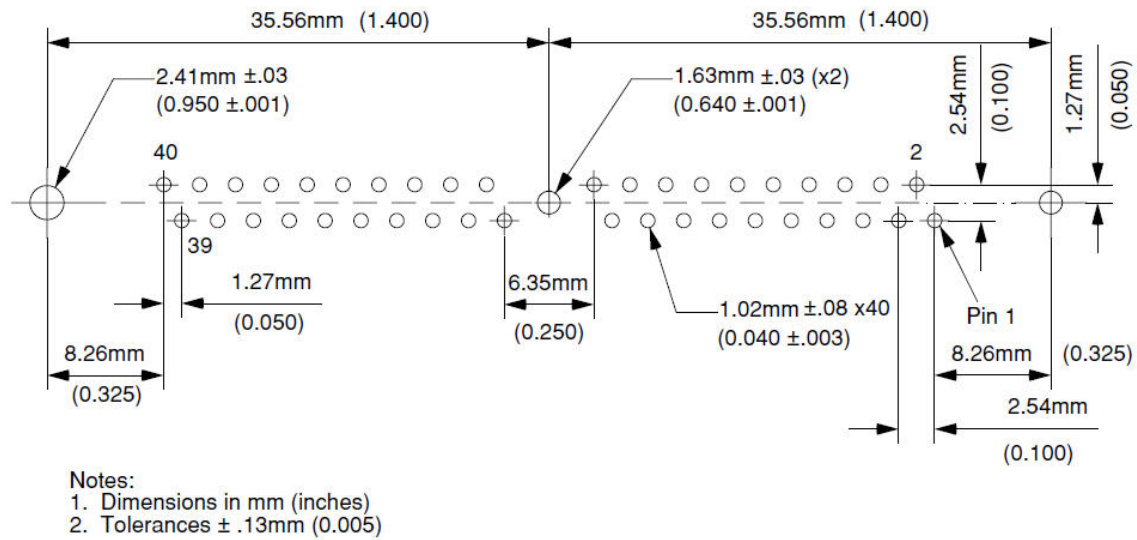
- [www.molex.com/customer.html?supplierPN=015821390](http://www.molex.com/customer.html?supplierPN=015821390)
- [www.molex.com/customer.html?supplierPN=015820793](http://www.molex.com/customer.html?supplierPN=015820793)

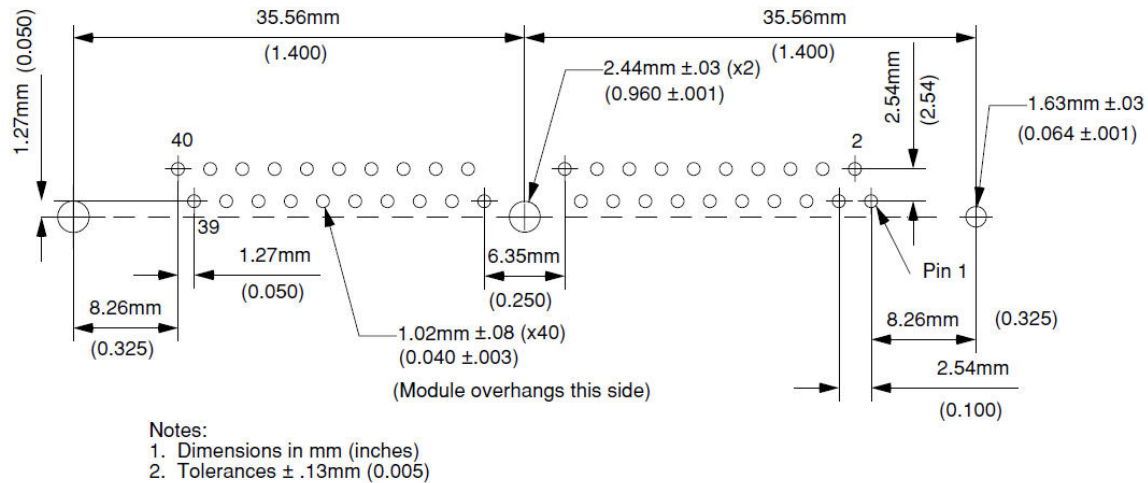
However, these Molex SIMM sockets are obsoleted and are unavailable for purchase from Molex. Echelon has a limited supply of these sockets (models 61101R and 61102R); contact Echelon Support for more information.

The following figures show recommended mechanical layouts for the RTR-10: **Figure 22** shows the vertical socket mechanical footprint, **Figure 23** shows the vertical socket pad layout, **Figure 24** shows the right-angle socket mechanical footprint, and **Figure 25** shows the right-angle socket pad layout.



**Figure 22.** RTR-10 PCB Footprint (Component Side, Vertical Mounting)





**Figure 25.** RTR-10 Recommended PCB Hole Pattern (Component Side, Horizontal Mounting)

Decisions about component placement on the motherboard must consider electromagnetic interference (EMI) and electrostatic discharge (ESD) issues; see Chapter 5, *LONWORKS Router Design Issues*.

## Series 5000 Router Mechanical Description

The mechanical description of the Series 5000 router chip is similar to the mechanical description of general Series 5000 chips, as described in the *Series 5000 Chip Data Book* and the data sheet for the Neuron 5000 Processor or FT 5000 Smart Transceiver.

**Figure 26** shows the mechanical specifications for a Series 5000 router chip.

# PACKAGE OUTLINE 48L QFN 7.0x7.0x0.9 mm

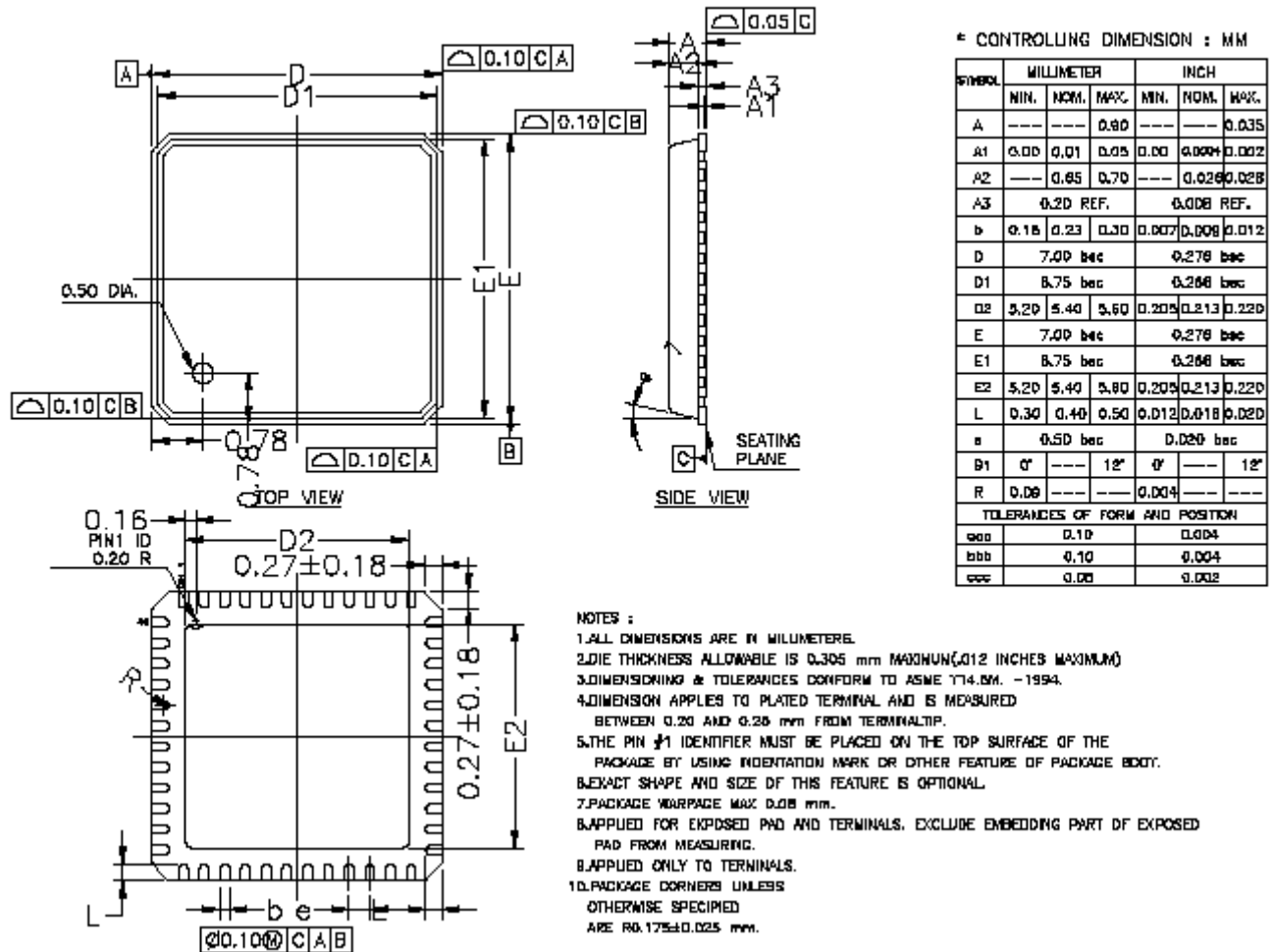


Figure 26. Series 5000 Router Mechanical Specifications



# 4

## Developing a LONWORKS Router

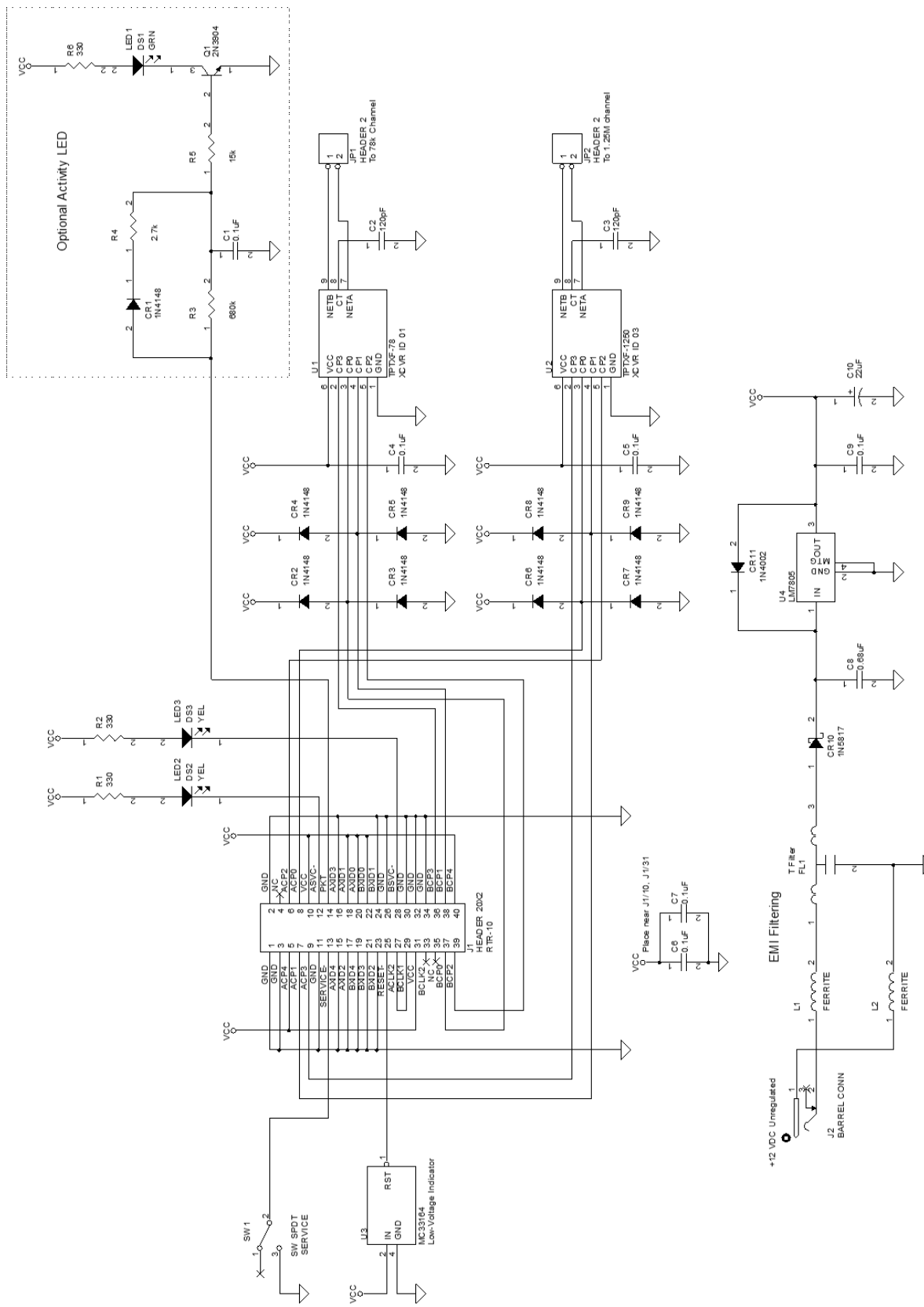
This chapter describes the process of developing a router based on the RTR-10 Router Core Module, the Router 5000 chip, or the FT Router 5000 chip.

---

## Developing a Router with the RTR-10 Module

To create a LONWORKS router with the RTR-10, perform the following steps:

1. Build a router motherboard according to the specifications described in Chapter 2, *LONWORKS Router Electrical Interfaces*, and the guidelines described in Chapter 5, *LONWORKS Router Design Issues*. The motherboard can be part of custom application hardware, or can be a standalone board. **Figure 27** shows a sample motherboard schematic for a TP/XF-78 to TP/XF-1250 twisted pair router. Additional transceiver interfaces are described in the next section.
2. Ensure that the communications parameters in the RTR-10 router are compatible with both of the transceivers. The transceivers listed in **Table 7** are supported directly by the RTR-10 router as predefined types. Set the transceiver ID lines to select the proper transceiver type. For custom transceivers, modify the communications parameters as described in *Using Custom Transceivers*.
3. Assemble the router, including the RTR-10 router, two transceivers, and a motherboard.
4. Install the router on a network as described in Chapter 6, *Installing a LONWORKS Router*. The network could be a development network for initial testing, a manufacturing network for configuration during manufacture, or a production network for field installation.



**Figure 27.** RTR-10 Motherboard Example Schematic

---

## Using Predefined Transceivers

The RTR-10 router includes built-in transceiver parameters for the transceivers listed in **Table 7**. When using any of these transceivers, the communications parameters are automatically programmed, as described in Chapter 2, *LONWORKS Router Electrical Interfaces*.

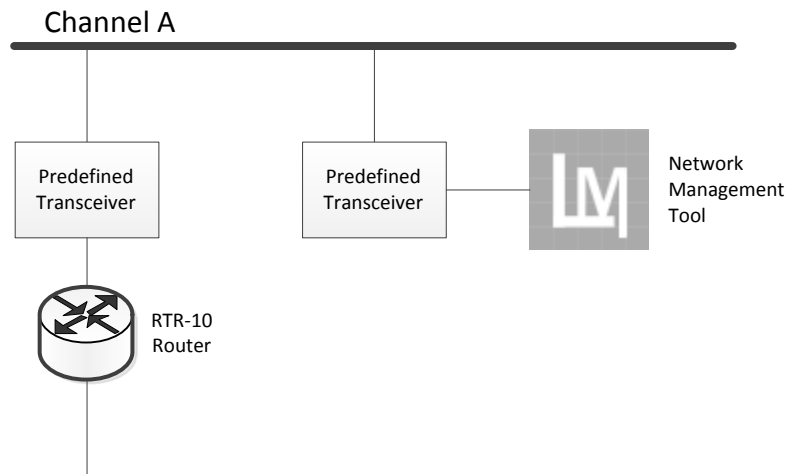
The user's guide for each transceiver contains documentation on the interface requirements. You also must set the transceiver ID input for each side of the RTR-10..

---

## Using Custom Transceivers

The RTR-10 router can be used with transceivers not listed in the table on page 25, but the communications parameters must be reprogrammed to match the custom transceiver. If one side of the router is a predefined transceiver type, this reprogramming can occur during manufacturing or during field installation. The first four steps of the following procedure describe how the custom communications parameters are programmed for one side. If both sides of the custom router will be custom transceiver types, additional configuration steps will be required, as described in steps 5 to 10.

1. Assuming that the predefined transceiver is Side A, attach a transceiver matching one of the predefined types to Side A of the RTR-10 and select the matching transceiver ID for Side A.
2. Select the custom transceiver type (ID 30, 0x1E) for Side B of the RTR-10 router.
3. Attach a network management tool, such as OpenLNS CT, with a compatible predefined transceiver to Channel A as shown in **Figure 28**.

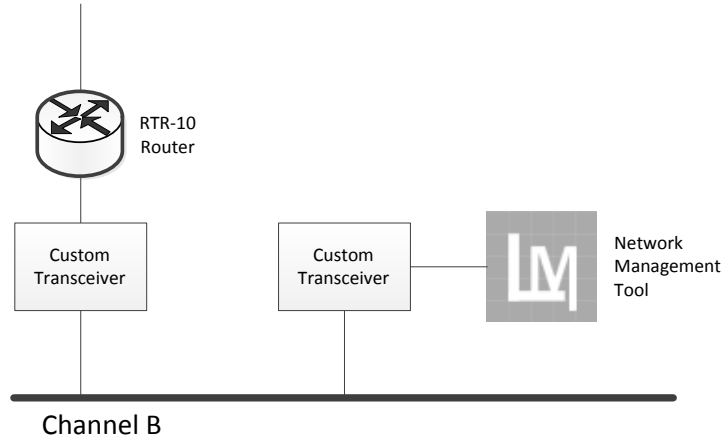


**Figure 28.** Configuring Side B

4. Configure the communications parameters on Side B of the RTR-10 router using the network management tool. Side A might be automatically reconfigured at the same time, depending on the network management tool. Installation procedures for the OpenLNS CT are described in Chapter 6, *Installing a LONWORKS Router*.

The preceding four steps complete the configuration when a single custom transceiver is used. Proceed with the following steps if two custom transceivers are to be used with the RTR-10 router.

5. Remove power from the RTR-10 router.
6. Disconnect the predefined transceiver from Side A.
7. Select the custom transceiver ID (type 30, 0x1E) on Side A.
8. Attach the selected custom transceiver to Side B as shown in **Figure 29**, leaving the Side B transceiver ID set to 30 (0x1E).



**Figure 29.** Configuring Side A

9. Attach a network services tool with a compatible custom transceiver to Channel B, as shown in **Figure 29**.
10. Configure the communications parameters on side A of the RTR-10 router using the network management tool. Side B might be automatically reconfigured at the same time depending on the network management tool.

---

## Developing a Router with the Router 5000 Chip

To create a LONWORKS router with the Router 5000, perform the following steps:

1. Build a router motherboard according to the specifications described in Chapter 2, *LONWORKS Router Electrical Interfaces*, and the guidelines described in Chapter 5, *LONWORKS Router Design Issues*. The motherboard can be part of custom application hardware, or can be a standalone board.
2. Program the serial EEPROM for each router half. The serial EEPROM holds the router configuration and communications parameters for the transceiver. See *Configuring a Series 5000 Half-Router* for more information about router configuration. You can program the EEPROMs either before assembly or in-circuit after assembly.
3. Assemble the router, including the two Router 5000 half-routers, two transceivers, and a motherboard.
4. Install the router on a network as described in Chapter 6, *Installing a LONWORKS Router*. The network could be a development network for

initial testing, a manufacturing network for configuration during manufacture, or a production network for field installation.

The following sections describe how to connect the Router 5000 to various transceiver types. After these descriptions is a set of example schematics for a simple TP/XF-1250 to EIA-485 twisted pair router.

---

## Using an External Transceiver with the Router 5000

You can develop a Router 5000 half-router using any of the following transceiver types:

- Echelon TPT Twisted Pair Transceiver Module for a TP/XF-1250 channel
- EIA-485 transceiver for various channel types
- Echelon FTT-10A Free Topology Twisted Pair Transceiver for a TP/FT-10 channel
- Echelon LONWORKS LPT-11 Link Power Twisted Pair Transceiver for a 78 kbps link-power channel

The following sections describe the basic electrical connections for each of these transceiver types.

You can also connect a Router 5000 half-router to a Series 3100 half-router for other transceiver and channel types, for example, a PL-20 channel. See *Connecting Half-Routers: Series 5000 and Series 3100* for more information.

## Using a TP/XF-1250 Transceiver

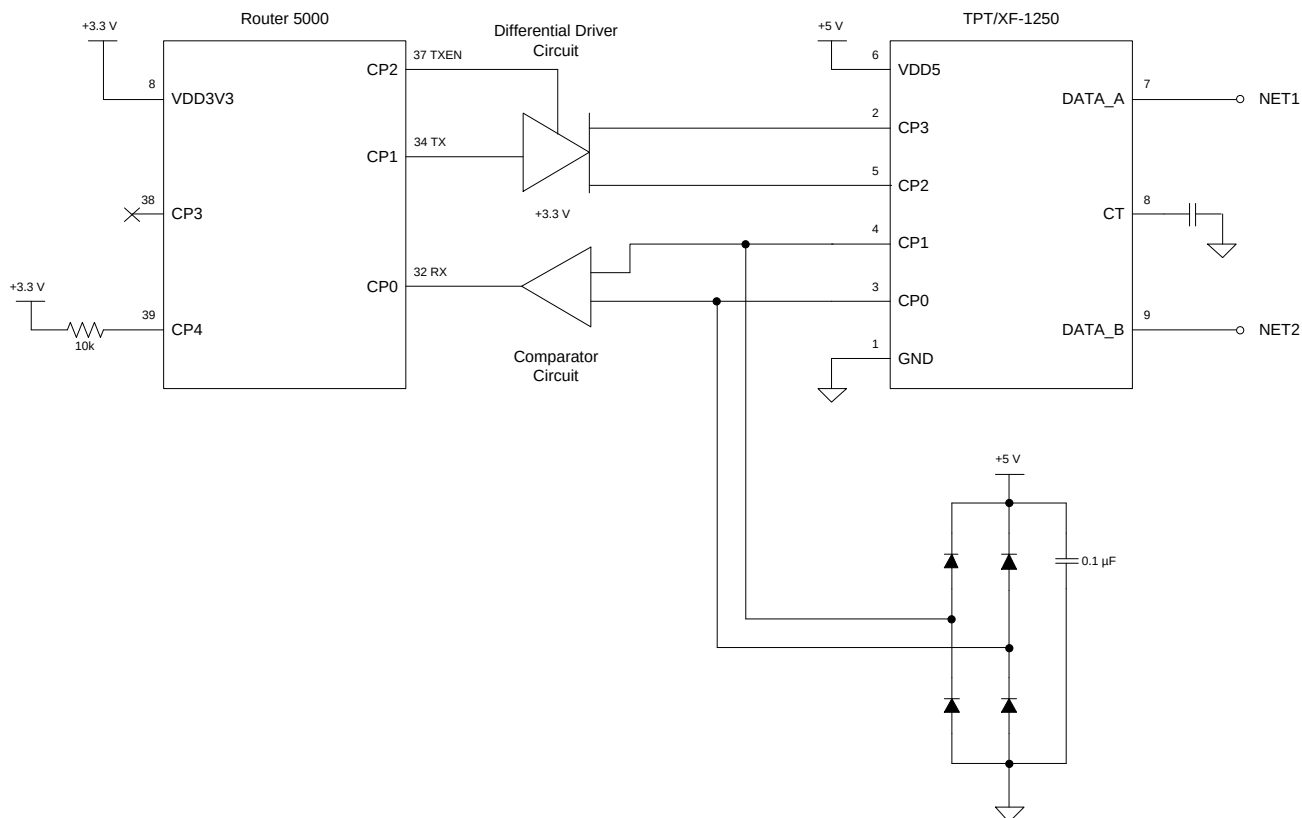
You can use a Router 5000 with an Echelon TPT Twisted Pair Transceiver Module for a TP/XF-1250 channel. However, because the Router 5000 does not include an on-chip differential transceiver, you must:

- Select “TP/XF-1250” as the transceiver type within the Hardware Template Editor of the NodeBuilder FX Development Tool or the Mini FX Evaluation Kit. Within this template, select “Neuron 5000” as the Neuron Chip Model. Using the TP/XF-1250 template causes the Neuron firmware to configure the Neuron 5000 Processor’s communications port to operate in 3.3 V single-ended mode.

**Important:** Select a clock multiplier of at least 2 (to use a 20 or 40 MHz system clock). However, if the other router-half uses a Series 3100 chip, do not specify a value higher clock multiplier value than 2. You can specify a value of 4 if both sides are Router 5000 chips. Do not specify a value of 8.

- Add a single-ended mode to differential mode converter circuit, as described in *Differential Driver Circuit*, and a differential comparator circuit as described in *Comparator Circuit*. These circuits convert the Router 5000’s 3.3 V single-ended mode signals to the 5 V differential mode signals required for the TPT/XF-1250 transceiver.

**Figure 30** shows the basic configuration for connecting a Router 5000 to a TPT/XF-1250 transceiver.



**Figure 30.** Connecting a Router 5000 to a TP/XF-1250 Transceiver

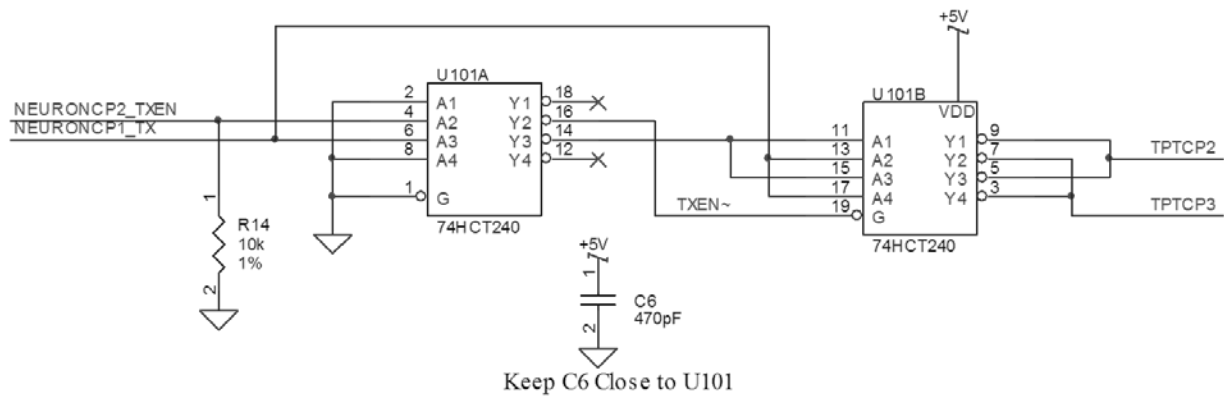
In the figure, the pullup resistor for the Router 5000's **CP4** pin is optional, but helps prevent contention on the **CP4** pin if the router is incorrectly configured to operate in special-purpose mode (for which the **CP4** pin is an output). The diode clamps for the TPT/XF-1250 transceiver's **CP0** and **CP1** signals are high-speed switching diodes, such as Fairchild Semiconductor® 1N4148 small-signal diodes. The value of the capacitor on the TPT/XF-1250 transceiver's transformer center tap (CT) pin depends on the device's PCB layout and EMI characteristics. A typical value is 100 pF rated for 1000 V.

See the *LonWorks TPT Twisted Pair Transceiver Module User's Guide* for information about the TPT/XF-1250 Transceiver.

## Differential Driver Circuit

**Figure 31** shows a differential driver circuit for connecting a Router 5000 to a TPT/XF-1250 transceiver. The differential driver circuit buffers the Router 5000's transmit (**CP1**) signal and transmit enable (**CP2**) signal to generate the TPT/XF-1250 transceiver's differential transmit signals (**CP2** and **CP3**).

The heart of the differential driver circuit is a pair of 4-bit buffers/drivers in a single 74HCT240 octal inverting buffer/line driver (such as the Texas Instruments™ SN74HCT240 Octal Buffer and Line Driver with 3-State Outputs).



**Figure 31.** Differential Driver Circuit

**Table 13.** Bill of Materials for the Differential Driver Circuit

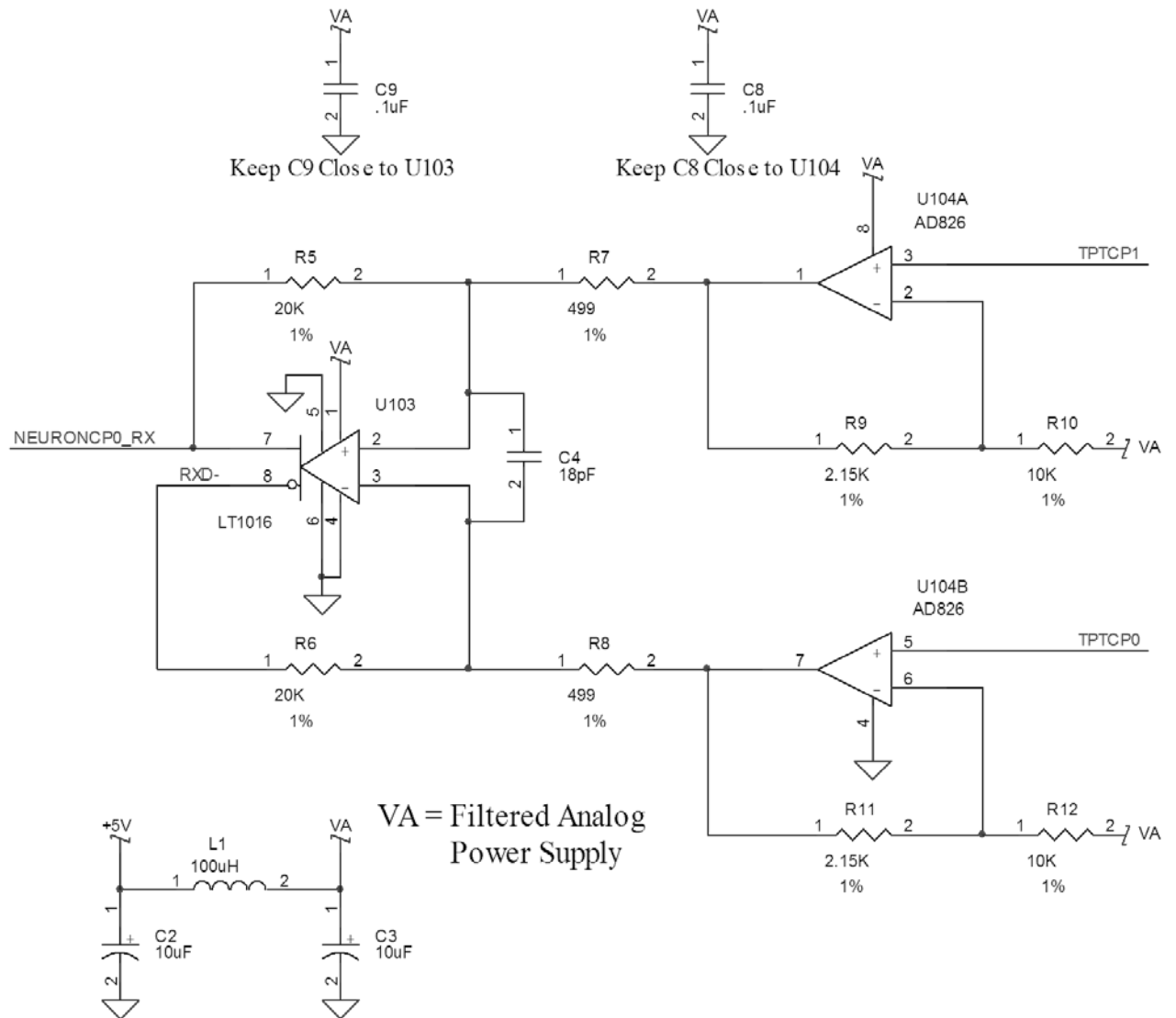
| Designator | Value              |
|------------|--------------------|
| C6         | 470 pF             |
| R14        | 10 k $\Omega$ , 1% |
| U101       | SN74HCT240         |

## Comparator Circuit

**Figure 32** shows a differential comparator circuit for connecting a Router 5000 to a TPT/XF-1250 transceiver. The differential comparator circuit drives the Router 5000's receive (**CP0**) signal based on the TPT/XF-1250 transceiver's differential receive signals (**CP0** and **CP1**).

The heart of the differential comparator circuit is a dual, high speed voltage feedback operational amplifier (such as an Analog Devices AD826 Low Cost, High Speed, Low Power Dual Operational Amplifier) and a high-speed comparator (such as a Linear Technology LT1016 Ultra Fast Precision 10ns Comparator). The operational amplifiers buffer the differential receive signal and form a low-pass filter. The comparator interfaces directly to TTL/CMOS logic while operating off the same 5 V power supply as the TPT/XF-1250 transceiver or a separate 5 V analog power supply (VA).

**Important:** Because capacitor **C4** with resistors **R7** and **R8** act as a low-pass filter for the differential signal, be sure to keep the traces between them and **U103** as short as possible. Excessive trace capacitance can lower the filter's cutoff frequency, which can cause signal loss from the TPT/XF-1250 transceiver.



**Figure 32.** Differential Comparator Circuit

**Table 14.** Bill of Materials for the Differential Comparator Circuit

| Designator | Value                                                                  |
|------------|------------------------------------------------------------------------|
| C2, C3     | 10 $\mu$ F                                                             |
| C4         | 18 pF                                                                  |
| C8, C9     | 0.1 $\mu$ F                                                            |
| L1         | 100 $\mu$ H, $\pm 20\%$ , $I_{sat} \geq 100$ mA, $DCR \leq 0.5 \Omega$ |
| R5, R6     | 20 k $\Omega$ , 1%                                                     |

| Designator | Value                |
|------------|----------------------|
| R7, R8     | 499 $\Omega$ , 1%    |
| R9, R11    | 2.15 k $\Omega$ , 1% |
| R10, R12   | 10 k $\Omega$ , 1%   |
| U103       | LT1016CN8            |
| U104       | AD826AN              |

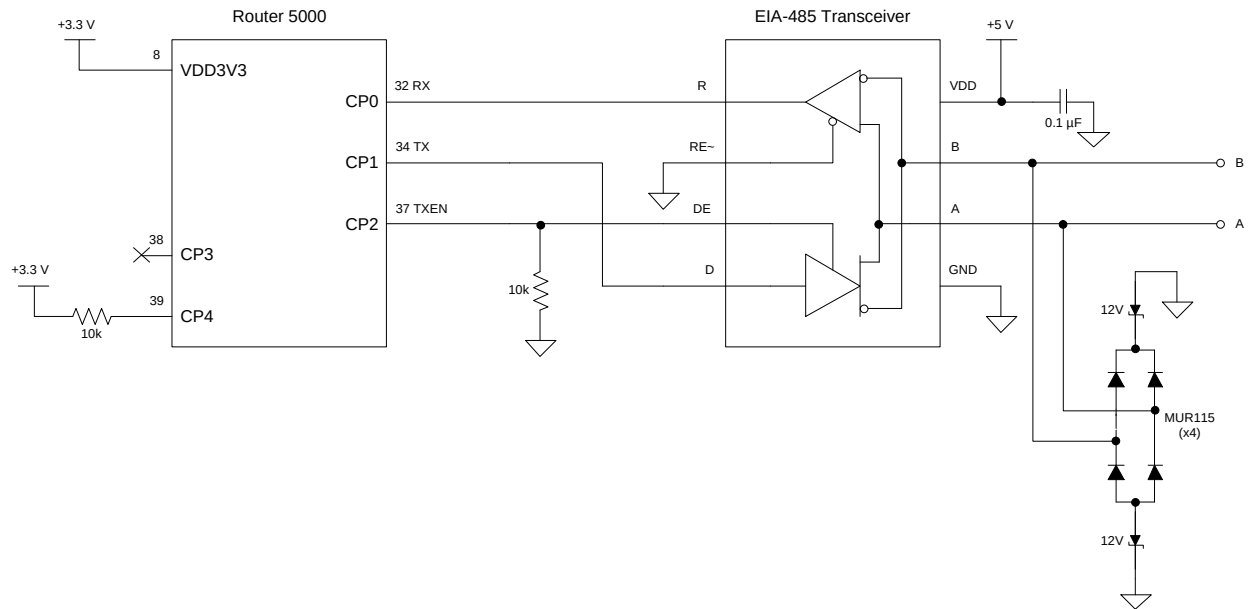
## Using an EIA-485 Transceiver

You can use the Router 5000 with commercially available EIA-485 transceivers. A number of wire types can be supported, along with multiple data rates (up to 1.25 Mbps), as listed in the *Series 5000 Chip Data Book*.

With an EIA-485 transceiver, the common-mode network voltage can range between  $-7$  V to  $+12$  V. To implement an EIA-485 device, the Router 5000's communications port runs in single-ended mode.

Available industry standards that describe EIA-485 specifications provide details on unit loads, data rate, wire size, and wire distances. To ensure interoperability between devices, the LONMARK<sup>®</sup> interoperability guidelines require a data rate of 39 kbps for devices that use EIA-485 transceivers. In addition, the EIA-485 transceiver must have TTL-compatible inputs for the connection to the 3.3 V Router 5000 chip. A typical circuit configuration, shown in **Figure 33**, can support up to 32 loads.

An EIA-485 network works best with a common power source. Individual device power sources can create problems when the network common-mode voltage exceeds  $-7$  V to  $+12$  V, or when ground faults cause damage to devices.



**Figure 33.** EIA-485 Twisted-Pair Interface (Uses Single-Ended Mode)

The EIA-485 specification requires a common ground reference for all transceivers. This common ground reference can be provided by adding a third conductor in the network cable or a separate connection to common ground at each device.

## Using an FTT-10A Transceiver

The FTT-10A Free Topology Twisted Pair Transceiver provides a simple, cost-effective method of adding a LONWORKS transceiver to any Neuron Chip-based control system. The FTT-10A transceiver supports polarity insensitive, free topology wiring, freeing the system installer from the need to wire using a bus topology. Free topology wiring reduces the time and expense of system installation by allowing the wiring to be installed in the most expeditious manner. It also simplifies network expansion by eliminating restrictions on wire routing, splicing, and device placement.

The FTT-10A transceiver consists of an isolation transformer that is integrated with a 78 kbps differential Manchester coded communication transceiver. Pins are provided for connections to the Router 5000 Communications Port (CP) and clock lines, +5 V power, and the twisted pair network. The FTT-10A transceiver provides automatic detection of the input clock frequency at 5, 10, and 20 MHz. The pins are keyed to prevent accidental reversal during mounting. The FTT-10A transceiver appears as a high impedance to the network when unpowered, and does not interfere with network communications when powered down.

The transceiver is housed in an encapsulated plastic shell which contains the transformer and signal processing electronics. The compact package is only 7.2 mm (0.28 inches) high, and is ideal for use in low profile applications such as DIN packs. The sealed housing protects the transceiver should conformal coating or other forms of environmental sealing be required on the printed circuit assembly.

The FTT-10A transceiver is compatible with Echelon's LPT-11 Link Power Transceiver, and these transceivers can communicate with each other on a single twisted pair cable. This capability provides an inexpensive means of interfacing to nodes whose current or voltage requirements would otherwise exceed the capacity of the link power segment. When equipped with an FTT-10A transceiver, these nodes can be operated from a local power supply without the need for additional electrical isolation from the link power network.

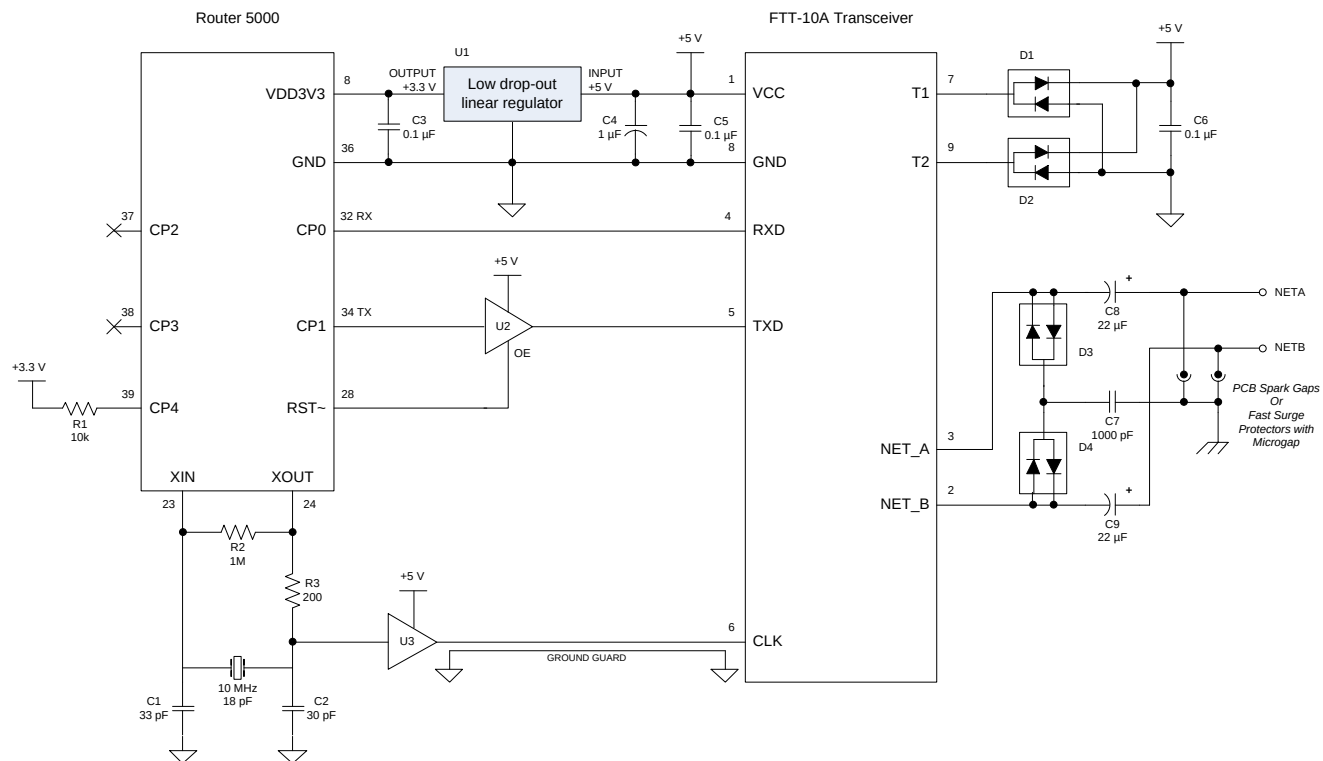
The FTT-10A transceiver also provides electrical isolation for I/O devices that are grounded, allowing such devices to be used on a link power network segment. In many applications, some I/O devices are grounded, either to meet functional requirements or safety regulations. The transformer of the FTT-10A transceiver electrically isolates the node from the segment, allowing devices connected to the node to be grounded without impairing communications.

The FTT-10A transceiver receives its clock input from the Router 5000 through its CMOS input **CLK** pin. This pin is driven by the **XOUT** output of the Router 5000, buffered with a standard bus buffer/line driver that supports TTL-compatible input and 5V CMOS output. Clock traces should be kept short ( $\leq 2$  cm) to minimize noise coupling. In addition, a logic ground guard must be added for the **CLK** trace to minimize clock noise and to help keep EMI levels low. However, this ground guard should not be used as a ground source for digital circuitry.

**Figure 34** shows the basic configuration for connecting a Router 5000 Processor to an FTT-10A Free Topology Twisted Pair Transceiver.

The major differences between connecting a Series 3100 Neuron Chip to an FTT-10A transceiver (see the *LONWORKS FTT-10A Free Topology Transceiver User's Guide*) and connecting a Router 5000 to an FTT-10A transceiver include:

- The connection between the FTT-10A **VCC** pin and the Router 5000 **VDD3V3** pin requires the addition of a low drop-out linear regulator to convert the +5 V output from the LPT-11 transceiver to the +3.3 V input for the Router 5000.
- The connection between the FTT-10A **TXD** pin and the Router 5000 **CP1** pin requires the addition of a non-inverting bus buffer/line driver that supports TTL-compatible input and 5V CMOS output. The output of the Router 5000 **RST~** pin is also connected to the buffer/line driver to allow the Router 5000 to propagate a device reset to the FTT-10A transceiver by setting the buffer/line driver to a tri-state impedance state. An example part for the buffer/line driver is an NXP 74AHCT1G126 bus buffer/line driver.
- The connection between the FTT-10A **CLK** pin and the Router 5000 **XOUT** pin requires the addition of a standard (inverting or non-inverting) bus buffer/line driver that supports TTL-compatible input and 5V CMOS output.



**Figure 34.** Connecting a Router 5000 to an FTT-10A Transceiver

**Table 15.** Example Bill of Materials for the FTT-10A Circuit

| Designator     | Value                   |
|----------------|-------------------------|
| C1             | 33 pF                   |
| C2             | 30 pF                   |
| C3, C5, C6     | 0.1 $\mu$ F             |
| C4             | 1 $\mu$ F               |
| C7             | 1000 pF                 |
| C8, C9         | 22 $\mu$ F, 50 V, polar |
| D1, D2, D3, D4 | BAV99 or 1N4148 (x2)    |
| R1             | 10 k $\Omega$           |
| R2             | 1 M $\Omega$            |
| R3             | 200 $\Omega$            |

| Designator | Value                                           |
|------------|-------------------------------------------------|
| U1         | TDA3663 (or similar LDO regulator)              |
| U2         | 74AHCT1G126                                     |
| U3         | 74AHCT1G04 (or a 74AHCT1G126 with OE tied high) |

**Important:** Because the Router 5000 **XOUT** pin drives the FTT-10A **CLK** signal, the value of **C2** does not match the value of **C1**. The value for **C2** is specified as 30 pF based on an input capacitance for the buffer/line driver of 3 pF at 25 °C (so that the total capacitance for the **XOUT** pin is 33 pF). For the 74AHCT1G126 part, input capacitance can vary over temperature, up to 10 pF. If your device is likely to experience extreme temperatures, consider changing the value of **C2** to 27 pF to allow for the change in capacitance over temperature.

See the *LONWORKS FTT-10A Free Topology Transceiver User's Guide* for additional information about selecting appropriate parts for capacitors **C8** and **C9**, for using fast switching rectifiers in place of diodes **D3** and **D4**, and for replacing the PCB spark gaps with fast surge protectors with microgaps in applications that must be conformally coated or potted.

See the *LONWORKS FTT-10A Free Topology Transceiver User's Guide* for PCB layout guidelines for the FTT-10A transceiver; see *PCB Layout Guidelines* and the *Series 5000 Chip Data Book* for PCB layout guidelines for the Router 5000.

## Using an LPT-11 Link Power Transceiver

The Echelon LONWORKS LPT-11 Link Power Twisted Pair Transceiver provides a simple, cost effective method for adding a network-powered LONWORKS transceiver to any Neuron Chip-based sensor, activator, display, lighting device, or general purpose I/O controller. The LPT-11 transceiver consists of a Single In-Line Package (SIP) that contains a 78 kbps differential Manchester coded communications transceiver, a switching power supply that draws power from the twisted-pair network, and connections for the Router 5000 Communications Port (CP) lines and for the twisted pair network. The LPT-11 transceiver eliminates the need to use a local power supply for each device, because device power is supplied by a central power supply over the same twisted wire pair that handles network communications. A single network segment can support up to 128 LPT-11 based devices.

The LPT-11 transceiver includes an integral switching power supply that can furnish +5 VDC at up to 100 mA. The LPT-11 transceiver derives its power directly from the switching power supply, leaving up to 100 mA of current for a Router 5000, application electronics, sensors, actuators, and displays. If a high-current or high-voltage device must be controlled, then the +5 VDC power can be used to trigger an isolating high-current triac, relay, or contactor.

The link-power system uses a single point of Earth ground, at the LPI-10 module, and all of the LPT-11 transceivers electrically float relative to the local ground. Differential transmission minimizes the effects of common-mode noise on signal transmission. If grounded sensors or actuators are used, then either the

communication port (CP) or the I/O lines of the Router 5000 must be electrically isolated.

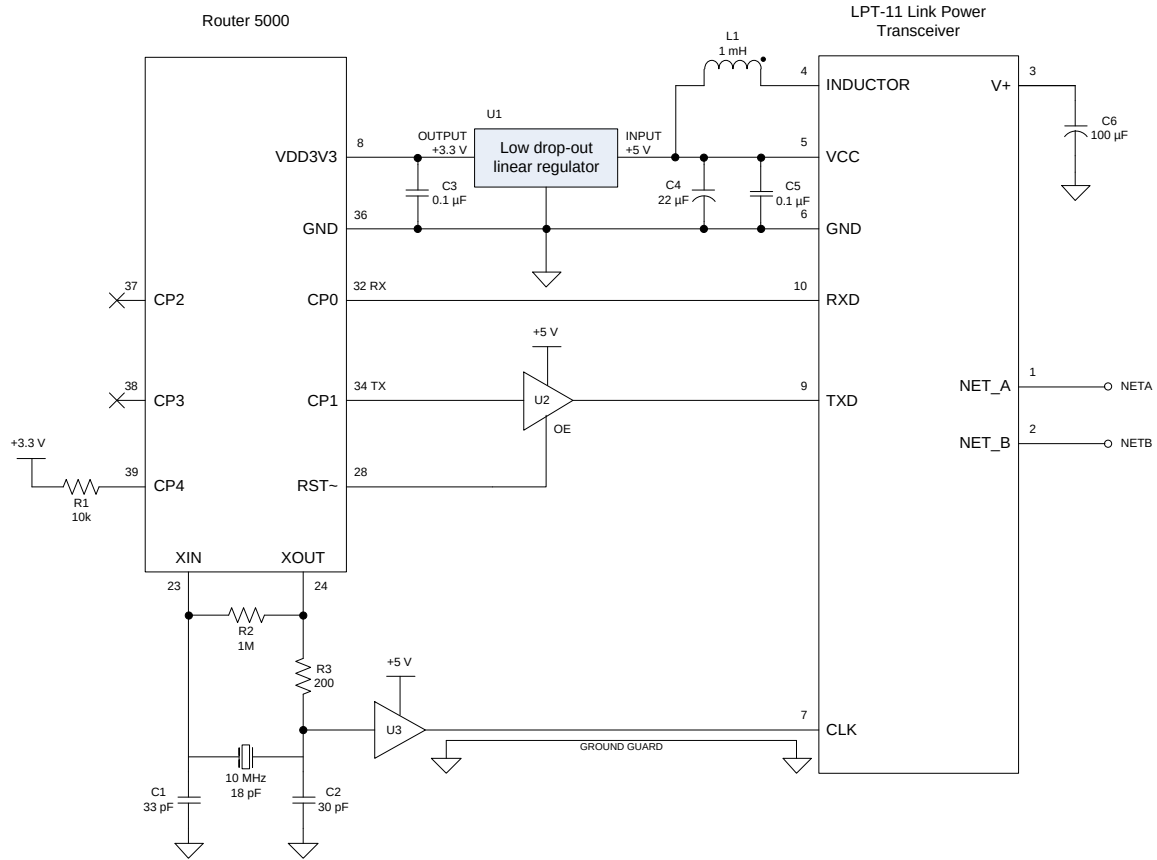
The LPT-11 transceiver receives its clock input from the Router 5000 through its CMOS input **CLK** pin. This pin is driven by the **XOUT** output of the Router 5000, buffered with a standard bus buffer/line driver that supports TTL-compatible input and 5V CMOS output. Clock traces should be kept short ( $\leq 2$  cm) to minimize noise coupling. In addition, a logic ground guard must be added for the **CLK** trace to minimize clock noise and to help keep EMI levels low. However, this ground guard should not be used as a ground source for digital circuitry.

The LPT-11 transceiver can operate at 20, 10, or 5 MHz. When coupled to a Router 5000, the LPT-11 transceiver operates at 10 MHz. The operating frequency is automatically detected on the LPT-11 transceiver's **CLK** pin.

**Figure 35** shows the basic configuration for connecting a Router 5000 to an LPT-11 Twisted-Pair Link Power Transceiver.

The major differences between connecting a Series 3100 Neuron Chip to an LPT-11 transceiver (see the *LONWORKS LPT-11 Link Power Transceiver User's Guide*) and connecting a Router 5000 to an LPT-11 transceiver include:

- The connection between the LPT-11 **VCC** pin and the Router 5000 **VDD3V3** pin requires the addition of a low drop-out linear regulator to convert the +5 V output from the LPT-11 transceiver to the +3.3 V input for the Router 5000.
- The connection between the LPT-11 **TXD** pin and the Router 5000 **CP1** pin requires the addition of a non-inverting bus buffer/line driver that supports TTL-compatible input and 5V CMOS output. The output of the Router 5000 **RST~** pin is also connected to the buffer/line driver to allow the Router 5000 to propagate a device reset to the LPT-11 transceiver by setting the buffer/line driver to a tri-state impedance state. An example part for the buffer/line driver is an NXP 74AHCT1G126 bus buffer/line driver.
- The connection between the LPT-11 **CLK** pin and the Router 5000 **XOUT** pin requires the addition of a standard (inverting or non-inverting) bus buffer/line driver that supports TTL-compatible input and 5V CMOS output.



**Figure 35.** Connecting a Router 5000 to an LPT-11 Link Power Transceiver

**Table 16.** Example Bill of Materials for the LPT-11 Circuit

| Designator | Value                                                                                               |
|------------|-----------------------------------------------------------------------------------------------------|
| C1         | 33 pF                                                                                               |
| C2         | 30 pF                                                                                               |
| C3, C5     | 0.1 µF                                                                                              |
| C4         | 22 µF, DCWV $\geq 10$ V, $I_{\text{ripple}} \geq 200$ mA <sub>rms</sub> @ 100 kHz, ESR $\leq 1.2$ Ω |
| C6         | 100 µF, DCWV $\geq 63$ V, $I_{\text{ripple}} \geq 100$ mA <sub>rms</sub> @ 100 kHz                  |
| L1         | 1 mH, DCR $\leq 4$ Ω, $I_{\text{sat}} \geq 200$ mA, $F_{\text{res}} \geq 800$ kHz                   |
| R1         | 10 kΩ                                                                                               |

| Designator | Value                                           |
|------------|-------------------------------------------------|
| R2         | 1 M $\Omega$                                    |
| R3         | 200 $\Omega$                                    |
| U1         | TDA3663 (or similar LDO regulator)              |
| U2         | 74AHCT1G126                                     |
| U3         | 74AHCT1G04 (or a 74AHCT1G126 with OE tied high) |

**Important:** Because the Router 5000 **XOUT** pin drives the LPT-11 **CLK** signal, the value of **C2** does not match the value of **C1**. The value for **C2** is specified as 30 pF based on an input capacitance for the buffer/line driver of 3 pF at 25 °C (so that the total capacitance for the **XOUT** pin is 33 pF). For the 74AHCT1G126 part, input capacitance can vary over temperature, up to 10 pF. If your device is likely to experience extreme temperatures, consider changing the value of **C2** to 27 pF to allow for the change in capacitance over temperature.

See the *LONWORKS LPT-11 Link Power Transceiver User's Guide* for additional information about selecting appropriate parts for capacitors **C3** and **C5** and for inductor **L1**.

See the *LONWORKS LPT-11 Link Power Transceiver User's Guide* for PCB layout guidelines for the LPT-11 transceiver; see *PCB Layout Guidelines* and the *Series 5000 Chip Data Book* for PCB layout guidelines for the Router 5000.

---

## Example Router 5000 Schematics

**Figure 36** a sample core schematic for two Router 5000 half routers. This core schematic could apply to any Router 5000 router. **Figure 37** shows a sample schematic for the transceivers used by each router half; in this example, TP/XF-1250 and EIA-485. The transceiver schematics also include the RX and TX activity indicator circuits described in *Network Activity Indicator – Router 5000*.

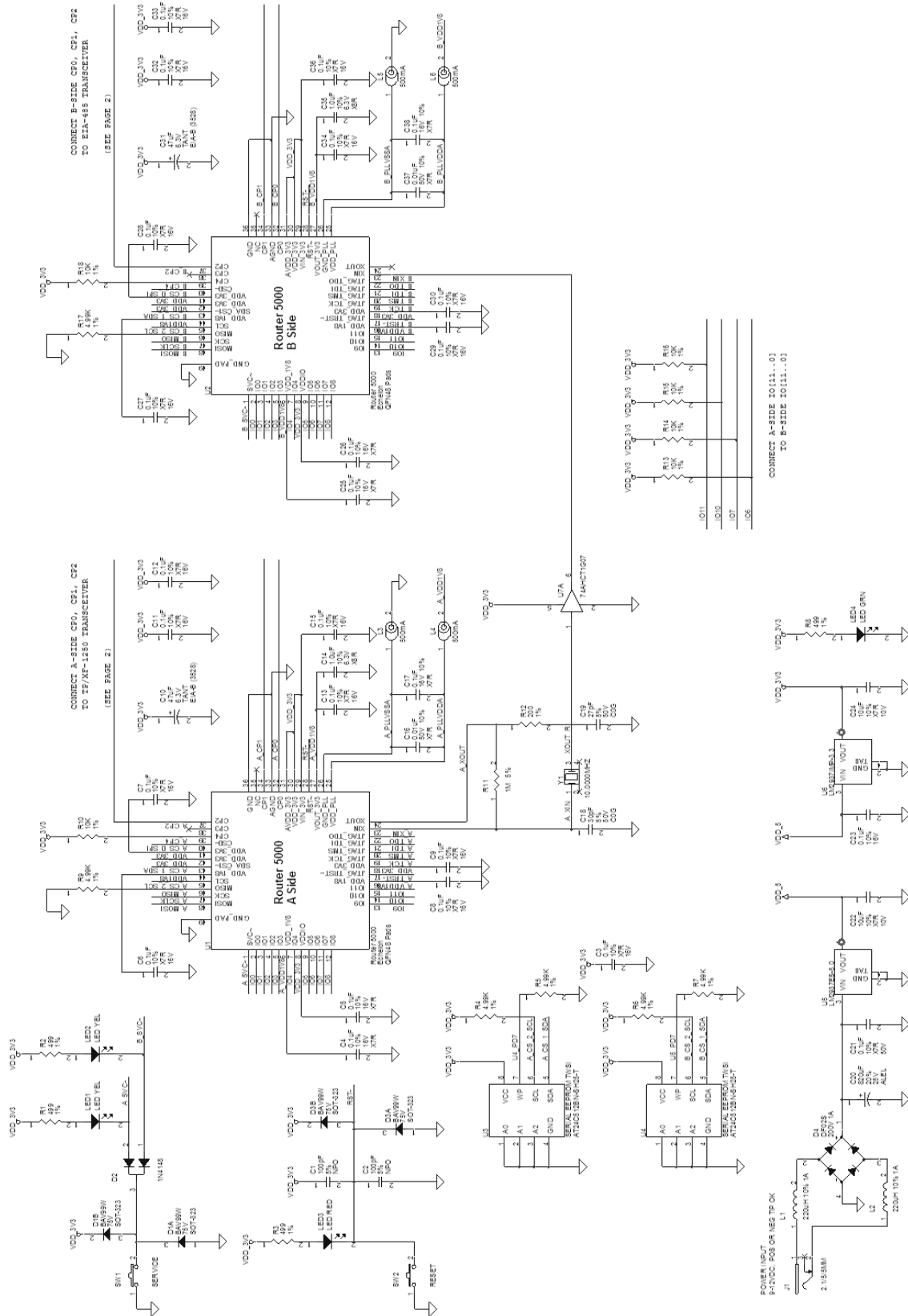


Figure 36. Router 5000 Motherboard Example Schematic – Core

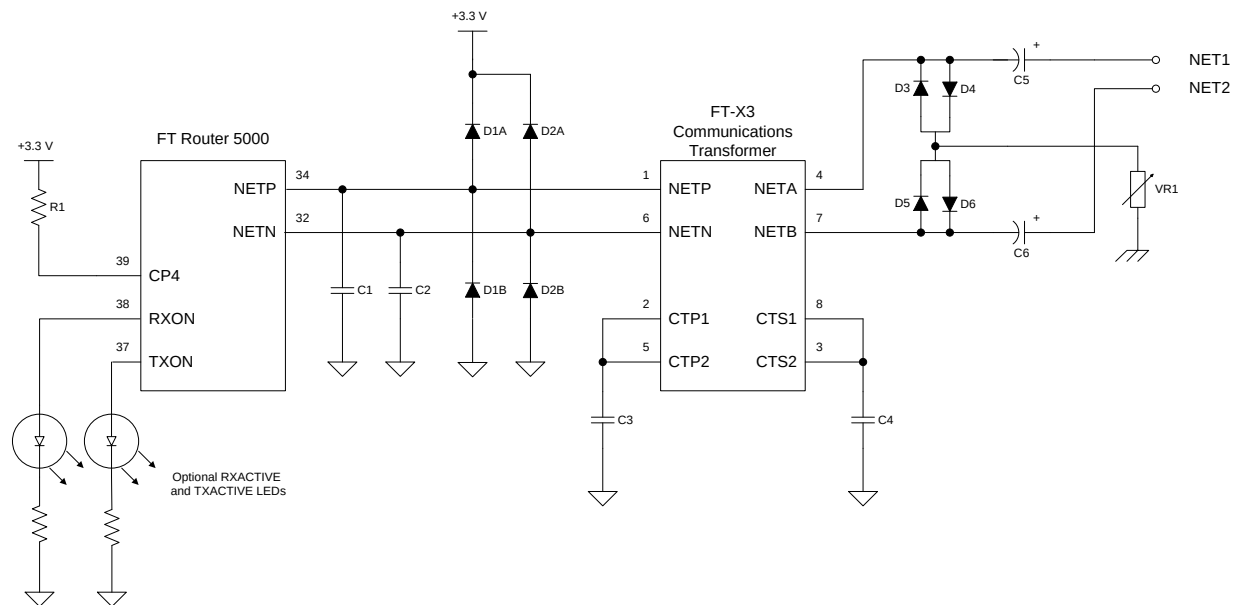


# Developing a Router with the FT Router 5000 Chip

To create a LONWORKS router with the FT Router 5000, perform the following steps:

1. Build a router motherboard according to the specifications described in Chapter 2, *LONWORKS Router Electrical Interfaces*, and the guidelines described in Chapter 5, *LONWORKS Router Design Issues*. The motherboard can be part of custom application hardware, or can be a standalone board.
2. Program the serial EEPROM for each router half. The serial EEPROM holds the router configuration and communications parameters for the transceiver. See *Configuring a Series 5000 Half-Router* for more information about router configuration. You can program the EEPROMs either before assembly or in-circuit after assembly.
3. Assemble the router, including the two FT Router 5000 half-routers, two FT-X3 Communications Transformers, and a motherboard.
4. Install the router on a network as described in Chapter 6, *Installing a LONWORKS Router*. The network could be a development network for initial testing, a manufacturing network for configuration during manufacture, or a production network for field installation.

**Figure 38** shows the preferred interconnection between the FT Router 5000 and the FT-X3 Communications Transformer; the figure also shows the associated transient protection circuitry. Connect pins 1 and 6 of the FT-X3 transformer to the FT Router 5000, as shown in the figure.



**Figure 38.** FT Router 5000 and FT-X3 Interconnections for the TP/FT-10 Network

**Table 17.** Bill of Materials for the FT Router 5000 and FT-X3 Interconnection

| Designator     | Value                                                             | Description                                                                                                         |
|----------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| R1             | 4.99 k $\Omega$                                                   | Pullup resistor                                                                                                     |
| VR1            | 470 V MOV, 5 mm, 40 pF (typical)                                  | Panasonic ERZV05D471, Digi-Key P7186-ND or equivalent                                                               |
| C1, C2         | 56 pF, 50 V                                                       | Common-mode noise immunity capacitors (for EN61000-4-6 Level 3)                                                     |
| C3, C4         | 100 pF, 5%                                                        | Optional center-tap capacitors                                                                                      |
| C5, C6         | 22 $\mu$ F, $\geq$ 50 V, polar                                    | DC blocking capacitors                                                                                              |
| D1, D2         | BAV99                                                             | ESD transient clamping diodes                                                                                       |
| D3, D4, D5, D6 | BAV99, 1N4148-equivalent<br><br>1N4934, 1N4935, FR1D, RS1D, RS1DB | Differential network clamping diodes:<br><br>For up to 2 kV Surge Protection<br><br>For up to 6 kV Surge Protection |

In **Figure 38**, diodes **D1** and **D2** are ESD transient clamping diodes. Capacitors **C1** and **C2** provide common-mode noise immunity for compliance with EN61000-4-6 Level 3. Capacitors **C5** and **C6** are used to provide DC voltage isolation for the FT 5000 Smart Transceiver when it is used on a link power network and to protect it in the event of a DC power fault on the network wires. The capacitors are required to meet LONMARK interoperability guidelines for the TP/FT-10 channel. These capacitors are not needed for devices that will be connected exclusively to non-link power networks and do not require protection against DC faults. Two polar capacitors are used to protect against the application of a DC voltage of either polarity, while providing a total capacitance of 11  $\mu$ F. Alternatively, a single non-polar capacitor of 10  $\mu$ F can be used in either of the two legs that connect to the network. The initial tolerance of the capacitor should be  $\pm 20\%$  or less, and degradation due to aging and temperature effects should not exceed 20% of the initial minimum value.

In some cases, adding capacitors (**C3** and **C4**) between the center tap pins of the FT-X3 Communications Transformer and ground can reduce EMI emissions. If used, **C3** should always be connected to logic ground. If used, **C4** can connect to either logic ground or Earth ground, depending on whether your device connects Earth and logic ground.

See the *Series 5000 Chip Data Book* for information about connecting a Series 5000 chip to a TP/FT-10 channel, including the FT Router 5000, and for information about the FT-X3 Communications Transformer.

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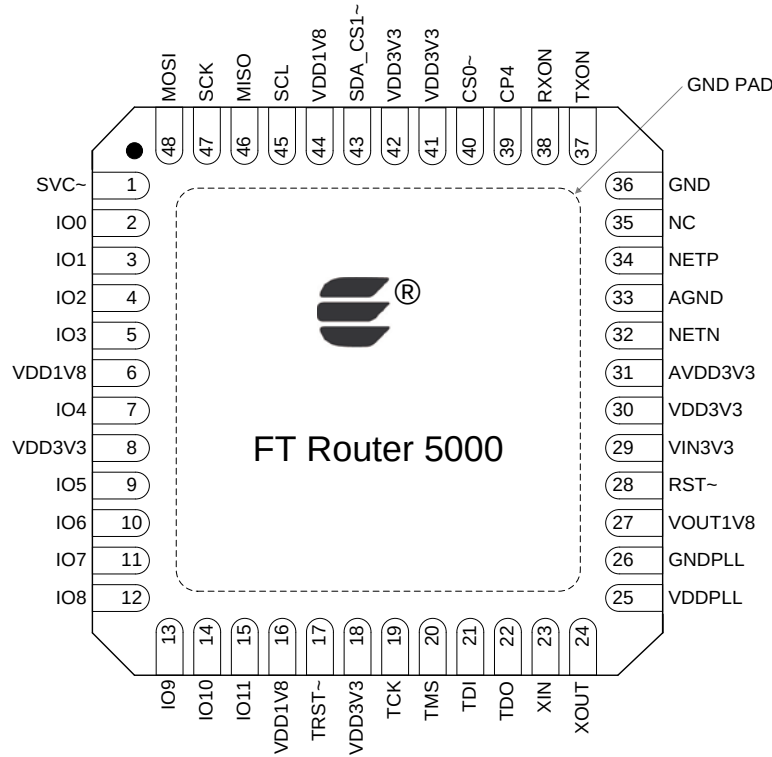
## Connecting Half-Routers: Series 5000 and Series 3100

You can connect a Series 5000 half-router to a Series 3100 half-router to provide routing functions for transceiver and channel types that the Series 5000 router chip does not directly support, for example, a PL-20 channel.

**Important:** Echelon provides special licensing for many of these other transceiver types, such as a Power Line Smart Transceiver; contact Echelon Support for additional information.

**Figure 39** shows the basic connections for a router based on a Series 3100 half-router and a Series 5000 half-router. These connections include:

- Power and ground connections — Use a low-dropout voltage regulator (+5 V input from the Series 3100 half-router power supply, and +3.3 V output for the Series 5000 router chip), such as an NXP TDA3663 very low dropout voltage/quiescent current 3.3 V voltage regulator. See *Power and Ground* for other power considerations for the Series 5000 router chip.
- I/O connections — Connect the Series 3100 **IO[11..0]** pins to the Series 5000 router **IO[11..0]** pins, with 10 k $\Omega$  resistors for the **IO6**, **IO7**, **IO10**, and **IO11** pins, as described in *IO[11..0]*. No level shifters are required.
- Reset pin connection — Connect the Series 3100 **RESET~** pin and the Series 5000 router **RST~** pin, along with the associated Reset button, LED, clamping diodes, and EMC capacitors, as described in *RST~*.
- Service pin connection — Connect the Series 3100 **SERVICE~** pin and the Series 5000 router **SVC~** pin, along with the associated Service button, LEDs, and clamping diodes, as described in *SVC~*.
- Clock circuitry — If the Series 3100 router-half requires a 10 MHz input crystal, you can use a single crystal for both router halves and drive the Series 3100 **CLK1** pin from the Series 5000 router **XOUT** pin, using a standard (inverting or non-inverting) bus buffer/line driver that supports TTL-compatible input and 5V CMOS output, such as an NXP 74AHCT1G04 (or a 74AHCT1G126 with OE tied high) — see **Figure 35** for an exemplar clock circuit. However, if the Series 3100 half-router requires an input clock frequency other than 10 MHz, or is a Power Line Series 3100 half-router, do not connect the Series 5000 router **XOUT** pin to the Series 3100 **CLK1** pin, but instead use separate input crystals for each router-half. See *FT Router 5000 Pinout*.
- *Figure 12* shows the pinout for the FT Router 5000 chip. The central rectangle in the figure represents the bottom pad (pin 49), which must be connected to ground.



**Figure 12.** FT Router 5000 Chip Pinout

Table 9 lists the pin assignments for the FT Router 5000 chip. All digital inputs are low-voltage transistor-transistor logic (LVTTL) compatible, 5 V tolerant, with low leakage. All digital outputs are slew-rate limited to reduce Electromagnetic Interference (EMI) concerns.

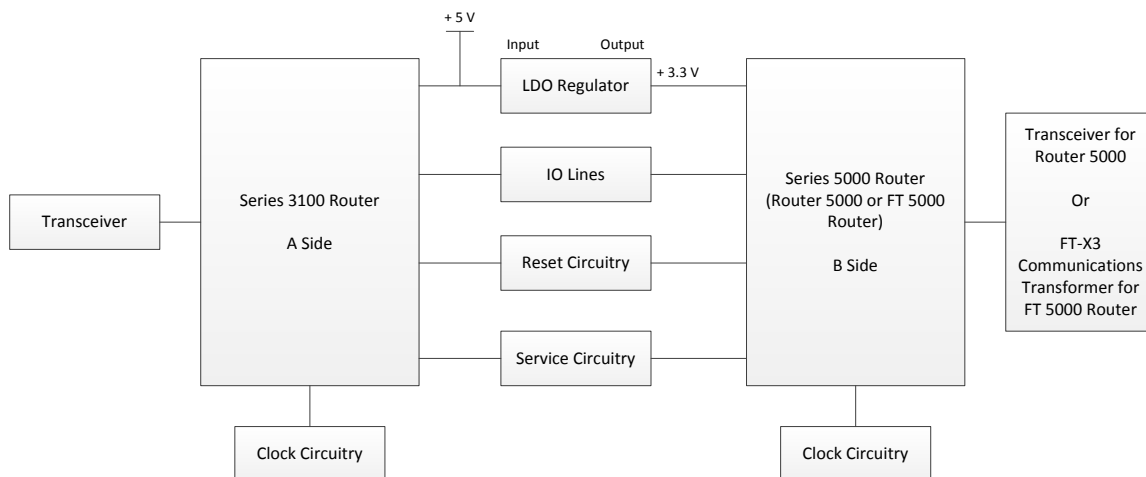
**Table 9.** FT Router 5000 Chip Pin Assignments

| Name   | Pin Number | Type        | Description                                            |
|--------|------------|-------------|--------------------------------------------------------|
| SVC~   | 1          | Digital I/O | Service (active low)                                   |
| IO0    | 2          | Digital I/O | IO0 (side A to side B)                                 |
| IO1    | 3          | Digital I/O | IO1 (side A to side B)                                 |
| IO2    | 4          | Digital I/O | IO2 (side A to side B)                                 |
| IO3    | 5          | Digital I/O | IO3 (side A to side B)                                 |
| VDD1V8 | 6          | Power       | 1.8 V Power Input<br>(from internal voltage regulator) |
| IO4    | 7          | Digital I/O | IO4 (side A to side B)                                 |
| VDD3V3 | 8          | Power       | 3.3 V Power                                            |
| IO5    | 9          | Digital I/O | IO5 (side A to side B)                                 |
| IO6    | 10         | Digital I/O | IO6 (side A to side B)                                 |
| IO7    | 11         | Digital I/O | IO7 (side A to side B)                                 |

| Name    | Pin Number | Type                   | Description                                                |
|---------|------------|------------------------|------------------------------------------------------------|
| IO8     | 12         | Digital I/O            | IO8 (side A to side B)                                     |
| IO9     | 13         | Digital I/O            | IO9 (side A to side B)                                     |
| IO10    | 14         | Digital I/O            | IO10 (side A to side B)                                    |
| IO11    | 15         | Digital I/O            | IO11 (not used for routers)                                |
| VDD1V8  | 16         | Power                  | 1.8 V Power Input<br>(from internal voltage regulator)     |
| TRST~   | 17         | Digital Input          | JTAG Test Reset (active low)                               |
| VDD3V3  | 18         | Power                  | 3.3 V Power                                                |
| TCK     | 19         | Digital Input          | JTAG Test Clock                                            |
| TMS     | 20         | Digital Input          | JTAG Test Mode Select                                      |
| TDI     | 21         | Digital Input          | JTAG Test Data In                                          |
| TDO     | 22         | Digital Output         | JTAG Test Data Out                                         |
| XIN     | 23         | Oscillator In          | Crystal oscillator input                                   |
| XOUT    | 24         | Oscillator Out         | Crystal oscillator output                                  |
| VDDPLL  | 25         | Power                  | 1.8 V Power Input<br>(from internal voltage regulator)     |
| GNDPLL  | 26         | Power                  | Ground                                                     |
| VOUT1V8 | 27         | Power                  | 1.8 V Power Output<br>(of internal voltage regulator)      |
| RST~    | 28         | Digital I/O            | Reset (active low)                                         |
| VIN3V3  | 29         | Power                  | 3.3 V Power Input                                          |
| VDD3V3  | 30         | Power                  | 3.3 V Power                                                |
| AVDD3V3 | 31         | Power                  | 3.3 V Power                                                |
| NETN    | 32         | Comm                   | Network Port (polarity insensitive)                        |
| AGND    | 33         | Ground                 | Ground                                                     |
| NETP    | 34         | Comm                   | Network Port (polarity insensitive)                        |
| NC      | 35         | N/A                    | Do Not Connect                                             |
| GND     | 36         | Ground                 | Ground                                                     |
| TXON    | 37         | Comm                   | TxActive for optional network activity LED                 |
| RXON    | 38         | Comm                   | RxActive for optional network activity LED                 |
| CP4     | 39         | Comm                   | Connect to VDD33 through a 4.99 k $\Omega$ pullup resistor |
| CS0~    | 40         | Digital I/O for Memory | SPI slave select 0 (active low)                            |
| VDD3V3  | 41         | Power                  | 3.3 V Power                                                |

| Name     | Pin Number | Type                   | Description                                            |
|----------|------------|------------------------|--------------------------------------------------------|
| VDD3V3   | 42         | Power                  | 3.3 V Power                                            |
| SDA_CS1~ | 43         | Digital I/O for Memory | I2C: serial data<br>SPI: slave select 1 (active low)   |
| VDD1V8   | 44         | Power                  | 1.8 V Power Input<br>(from internal voltage regulator) |
| SCL      | 45         | Digital I/O for Memory | I2C serial clock                                       |
| MISO     | 46         | Digital I/O for Memory | SPI master input, slave output (MISO)                  |
| SCK      | 47         | Digital I/O for Memory | SPI serial clock                                       |
| MOSI     | 48         | Digital I/O for Memory | SPI master output, slave input (MOSI)                  |
| PAD      | 49         | Ground Pad             | Ground                                                 |

- Clock Pins (XIN and XOUT) show information about the Series 5000 router clock requirements and how to use the Series 5000 router **XOUT** pin to drive an external clock.



**Figure 39.** Basic Connections for Series 3100 and Series 5000 Half-Routers

See the *FT 3120 / FT 3150 Smart Transceiver Databook* or the *PL 3120/PL 3150/PL 3170 Power Line Smart Transceiver Data Book* for more information about Series 3100 chips.

## Configuring a Series 5000 Half-Router

Before programming, a Router 5000 uses its default communications parameters, which define a simplified single-ended mode 78 kbps channel. Like the Router 5000, an FT Router 5000's default communications parameters define a simplified single-ended mode 78 kbps channel – a TP/FT-10 channel. For the

Router 5000, the default communications parameters allow you to load an application image over a 78 kbps network, for example during device manufacturing. Devices that use a 78 kbps transceiver (such as a 78 kbps EIA-485 transceiver or an LPT-11 Link Power Transceiver) can use the default communications parameters within development or manufacturing test networks. For production networks (networks with many devices), you should ensure that each device has communications parameters defined for the channel; use the NodeBuilder FX Development Tool to develop applications with the correct communications parameters.

Note that devices defined for a TP/XF-1250 channel cannot use the default communications parameters; each router-half's external serial non-volatile memory must be programmed with the correct communications parameters before connecting to the network.

To create a Series 5000 router configuration that you can program into the Series 5000 router's EEPROM serial memory, use the NodeBuilder FX Development Tool to create an **NME** file:

1. Define a Hardware Template, as described in *NodeBuilder Hardware Template*.
2. Define a Device Template, as described in *NodeBuilder Device Template*.
3. Define the buffer configuration for the router-half, as described in *Buffer Configurations*.
4. Use a Neuron C source file to define a well-formed **NME** router image, such as the one described in *Example Neuron C Source*.

Because the NodeBuilder FX Development Tool does not produce an **NME** file that allows you control the router mode (configured, learning, repeater) and the routing tables, the example Neuron C source code defines a minimal application, with 15 address table entries, zero NV entries, and zero NV alias entries. The example Neuron C source code places an unused checksum byte out in the EEPROM space that lies in the Domain:0 Group:80-87 forwarding table. If the router will be commissioned by an LNS Server, it will clear this byte and set it to the needed value. If you want to create a preconfigured router image (see *Creating a Pre-Configured NME File*), be sure to clear this byte during in the pre-configuration step.

The router firmware does not use the specific configuration checksum defined in example Neuron C source code.

---

## NodeBuilder Hardware Template

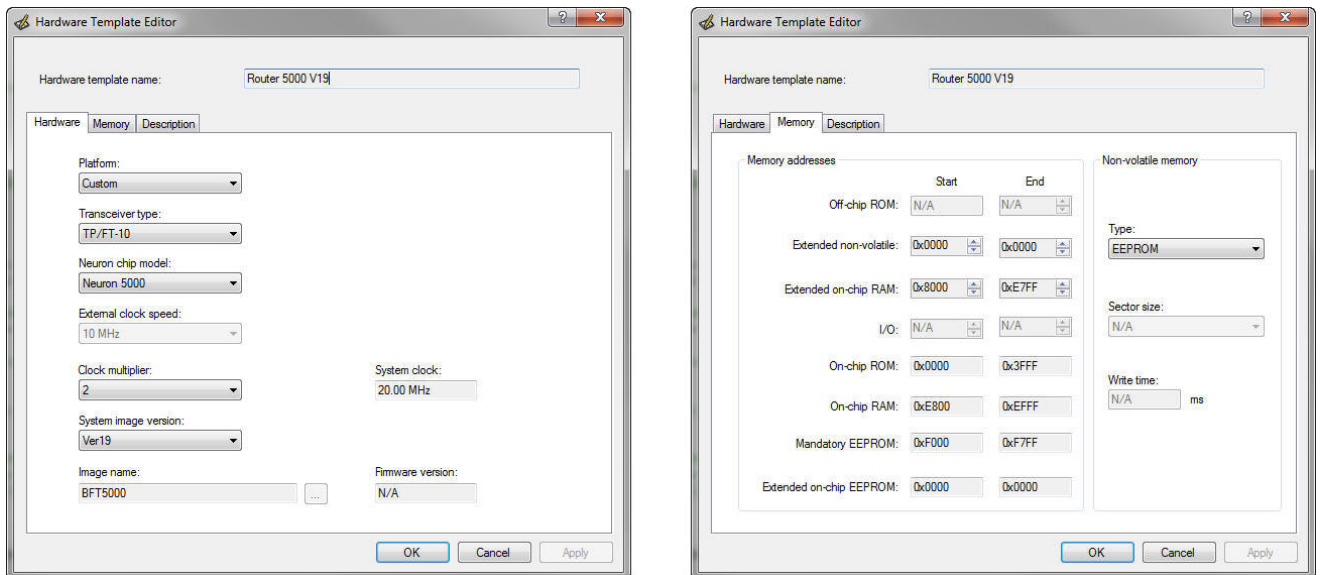
The hardware template defines the target transceiver configuration in the **NME** file produced by the NodeBuilder FX Development Tool. The settings for this template are (see **Figure 40** for an example hardware template for a Router 5000 device):

- Platform: Custom
- Transceiver Type:
  - Depends on transceiver for a Router 5000
  - TP/FT-10 for an FT Router 5000
- Neuron Chip Model:

- Neuron 5000 for a Router 5000
- FT 5000 for an FT Router 5000
- Clock Multiplier: 2 (Recommended)

**Important:** If the other router-half uses a Series 3100 chip, do not specify a value higher clock multiplier value than 2. You can specify a value of 4 if both sides are Series 5000 router chips. Do not specify a value of 8.

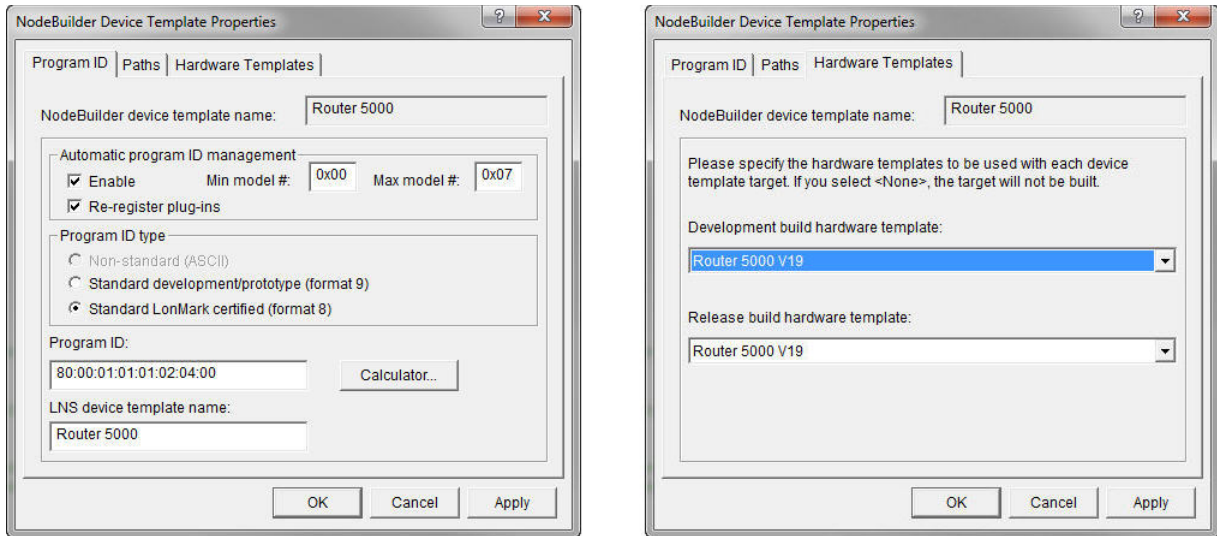
- System image version: Ver19
- Memory – Extended non-volatile: None
- Extended on-chip RAM: 0x8000 – 0xE7FF
- All other memory options: (Leave as default values)



**Figure 40.** Example NodeBuilder Hardware Template for the Router 5000

## *NodeBuilder Device Template*

The device template should include a standard Program ID, such as “80:00:01:01:01:02:04:01”, where the channel type field varies according to the transceiver type. **Figure 41** shows an example device template for a Router 5000 device.



**Figure 41.** Example NodeBuilder Device Template for the Router 5000

You can also use this template to export a specific domain configuration (limited to domain 0) along with a receive transaction timer (typically, 768 ms) and a location string.

## Buffer Configurations

The NodeBuilder FX Development Tool issues an error if you try to build a target with too large a buffer configuration. For Series 3100 routers, buffering is constrained by available RAM, but a Series 5000 router has sufficient RAM for buffering (see *Router 5000 and FT Router 5000 Message Buffers*). However, large buffer counts (for example, greater than 15) can create conditions where messages can become backed up in the router.

When defining the NET buffer sizes, you must consider the other router-half: the input buffer size for Side A should be as large as Side B's output buffer size. Likewise, the input buffer sizes for Side B should be as large as Side A's output buffer size. When a router-half forwards a packet to a router half that cannot accommodate the size of the packet, that packet is dropped.

## Creating a Pre-Configured NME File

To create a pre-configured router image:

1. Use a programming tool to program the **NME** file produced by the NodeBuilder FX Development Tool into the router-half.
2. Bring the router-half up and configure it to the desired state.
3. Use a programming tool to extract the configured router image (read the 2 KB EEPROM image), and save it for subsequent device programming.

**Note:** The Series 5000 router firmware ignores all versioning information and application code components in the **NME** file.

---

## Example Neuron C Source

This section shows an example Neuron C file for Series 5000 router development. This file primarily controls the router's buffering, but it also contains important declarations to set up the parallel IO configuration and explicit addressing.

```
//
// Copyright (c) 2011 Echelon Corporation.
// All Rights Reserved.

#include <control.h>
#include <msg_addr.h>

// Basic application configuration space:
#pragma num_domain_entries      2
#pragma num_alias_table_entries 0
#pragma receive_trans_count     3
#pragma disable_snvt_si
#pragma run_unconfigured

// Router buffer configurations: APP
#pragma app_buf_out_size        42
#pragma app_buf_in_size         42
#pragma app_buf_out_count       1
#pragma app_buf_in_count        2

// Router buffer configurations: NET
#pragma net_buf_out_size        255
#pragma net_buf_in_size         255
#pragma net_buf_in_count        7
#pragma net_buf_out_count       7

#pragma app_buf_out_priority_count 1
#pragma net_buf_out_priority_count 2

// This pad covers the router EE data.
const unsigned int code_pad[200] = {0};

// Make some room for router configurations
#pragma num_addr_table_entries  15

// Force explicit addressing on by referencing a dummy
// message address
// This code never actually runs.
// Don't use #pragma micro_interface for routers because
// setting that bit causes SI data issues with installers.
// Place this code AFTER code_pad[].
msg_tag NMtag;
void send_msg_dummy(void) {
    msg_out.tag = NMtag;
    msg_out.dest_addr.snode.type = SUBNET_NODE;
    msg_send();
}

IO_0 parallel slave pios1;

// The Transceiver ID is declared here to allocate
// space for it in the link.
```

```
eeeprom unsigned int mip_eevars[2] = {
    0x00,          // M/S Designation.
    0x00          // TXID, always last (not used).
};
#pragma ignore_notused code_pad
#pragma ignore_notused pios1
#pragma ignore_notused mip_eevars
#pragma ignore_notused send_msg_dummy
```

# 5

## LONWORKS Router Design Issues

This chapter examines a number of design issues, including a discussion of PCB layout, electromagnetic interference (EMI), and electrostatic discharge (ESD), for LONWORKS routers.

---

## PCB Layout Guidelines

Printed circuit board (PCB) layout for a Router 5000 or FT Router 5000 is similar to layout for a Neuron 5000 Processor or FT 5000 Smart Transceiver, and should include the following general features:

- **Star-Ground Configuration:** Arrange the various blocks of the device that directly interface with off-board connections (the network, any external I/O, and the power supply cable) so that they are together along one edge of the PCB.
- **ESD Keepout Area:** Consider the area around the network connection traces and components as “ESD Hot”. The PCB layout should be designed so that substantial ESD hits from the network discharge directly to the star-ground center point.
- **Clamp Diodes:** For transceivers that use differential receive signals (such as the TP/XF-1250 transceiver) use four diodes to clamp the transceiver’s differential receive signals to ground during ESD and surge transients. For the FT Router 5000, use diodes to clamp the FT Router 5000 side of the FT-X3 transformer between  $V_{DD33}$  and ground.
- **Ground Return for a Series 5000 router:** A Router 5000 has internal protection circuitry built into its **CP[4..0]** pins, and an FT Router 5000 has internal protection circuitry built into its **NETP** and **NETN** pins. When an ESD or surge transient comes in from the network, the portion of the transient that makes it to the Series 5000 router is clamped to the chip’s  $V_{DD33}$  power pins and ground pins. Be sure to provide a short and wide ground path from the Series 5000 router back to the center of the star ground.
- **Ground Planes:** As ground is routed from the center of the star out to the function blocks on the board; planes or very wide traces should be used to lower the inductance (and therefore the impedance) of the ground distribution system.
- **$V_{DD33}$  Decoupling Capacitors:** A good rule of thumb is to provide at least one  $V_{DD33}$  decoupling capacitor to ground for each  $V_{DD33}$  power pin on an IC in the design. For SMT devices like a Series 5000 router, each decoupling capacitor should be placed on the top layer with the chip, and placed as close as possible to the chip to minimize the length of  $V_{DD33}$  trace between the capacitor and the chip’s  $V_{DD33}$  pad.

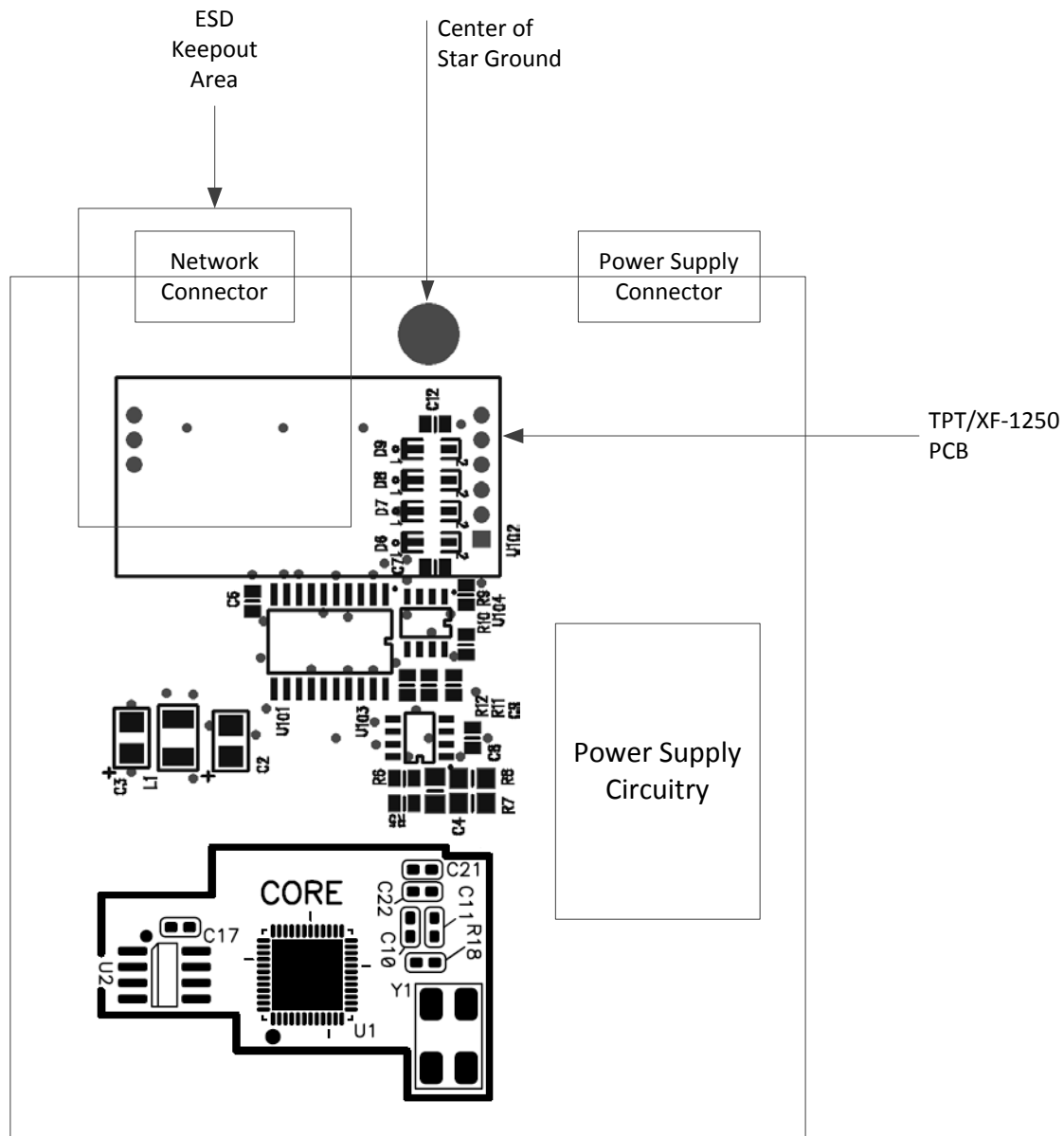
---

### Router 5000

**Figure 42** shows a portion of the top layer of a 4-layer PCB layout for the Router 5000 half-router for a TP/XF-1250 transceiver, including the differential driver circuit, and the comparator circuit, and the other building blocks of a PCB design. The figure shows a rectangle for the placement of the TPT/XF-1250 transceiver PCB, which is mounted above the main board.

See the *Connecting a Neuron 5000 Processor to an External Transceiver* Engineering Bulletin for more information about PCB layout considerations for connecting external transceivers to Neuron 5000 Processors, including the Router 5000. See Chapters 3 and 4 of the *Series 5000 Chip Data Book* for additional

information about PCB layout and electromagnetic compatibility (EMC) design guidelines for a Series 5000 Chip, including the Router 5000.



**Figure 42.** Example PCB Layout for a Router 5000 with a TPT/XF-1250 Transceiver

In the figure, the area marked CORE represents the essential circuitry for the Router 5000, its serial EEPROM memory chip, its crystal, and associated capacitors and resistors. The figure does not show I/O or other connections to the other side of the router.

The differential driver circuit is shown as **U101** and associated parts. The comparator circuit is shown as **U103**, **U104**, and associated parts. The TPT/XF-1250 transceiver is shown as **U102**, although the transceiver itself resides on a separate sub-assembly PCB, above the main board and is connected to it by two headers (one 6-pin header and one 3-pin header). Below the TPT/XF-1250

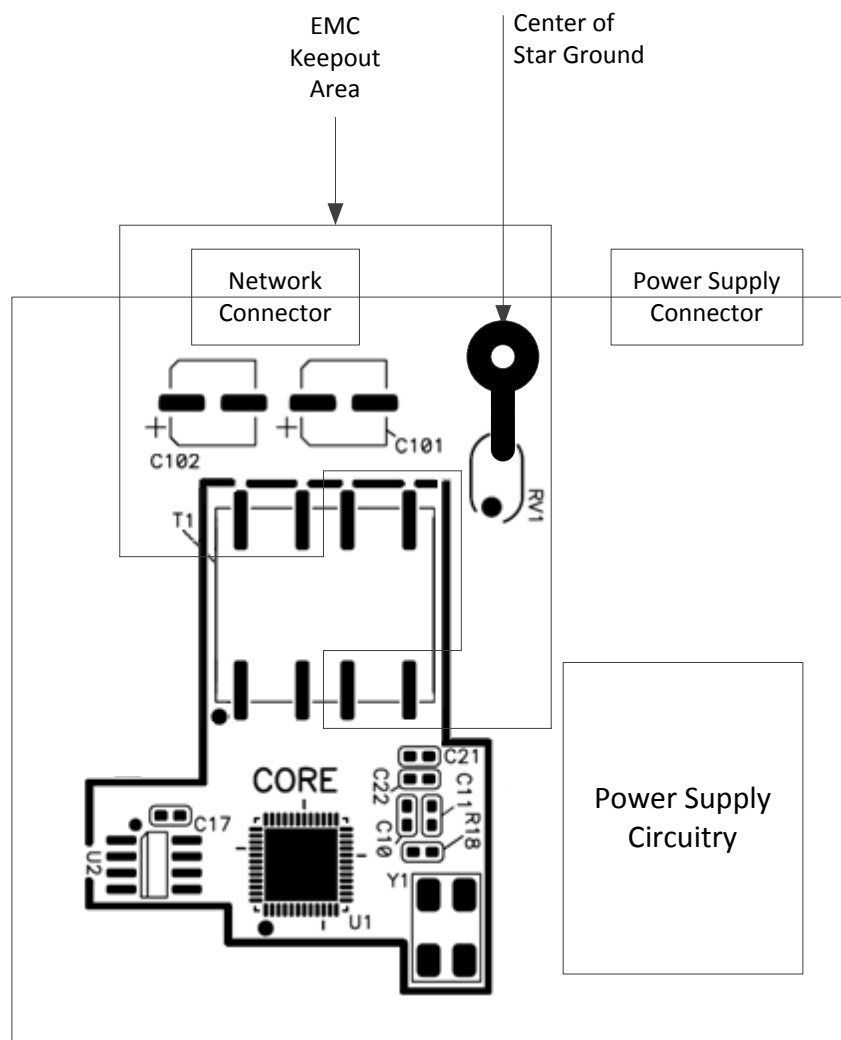
transceiver PCB are the clamping diodes (**D6-D9**) for the transceiver's receive signals.

---

## FT Router 5000

**Figure 43** shows a portion of the top layer of a 4-layer PCB layout for an FT Router 5000 half-router, including the FT-X3 Communications Transformer and the other building blocks of a PCB design.

See Chapters 3 and 4 of the *Series 5000 Chip Data Book* for additional information about PCB layout and electromagnetic compatibility (EMC) design guidelines for a Series 5000 Chip, including the FT Router 5000.



**Figure 43.** Example PCB Layout for an FT Router 5000

In the figure, the area marked CORE represents the essential circuitry for the FT Router 5000, the FT-X3 Communications Transformer, the router chip's serial EEPROM memory chip, its crystal, and associated capacitors and resistors. The figure does not show I/O or other connections to the other side of the router.

---

## EMI Design Issues

The high-speed digital signals associated with microcontroller designs can generate unintentional Electromagnetic Interference (EMI). High-speed voltage changes generate RF currents that can cause radiation from a product with a length of wire or piece of metal that can serve as an antenna.

Products that use the RTR-10 router will generally need to demonstrate compliance with EMI limits enforced by various regulatory agencies. In the USA, the FCC requires that unintentional radiators comply with Part 15 level “A” for industrial products, and level “B” for products that can be used in residential environments. Similar regulations are imposed in most countries throughout the world. For more information about such regulations, see European EMC standards, such as VDE 0871, Class “B” 1984, and CISPR Publications 22.

Echelon has designed the RTR-10 router with low enough RF noise levels for design into level “B” products. Echelon encourages level “B” compliance for all LONWORKS compatible products.

Echelon has performed immunity tests for CE Marking on Series 5000 devices, including the Router 5000 and FT Router 5000, and has also performed additional tests to ensure immunity and low emissions. Specifically, Echelon has performed the following immunity tests:

- Electrostatic discharge (ESD) testing (both air and contact discharge) for compliance with Comité Européen de Normalisation<sup>2</sup> (CEN), standard EN 61000-4-2
- Radiated radio frequency (RF) immunity testing for compliance with CEN standard EN 61000-4-3
- Burst testing for compliance with CEN standard EN 61000-4-4
- Surge testing for compliance with CEN standard EN 61000-4-5
- Conducted RF Immunity testing for compliance with CEN standard EN 61000-4-6

You need to perform your own immunity testing for Series 5000 router devices that you design and build. See the *Series 5000 Chip Data Book* for additional information about electromagnetic compatibility (EMC) design guidelines for a Series 5000 Chip, including the Router 5000 and FT Router 5000.

---

## Designing Systems for EMC (Electromagnetic Compatibility)

The RTR-10 router has been designed so that products using it should be able to meet both FCC and VDE level “B” limits. Careful system design is important to guarantee that an RTR-10 router-based product will achieve the desired EMC.

---

### *EMC Design Tips*

The following general design tips can help ensure successful EMC for your RTR-10 or Series 5000 router devices:

---

<sup>2</sup> European Committee for Standardization

- Most of the RF noise originates in the CPU portion of the RTR-10 router—which effectively means the entire board. Most of the RF noise originates with the Series 5000 router rchip.
- Most of the EMI will be radiated by the network cable and the power cable.
- Filtering is generally necessary to keep RF noise from getting out on the power cable.
- EMI radiators should be kept away from the RTR-10 router or Series 5000 router chip to prevent internal RF noise from coupling onto the radiators.
- The RTR-10 router must be well grounded to ensure that its built-in EMI filtering works properly. Likewise, a Series 5000 router must be well grounded.
- Early EMI testing of prototypes at a certified outdoor range is an extremely important step in the design of level “B” products. This testing ensures that grounding and enclosure design questions are addressed early enough to avoid most last-minute changes.

---

## *ESD Design Issues*

Electrostatic Discharge (ESD) is encountered frequently in industrial and commercial use of electronic systems. Reliable system designs must consider the effects of ESD and take steps to protect sensitive components. Static discharges occur frequently in low-humidity environments when operators touch electronic equipment. The static voltages generated by humans can easily exceed 10 kV. Keyboards, connectors, and enclosures provide paths for static discharges to reach ESD-sensitive components, such as the Neuron Chip in the RTR-10 or a Series 5000 router.

---

## *Designing Systems for ESD Immunity*

ESD hardening includes the following techniques:

- Provide adequate creepage and clearance distances to prevent ESD hits from reaching sensitive circuitry
- Provide low-impedance paths for ESD hits to ground
- Use diode clamps or transient voltage suppression devices for accessible, sensitive circuits

The best protection from ESD damage is circuit inaccessibility. If all circuit components are positioned away from package seams, the static discharges can be prevented from reaching ESD-sensitive components. There are two measures of “distance” to consider for inaccessibility: creepage and clearance.

- Creepage is the shortest distance between two points along the contours of a surface.
- Clearance is the shortest distance between two points through the air.

An ESD hit generally arcs farther along a surface than it will when passing straight through the air. For example, a 20 kV discharge will arc about 10 mm (0.4 inches) through dry air, but the same discharge can travel over 20 mm (0.8

inches) along a clean surface. Dirty surfaces can allow arcing over even longer creepage distances.

When ESD hits to circuitry cannot be avoided through creepage, clearance, and ground guarding techniques (that is, at external connector pins), explicit clamping of the exposed lines is required to shunt the ESD current. In general, exposed lines require diode clamps to the power supply rails or Zener clamps to chassis ground to shunt the ESD current to ground while clamping the voltage low enough to prevent circuit damage. The Neuron Chip's communications port lines are connected directly to the RTR-10 edge connector without any ESD protection beyond that provided by the chip itself. For a Series 5000 router device, consider how the communications port lines are connected to other parts of the router device. If these lines will be exposed to ESD in a custom router, protection must be added to the router motherboard.



# 6

## Installing a LONWORKS Router

This chapter describes how to install a LONWORKS router.

---

## Introduction

To install a LONWORKS router, perform the following steps:

1. Define a network topology.
2. Physically attach the router to a LONWORKS network.
3. Connect power to the router.
4. Logically install the router on the network.
5. Test the router installation.

The following sections describe these steps in more detail.

---

## Defining a Network Topology

There are many possible network topologies when using routers. The first rule for initial integration is that if a network management tool is used for installation, then a physical or logical path must exist between the network management tool and the router targeted for installation:

- A physical path is created if the network management tool is connected to the same media as one side of the LONWORKS router.
- A logical path is created if one or more active installed routers exist between the LONWORKS router and the network management tool.

The routers creating the logical path can be LONWORKS routers, custom routers based on the RTR-10 Router Core Module, or custom routers based on a Series 5000 router. The routers in the logical path must be installed, loaded, and online before you can add the new router to the network.

When installing routers on a development network, you can use the LonScanner™ Protocol Analyzer to verify that a path exists to a router to be installed. To verify the existence of a logical path, press the service switch of a powered router. If a physical or logical path to the protocol analyzer exists, this action increments the packets received count. A detailed view of the packet log resulting from the previous action should show a code of 0x7F, the message code for an unsolicited service pin message.

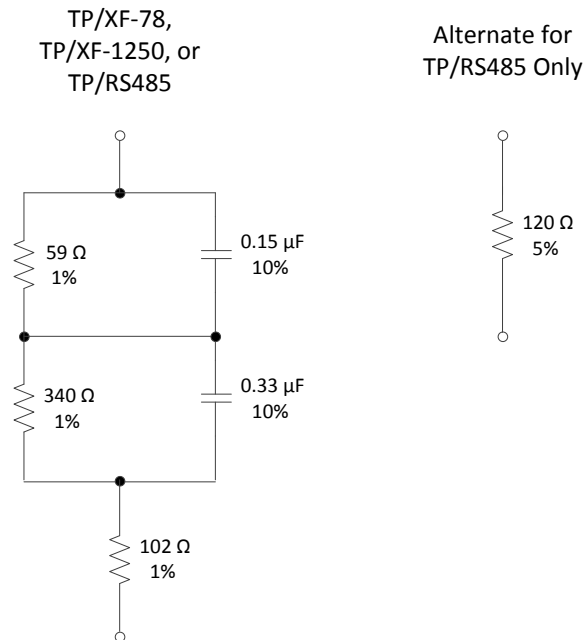
---

## Attaching the Router to a Network

The next step in installation is to physically attach the router to two channels in a LONWORKS network. It is important to insure that each channel has only one transceiver type attached to it. Mixing signals from different transceivers defeats the collision avoidance algorithms, and therefore severely degrades network performance.

The wire used for the network affects the overall system performance with respect to distance, stub length, and total number of devices supported for a single channel. See the *Junction Box and Wiring Guidelines for Twisted Pair LONWORKS Networks* engineering bulletin (005-0023-01) for information about supported cable and wire types for each type of twisted-pair channel.

Proper electrical termination is essential for each twisted-pair channel. Failure to terminate the network can degrade performance, and in some cases, eliminate a device’s ability to communicate with other devices. For TP/XF and TP/RS485 channels, use the terminator circuits shown in **Figure 44**. You can also use the terminators provided with the NodeBuilder FX Development Tool.

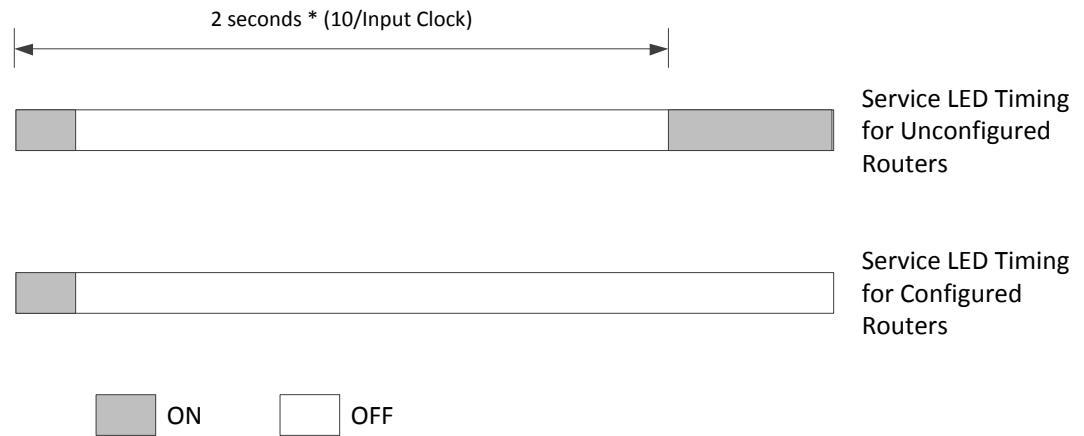


**Figure 44.** Network Termination Circuits for TP/XF and TP/RS485 Networks

# Connecting Power

After the router is physically attached to the desired channels, power must be supplied.

When power is connected to a router, the Service LED for each side changes state as described in **Figure 45**. After a router is powered and configured, the Service LEDs stay off, unless the service request button is pressed.



**Figure 45.** Router Service LED Timing

---

## Installing the Router on a Network

After a router is physically attached to a network, and powered-up, it must be logically installed on the network. You can install a router using a network management tool, such as OpenLNS CT.

Alternatively, you can use a custom network management tool that uses the router network management messages defined in Chapter 7; this method is recommended only for very simple networks with no more than a few routers because of the complexity of calculating timing parameters and forwarding tables for complex networks.

---

### *Router Installation with OpenLNS CT*

The OpenLNS Commissioning Tool (OpenLNS CT) is an end-user tool that supports installation of routers and application devices. See the *OpenLNS CT User's Guide* for a description of router installation.

**Important:** Before commissioning the router from OpenLNS CT, be sure that you have programmed both halves of the router (for example, see *Configuring a Series 5000 Half-Router*) and that the reset signals for both halves are tied together (for example, see *RST~*).

---

### *Router Installation with Network Management Messages*

You can install routers using the network management messages described in Chapter 7, *Network Management Messages*, and in Appendix B of the *Neuron Chip Data Book*, but this process is only recommended for simple networks with few routers. The process is similar to application device installation described in the *OpenLNS CT User's Guide*.

To install a router with network management messages, follow these steps:

1. Change the router state to Unconfigured using the *Set Node Mode* network management message.
2. Assign one or two domains, subnets, and node IDs to both sides of the router using the *Update Domain* network management message.

When installing the router in one domain, the same domain must be assigned to both sides.

When installing the router in two domains, the same domain must be assigned as the first domain on both sides, and the same domain must be assigned as the second domain on both sides.

3. Select a routing algorithm for both sides of the router using the *Write Memory* network management message. Both sides must be set to use the same algorithm.
4. For configured routers, load the group and subnet routing tables on both sides of the router using the *Group or Subnet Table Download* network management message. There are 255 forwarding flags for subnets and 255 forwarding flags for groups on each side for each domain.

5. Initialize the routing tables using the *Set Router Mode* network management message.
6. Change the router state on both sides of the router to Configured, on-line using the *Set Node Mode* network management message.

---

## Testing Router Installation

After a router has been installed, you can use the *Query Status* network diagnostic message to ensure that it is operational. If no response is received, query all intermediate routers to determine where the fault occurred. If the router has been installed with OpenLNS CT, use the Test command (described under “Testing Devices” in the *OpenLNS CT User’s Guide*) to query router status.

See the description of the *Query Status* message in the *Standard Messages* section for a description of the error codes returned by the *Query Status* message.



# 7

## Network Management Messages

This chapter describes network management messages for LONWORKS routers. These messages are used for router installation, as described in Chapter 6, *Installing a LONWORKS Router*.

---

## Introduction

As described in Chapter 6, routers are installed using network management messages. These messages are sent as explicit messages by a network management tool, such as OpenLNS CT. Routers respond to many of the same messages as any LONWORKS device, but also have an additional set of router-specific messages, as listed in **Table** .

**Table 19.** Network Messages

| Network Message Type | Request Code | Success Response | Failed Response |
|----------------------|--------------|------------------|-----------------|
| Network Diagnostic   | 0x50 to 0x5F | 0x31 to 0x3F     | 0x11 to 0x1F    |
| Network Management   | 0x60 to 0x7D | 0x21 to 0x3D     | 0x01 to 0x1D    |
| Router Configuration | 0x74 to 0x7E | 0x34 to 0x3E     | 0x14 to 0x1E    |

Several router options are set using the *Write Memory* network management message. These router options include specification of the routing algorithm, buffer sizes, and non-priority output buffer queue count.

---

## Standard Messages

Routers accept the standard network diagnostic and network management messages listed in **Table** and **Table** . These messages are described in Appendix B of the *Neuron Chip Data Book*.

**Table 20.** Network Diagnostic Messages

| Network Diagnostic Message | Request Code | Success Response | Failed Response |
|----------------------------|--------------|------------------|-----------------|
| <i>Query Status</i>        | 0x51         | 0x31             | 0x11            |
| <i>Proxy Command</i>       | 0x52         | 0x32             | 0x12            |
| <i>Clear Status</i>        | 0x53         | 0x33             | 0x13            |
| <i>Query XCVR Status</i>   | 0x54         | 0x34             | 0x14            |

**Table 21.** Network Management Messages

| Network Management Message | Request Code | Success Response | Failed Response |
|----------------------------|--------------|------------------|-----------------|
| <i>Query ID</i>            | 0x61         | 0x21             | 0x01            |
| <i>Respond to Query</i>    | 0x62         | 0x22             | 0x02            |

| Network Management Message  | Request Code | Success Response | Failed Response |
|-----------------------------|--------------|------------------|-----------------|
| <i>Update Domain</i>        | 0x63         | 0x23             | 0x03            |
| <i>Leave Domain</i>         | 0x64         | 0x24             | 0x04            |
| <i>Update Key</i>           | 0x65         | 0x25             | 0x05            |
| <i>Query Domain</i>         | 0x6A         | 0x2A             | 0x0A            |
| <i>Set Node Mode</i>        | 0x6C         | 0x2C             | 0x0C            |
| <i>Read Memory</i>          | 0x6D         | 0x2D             | 0x0D            |
| <i>Write Memory</i>         | 0x6E         | 0x2E             | 0x0E            |
| <i>Checksum Recalculate</i> | 0x6F         | 0x2F             | 0x0F            |
| <i>Memory Refresh</i>       | 0x71         | 0x31             | 0x11            |

The following exceptions apply to standard network management messages when used with routers:

- The *Query Status* network diagnostic message reports two errors that are unique to the router (159 [0x9F] and 164 [0xA4]). These errors are listed in Appendix B of the *Neuron Chip Data Book*.
- The *Set Node Mode* network management message is automatically processed by both sides of a router when it is used to place the router offline and online (the APPL\_OFFLINE and APPL\_ONLINE options).
- When the *Set Node Mode* message is used to place a router offline, the router stops forwarding, and all messages not addressed to the router are dropped.
- The router does not respond to *Set Node Mode* messages that use a broadcast address. Thus, broadcast *Restart* or *Offline* messages do not stop the router and prevent the same broadcast message from reaching destinations on the other side of the router. Routers must therefore be restarted or taken offline using a *Set Node Mode* message addressed directly to the router.

---

## Router-Specific Messages

Router-specific network management messages are listed in **Table** .

**Table 22.** Router-Specific Network Management Messages

| Network Management Message            | Request Code | Success Response | Failed Response |
|---------------------------------------|--------------|------------------|-----------------|
| <i>Set Router Mode</i>                | 0x74         | 0x34             | 0x14            |
| <i>Group or Subnet Table Clear</i>    | 0x75         | 0x35             | 0x15            |
| <i>Group or Subnet Table Download</i> | 0x76         | 0x36             | 0x16            |
| <i>Group Forward</i>                  | 0x77         | 0x37             | 0x17            |
| <i>Subnet Forward</i>                 | 0x78         | 0x38             | 0x18            |
| <i>Group No Forward</i>               | 0x79         | 0x39             | 0x19            |
| <i>Subnet No Forward</i>              | 0x7A         | 0x3A             | 0x1A            |
| <i>Group or Subnet Table Report</i>   | 0x7B         | 0x3B             | 0x1B            |
| <i>Router Status</i>                  | 0x7C         | 0x3C             | 0x1C            |
| <i>Far Side Escape Code</i>           | 0x7D         | —                | —               |

---

## Router-Specific Network Management Messages

This section describes the router-specific network management messages listed in **Table .**

### Set Router Mode

This message instructs the router to perform one of several router-related tasks:

- The NORMAL option returns the router from the TEMP\_BRIDGE mode.
- The INIT\_RTR\_TABLE option copies all forwarding tables from EEPROM into the RAM tables for a configured router, or, sets all RAM tables to flood for a learning router (this is the same action that occurs after node reset).
- The TEMP\_BRIDGE option causes the router to temporarily forward **all** messages in the domain (until the next reset or *Set Router Mode* message with the NORMAL option).

**Note:** The standard *Set Node Mode* message can be used to take the entire router offline and online.

The *Set Router Mode* message affects both router sides. This message uses the Request-Response protocol.

```

typedef enum {
    NORMAL = 0,          // Not a temporary bridge.
    INIT_RTR_TABLE = 1,  // Copy forwarding tables from EEPROM
                        // for configured routers.
                        // Initialize forwarding tables for
                        // learning routers.
    TEMP_BRIDGE = 2      // Temporarily a bridge until next reset
                        // or NORMAL router mode request.
} rtr_mode;

typedef rtr_mode NM_rtr_mode_request;

```

## Group or Subnet Table Clear

This message clears all entries in either the group or subnet forwarding table for a single domain for a single router side. The message is segmented to cover eight byte sections to prevent lengthy EEPROM write operations.

This message uses the Request-Response protocol. The configuration checksum in EEPROM is updated.

```

typedef struct {
    unsigned group_or_subnet : 1; // 1 => Group, 0 => Subnet
    unsigned domain_index : 1;
    unsigned unused : 4;
    unsigned index_times_8 : 2;
} NM_rtr_table_clear_request;

```

## Group or Subnet Table Download

This message configures the entire group or subnet forwarding table in EEPROM for the specified domain for a single router side. The download function is segmented into eight-byte sections.

The least significant bit (LSB) of the table field maps to the lowest subnet or group ID in the current set of table entries defined by the `index_times_8` field. A value of '1' specifies that forwarding be enabled for the corresponding group or subnet; a value of '0' disables forwarding. Subnet 0 is used for special protocol functions and is never marked for forwarding.

This message uses the Request-Response protocol. The configuration checksum in EEPROM is updated. Each byte in the table entry includes routing flags for eight subnets or groups.

```

typedef struct {
    unsigned group_or_subnet : 1; // 1 => Group, 0 => Subnet
    unsigned domain_index : 1;
    unsigned unused : 4;
    unsigned index_times_8 : 2;
    unsigned table[8];          // Table data
} NM_rtr_table_downld_request;

```

## Group Forward

This message sets the forwarding flag in the forwarding table for a given group in the specified domain. If the `ram_or_eeprom` field is set, both the RAM and

EEPROM flags are set, otherwise only the RAM flag is set, allowing temporary forwarding for a given group.

This message uses the Request-Response protocol. The configuration checksum in EEPROM is updated if EEPROM is changed.

```
typedef struct {
    unsigned unused1 : 1;
    unsigned domain_index : 1;
    unsigned unused2 : 5;
    unsigned ram_or_eeprom : 1; // 0 => RAM, 1 => RAM+EEPROM
    unsigned group;
} NM_rtr_group_fwd_request;
```

## Subnet Forward

This message sets the forwarding flag in the forwarding table for a given subnet in the specified domain. If the `ram_or_eeprom` field is set, both the RAM and EEPROM flags are set, otherwise only the RAM flag is set, allowing temporary forwarding for a given subnet.

This message uses the Request-Response protocol. The configuration checksum in EEPROM is updated if EEPROM is changed.

```
typedef struct {
    unsigned unused1 : 1;
    unsigned domain_index : 1;
    unsigned unused2 : 5;
    unsigned ram_or_eeprom : 1; // 0 => RAM, 1 => RAM+EEPROM
    unsigned subnet;
} NM_rtr_subnet_fwd_request;
```

## Group No Forward

This message clears the forwarding flag in the forwarding table for a given group in the specified domain. If the `ram_or_eeprom` field is set, both the RAM and EEPROM flags are cleared, otherwise only the RAM flag is cleared, allowing temporary control of forwarding for a given group (see the *Router Status* message).

This message uses the Request-Response protocol. The configuration checksum in EEPROM is updated if EEPROM is changed.

```
typedef struct {
    unsigned unused1 : 1;
    unsigned domain_index : 1;
    unsigned unused2 : 5;
    unsigned ram_or_eeprom : 1; // 0 => RAM, 1 => RAM+EEPROM
    unsigned group;
} NM_rtr_group_nofwd_request;
```

## Subnet No Forward

This message clears the forwarding flag in the forwarding table for a given subnet in the specified domain. If the `ram_or_eeprom` field is set, both the RAM and EEPROM flags are cleared, otherwise only the RAM flag is cleared, allowing temporary control of forwarding for a given subnet.

This message uses the Request-Response protocol. The configuration checksum in EEPROM is updated if EEPROM is changed.

```
typedef struct {
    unsigned unused1 : 1;
    unsigned domain_index : 1;
    unsigned unused2 : 5;
    unsigned ram_or_eeprom : 1; // 0 => RAM, 1 => RAM+EEPROM
    unsigned subnet;
} NM_rtr_subnet_nofwd_request;
```

## Group or Subnet Table Report

This message reports the current settings of either group or subnet forwarding tables in EEPROM or RAM for the specified domain for a single router side. The report function is segmented into eight byte sections.

**Important:** This message is supported by router firmware version 5 or later only. Use of this message with earlier versions of the router firmware cause the router to become inoperable.

This message uses the Request-Response protocol.

```
typedef struct {
    unsigned group_or_subnet : 1; // 1 => Group, 0 => Subnet
    unsigned domain_index : 1;
    unsigned ram_or_eeprom : 1; // 0 => RAM, 1 => EEPROM
    unsigned unused : 3;
    unsigned index_times_8 : 2;
} NM_rtr_table_report_request;

typedef struct {
    unsigned table[8]; // Table data
} NM_rtr_table_report_response;
```

## Router Status

This message is used to report the router configuration and flood/normal modes.

This message uses the Request-Response protocol.

```
typedef enum {
    CONFIGURED = 0,
    LEARNING = 1,
    BRIDGE = 2,
    REPEATER = 3
} algorithm;

typedef enum {
    NORMAL = 0, // Not a temporary bridge.
    INIT_RTR_TABLE = 1, // Copy forwarding tables from EEPROM
                        // for configured routers.
                        // Initialize forwarding tables for
                        // learning routers.
    TEMP_BRIDGE = 2 // Temporarily a bridge until next reset.
} rtr_mode;
```

```
typedef struct {
    algorithm type; // CONFIGURED, LEARNING, BRIDGE,
                  // or REPEATER
    rtr_mode mode; // TEMP_BRIDGE or NORMAL
} NM_rtr_status_response;
```

## Far Side Escape Code

When this message code is placed in the message, and is followed by any network management or network diagnostic message (except the escape message itself), that message is passed to the other (far) router side for processing. Any responses are returned in the normal manner. This command allows network management of the router side that is not directly addressable from a network management tool.

The far side escape code is not required for the *Set Node Mode* network management message when it is used to place the router offline and online (the APPL\_OFFLINE and APPL\_ONLINE options). The offline and online commands are automatically forwarded.

```
byte code; /* Destination: NM, code: 0x7E */
```

---

## Router Options Set with Write Memory

The *Write Memory* network management message is used to change the routing algorithm, buffer sizes, and buffer queue counts. To change these parameters, perform the following steps:

1. Change the parameters using the *Write Memory* network management message, as described in the following sections.
2. Reset the router using the *Set Node Mode* network management message.

## Set Routing Algorithm

The routing algorithm is selected using a *Write Memory* network management message with the following parameters:

```
mode    = CONFIG_RELATIVE (2)
offset  = 0x0037;
count   = 1;
form    = CNFG_CS_RECALC (4)
data    = routing_algorithm;
```

The `routing_algorithm` value is a byte of type `algorithm`:

```
typedef enum {
    CONFIGURED = 0,
    LEARNING = 1,
    BRIDGE = 2,
    REPEATER = 3
} algorithm;
```

## Set Buffer Size

The buffer sizes are selected using a *Write Memory* network management message with the following parameters:

```

mode    = READ_ONLY_RELATIVE (1)
offset  = 0x0019;
count   = 1;
form    = BOTH_CS_RECALC (1)
data    = buffer_sizes;

```

The `buffer_sizes` value contains two nibble fields that control the size of both the input and output buffers. The output size value also controls the priority output buffer size. The default size is 66 bytes (or `SIZE_66 = 0xB`).

When changing this value, you should set both nibble fields to the same value. Different values can be used if the maximum packet size is different for the two directions through the router. The default setting for this byte is 0xBB. The total number of bytes assigned to the buffer queues for a RTR-10 must not exceed 1254 bytes, as described in *Message Buffers*. A buffer size of less than 66 is not recommended because the router will not be able to forward network management messages if the buffers are too small.

The size values are represented by a code of type `buffer_size_entry`:

```

typedef enum {
    SIZE_20 = 0x2;
    SIZE_21 = 0x3;
    SIZE_22 = 0x4;
    SIZE_24 = 0x5;
    SIZE_26 = 0x6;
    SIZE_30 = 0x7;
    SIZE_34 = 0x8;
    SIZE_42 = 0x9;
    SIZE_50 = 0xA;
    SIZE_66 = 0xB;
    SIZE_82 = 0xC;
    SIZE_114 = 0xD;
    SIZE_146 = 0xE;
    SIZE_210 = 0xF;
    SIZE_255 = 0x0;
} buffer_size_entry;

```

## Set Priority Output Buffer Queue Count

The priority output buffer queue count is selected using a *Write Memory* network management message with the following parameters:

```

mode    = READ_ONLY_RELATIVE (1)
offset  = 0x001A;
count   = 1;
form    = BOTH_CS_RECALC (1)
data    = queue_count;

```

The `queue_count` value contains two nibble fields. The most significant nibble controls the number of priority output buffers. The least significant nibble must be zero. The total number of bytes assigned to the buffer queues for a RTR-10 must not exceed 1254 bytes, as described *Message Buffers*.

The most significant nibble of `queue_count` is represented by a code of type `queue_count_entry`:

```
typedef enum {
    COUNT_1 = 0x2;
    COUNT_2 = 0x3;
    COUNT_3 = 0x4;
    COUNT_5 = 0x5;
    COUNT_7 = 0x6;
    COUNT_11 = 0x7;
    COUNT_15 = 0x8;
    COUNT_23 = 0x9;
    COUNT_31 = 0xA;
    COUNT_47 = 0xB;
    COUNT_63 = 0xC;
} queue_count_entry;
```

## Set Input and Non-Priority Buffer Queue Count

The buffer queue counts are selected using a *Write Memory* network management message with the following parameters:

```
mode    = READ_ONLY_RELATIVE (1)
offset  = 0x001C;
count   = 1;
form    = BOTH_CS_RECALC (1)
data    = queue_counts;
```

The `queue_counts` value contains two nibble fields that control the count of both the input and non-priority output buffer queues. The least significant nibble controls the number of input buffers and the most significant nibble controls the number of nonpriority output buffers. The default for this field is 15 non-priority output buffers (COUNT\_15) and 2 input buffers (COUNT\_2). The total number of bytes assigned to the buffer queues for a RTR-10 must not exceed 1254 bytes, as described *Message Buffers*. The queue count for both queues is represented by the source code as that for the *Set Priority Output Buffer Queue Count* command.

# A

## Communications Parameters for LONWORKS Routers

LONWORKS routers are initially programmed with communications parameters as listed in this appendix. Parameters for LONMARK approved transceivers correspond to the parameters defined by the *LONWORKS Interoperability Guidelines*. Parameters specified as “Configurable” can be changed by a network services tool.

These parameters only apply to routers with router firmware version 5 or newer. The firmware version number for a router can be determined with the *Query Status* network diagnostic message.

Communications parameters for routers with version 4 or older firmware should be re-installed to ensure that the standard interoperable parameters are used.

# Communications Parameters

**Table 24**, **Table 25**, and **Table 26** together list the communications parameters for LONWORKS routers.

**Table 23.** Communications Parameters, Part 1

| Parameter                            | TP/XF-78                         | TP/XF-1250                       | TP/FT-10                         | TP/RS485-39                      |
|--------------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| Transceiver ID                       | 1 (0x01)                         | 3 (0x03)                         | 4 (0x04)                         | 5 (0x5)                          |
| Media                                | Isolated Twisted Pair            | Isolated Twisted Pair            | Free Topology or Link Power      | EIA-485 Twisted Pair             |
| Neuron Chip to Transceiver Interface | Differential                     | Differential                     | Single Ended                     | Single Ended                     |
| Interface Bit Rate                   | 78 kbps                          | 1.25 Mbps                        | 78 kbps                          | 39 kbps                          |
| Input Clock                          | 10 MHz                           | 10 MHz                           | 10 MHz                           | 10 MHz                           |
| Minimum Clock                        | Configurable; default = 5 MHz    | Configurable; default = 10 MHz   | Configurable; default = 5 MHz    | Configurable; default = 5 MHz    |
| Number of Priority Slots             | Configurable; default = 4 slots  | Configurable; default = 16 slots | Configurable; default = 4 slots  | Configurable; default = 4 slots  |
| Average Packet Size                  | Configurable; default = 15 bytes | Configurable; default = 15 bytes | Configurable; default = 15 bytes | Configurable; default = 15 bytes |
| Oscillator Accuracy                  | 200 ppm                          | 200 ppm                          | 200 ppm                          | 200 ppm                          |
| Oscillator Wakeup                    | 0 $\mu$ sec                      | 0 $\mu$ sec                      | 0 $\mu$ sec                      | 0 $\mu$ sec                      |
| Collision Detect (CD)                | No                               | No                               | No                               | No                               |
| CD Term after Preamble               | N/A                              | N/A                              | N/A                              | N/A                              |
| CD through Packet End                | N/A                              | N/A                              | N/A                              | N/A                              |
| Bit Sync Threshold                   | 5 bits                           | 7 bits                           | 4 bits                           | 4 bits                           |
| Hysteresis                           | 2                                | 0                                | N/A                              | N/A                              |
| Filter                               | 1                                | 0                                | N/A                              | N/A                              |
| Network Bit Rate                     | 78 kbps                          | 1.25 Mbps                        | 78 kbps                          | 39 kbps                          |

| Parameter              | TP/XF-78    | TP/XF-1250  | TP/FT-10    | TP/RS485-39 |
|------------------------|-------------|-------------|-------------|-------------|
| Alternate Rate         | N/A         | N/A         | N/A         | N/A         |
| Wakeup Pin Direction   | N/A         | N/A         | N/A         | N/A         |
| XCVR Controls Preamble | N/A         | N/A         | N/A         | N/A         |
| General Purpose Data   | N/A         | N/A         | N/A         | N/A         |
| Allow Node Override    | N/A         | N/A         | N/A         | N/A         |
| Receive Start Delay    | 2.9 bits    | 14.0 bits   | 9.0 bits    | 2.0 bits    |
| Receive End Delay      | 0.0 bits    | 0.0 bits    | 0.0 bits    | 0.0 bits    |
| Indeterminate Time     | 24.0 bits   | 25.0 bits   | 24.0 bits   | 4.0 bits    |
| Min Interpacket Time   | 0.0 bits    | 0.0 bits    | 0.0 bits    | 0.0 bits    |
| Turnaround Time        | 0 $\mu$ sec | 0 $\mu$ sec | 0 $\mu$ sec | 0 $\mu$ sec |
| Missed Preamble        | 1.0 bits    | 4.0 bits    | 4.0 bits    | 1.0 bits    |
| Preamble Length        | N/A         | N/A         | N/A         | N/A         |
| Use Raw Data           | No          | No          | No          | No          |

**Table 24.** Communications Parameters, Part 2

| Parameter                            | RF-10                         | PL-10                         | PL-20C                           | PL-20N                           |
|--------------------------------------|-------------------------------|-------------------------------|----------------------------------|----------------------------------|
| Transceiver ID                       | 7 (0x07)                      | 9 (0x09)                      | 16 (0x10)                        | 17 (0x11)                        |
| Media                                | 49 MHz Radio Frequency        | Power Line                    | Power Line                       | Power Line                       |
| Neuron Chip to Transceiver Interface | Single Ended                  | Special Purpose               | Special Purpose                  | Special Purpose                  |
| Interface Bit Rate                   | 4.9 kbps                      | 625 kbps                      | 156.3 kbps                       | 156.3 kbps                       |
| Input Clock                          | 5 MHz                         | 10 MHz                        | 10 MHz                           | 10 MHz                           |
| Minimum Clock                        | Configurable; default = 5 MHz | Configurable; default = 5 MHz | Configurable; default = 1.25 MHz | Configurable; default = 1.25 MHz |

| Parameter                | RF-10                                  | PL-10                                  | PL-20C                                 | PL-20N                                 |
|--------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|
| Number of Priority Slots | Configurable;<br>default = 4<br>slots  | Configurable;<br>default = 8<br>slots  | Configurable;<br>default = 8<br>slots  | Configurable;<br>default = 8<br>slots  |
| Average Packet Size      | Configurable;<br>default = 15<br>bytes | Configurable;<br>default = 15<br>bytes | Configurable;<br>default = 15<br>bytes | Configurable;<br>default = 15<br>bytes |
| Oscillator Accuracy      | 200 ppm                                | 200 ppm                                | 200 ppm                                | 200 ppm                                |
| Oscillator Wakeup        | 0 $\mu$ sec                            | 0 $\mu$ sec                            | 0 $\mu$ sec                            | 0 $\mu$ sec                            |
| Collision Detect (CD)    | No                                     | N/A                                    | N/A                                    | N/A                                    |
| CD Term after Preamble   | N/A                                    | N/A                                    | N/A                                    | N/A                                    |
| CD through Packet End    | N/A                                    | N/A                                    | N/A                                    | N/A                                    |
| Bit Sync Threshold       | 7 bits                                 | N/A                                    | N/A                                    | N/A                                    |
| Hysteresis               | N/A                                    | N/A                                    | N/A                                    | N/A                                    |
| Filter                   | N/A                                    | N/A                                    | N/A                                    | N/A                                    |
| Network Bit Rate         | 4.9 kbps                               | 9412 bps                               | 3987 bps                               | 3987 bps                               |
| Alternate Rate           | N/A                                    | 0 bps                                  | N/A                                    | N/A                                    |
| Wakeup Pin Direction     | N/A                                    | Output                                 | Output                                 | Output                                 |
| XCVR Controls Preamble   | N/A                                    | Yes                                    | Yes                                    | Yes                                    |
| General Purpose Data     | N/A                                    | 00 0A 00 00<br>00 00 00                | 4A 00 00 00<br>00 00 00                | 0E 01 00 00<br>00 00 00                |
| Allow Node Override      | N/A                                    | Yes                                    | No                                     | No                                     |
| Receive Start Delay      | 2.0 bits                               | 1.0 bit                                | 6.8 bits                               | 6.8 bits                               |
| Receive End Delay        | 0.0 bits                               | 10.4 bits                              | 1.6 bits                               | 1.6 bits                               |
| Indeterminate Time       | 9.8 bits                               | 0.0 bits                               | 0.0 bits                               | 0.0 bits                               |
| Min Interpacket Time     | 0.0 bits                               | 0.0 bits                               | 17.5 bits                              | 17.5 bits                              |
| Turnaround Time          | 0 $\mu$ sec                            | N/A                                    | N/A                                    | N/A                                    |
| Missed Preamble          | 9.0 bits                               | N/A                                    | N/A                                    | N/A                                    |

| Parameter       | RF-10 | PL-10     | PL-20C    | PL-20N    |
|-----------------|-------|-----------|-----------|-----------|
| Preamble Length | N/A   | 36.7 bits | 33.5 bits | 33.5 bits |
| Use Raw Data    | No    | No        | No        | No        |

**Table 25.** Communications Parameters, Part 3

| Parameter                            | PL-30                            | TP/RS485-625                     | TP/RS485-1250                    | TP/RS485-78                      |
|--------------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| Transceiver ID                       | 18 (0x12)                        | 10 (0x0A)                        | 11 (0x0B)                        | 12 (0x0C)                        |
| Media                                | Power Line                       | EIA-485 Twisted Pair             | EIA-485 Twisted Pair             | EIA-485 Twisted Pair             |
| Neuron Chip to Transceiver Interface | Special Purpose                  | Single Ended                     | Single Ended                     | Single Ended                     |
| Interface Bit Rate                   | 625 kbps                         | 625 kbps                         | 1.25 Mbps                        | 78 kbps                          |
| Input Clock                          | 10 MHz                           | 10 MHz                           | 10 MHz                           | 10 MHz                           |
| Minimum Clock                        | Configurable; default = 5 MHz    | Configurable; default = 5 MHz    | Configurable; default = 5 MHz    | Configurable; default = 5 MHz    |
| Number of Priority Slots             | Configurable; default = 12 slots | Configurable; default = 4 slots  | Configurable; default = 16 slots | Configurable; default = 4 slots  |
| Average Packet Size                  | Configurable; default = 15 bytes | Configurable; default = 15 bytes | Configurable; default = 15 bytes | Configurable; default = 15 bytes |
| Oscillator Accuracy                  | 200 ppm                          | 200 ppm                          | 200 ppm                          | 200 ppm                          |
| Oscillator Wakeup                    | 0 $\mu$ sec                      | 0 $\mu$ sec                      | 0 $\mu$ sec                      | 0 $\mu$ sec                      |
| Collision Detect (CD)                | N/A                              | No                               | No                               | No                               |
| CD Term after Preamble               | N/A                              | N/A                              | N/A                              | N/A                              |
| CD through Packet End                | N/A                              | N/A                              | N/A                              | N/A                              |
| Bit Sync Threshold                   | N/A                              | 4 bits                           | 4 bits                           | 4 bits                           |
| Hysteresis                           | N/A                              | N/A                              | N/A                              | N/A                              |
| Filter                               | N/A                              | N/A                              | N/A                              | N/A                              |

| Parameter              | PL-30                   | TP/RS485-625 | TP/RS485-1250 | TP/RS485-78 |
|------------------------|-------------------------|--------------|---------------|-------------|
| Network Bit Rate       | 1882 bps                | 625 kbps     | 1.25 Mbps     | 78 kbps     |
| Alternate Rate         | N/A                     | N/A          | N/A           | N/A         |
| Wakeup Pin Direction   | Output                  | N/A          | N/A           | N/A         |
| XCVR Controls Preamble | Yes                     | N/A          | N/A           | N/A         |
| General Purpose Data   | 00 8A 00 00<br>00 00 00 | N/A          | N/A           | N/A         |
| Allow Node Override    | Yes                     | N/A          | N/A           | N/A         |
| Receive Start Delay    | 1.0 bit                 | 2.0 bits     | 2.0 bits      | 2.0 bits    |
| Receive End Delay      | 10.4 bits               | 0.0 bits     | 0.0 bits      | 0.0 bits    |
| Indeterminate Time     | 0.0 bits                | 4.0 bits     | 4.0 bits      | 4.0 bits    |
| Min Interpacket Time   | 0.0 bits                | 0.0 bits     | 0.0 bits      | 0.0 bits    |
| Turnaround Time        | N/A                     | 0 $\mu$ sec  | 0 $\mu$ sec   | 0 $\mu$ sec |
| Missed Preamble        | N/A                     | 1.0 bit      | 1.0 bit       | 1.0 bit     |
| Preamble Length        | 36.7 bits               | N/A          | N/A           | N/A         |
| Use Raw Data           | No                      | No           | No            | No          |

**Table 26.** Communications Parameters, Part 4

| Parameter                            | FO-10        | DC-78          | DC-625         | DC-1250        |
|--------------------------------------|--------------|----------------|----------------|----------------|
| Transceiver ID                       | 24 (0x18)    | 27 (0x1B)      | 28 (0x1C)      | 29 (0x1D)      |
| Media                                | Fiber Optic  | Direct Connect | Direct Connect | Direct Connect |
| Neuron Chip to Transceiver Interface | Single Ended | Differential   | Differential   | Differential   |
| Interface Bit Rate                   | 1.25 Mbps    | 78 kbps        | 625 kbps       | 1.25 Mbps      |
| Input Clock                          | 10 MHz       | 10 MHz         | 10 MHz         | 10 MHz         |

| Parameter                | FO-10                                  | DC-78                                  | DC-625                                 | DC-1250                                |
|--------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|
| Minimum Clock            | Configurable;<br>default = 10<br>MHz   | Configurable;<br>default = 10<br>MHz   | Configurable;<br>default = 10<br>MHz   | Configurable;<br>default = 10<br>MHz   |
| Number of Priority Slots | Configurable;<br>default = 16<br>slots | Configurable;<br>default = 0<br>slots  | Configurable;<br>default = 0<br>slots  | Configurable;<br>default = 0<br>slots  |
| Average Packet Size      | Configurable;<br>default = 15<br>bytes | Configurable;<br>default = 15<br>bytes | Configurable;<br>default = 15<br>bytes | Configurable;<br>default = 15<br>bytes |
| Oscillator Accuracy      | 200 ppm                                | 200 ppm                                | 200 ppm                                | 200 ppm                                |
| Oscillator Wakeup        | 0 $\mu$ sec                            | 0 $\mu$ sec                            | 0 $\mu$ sec                            | 0 $\mu$ sec                            |
| Collision Detect (CD)    | Yes                                    | No                                     | No                                     | No                                     |
| CD Term after Preamble   | Yes                                    | N/A                                    | N/A                                    | N/A                                    |
| CD through Packet End    | Yes                                    | N/A                                    | N/A                                    | N/A                                    |
| Bit Sync Threshold       | 4 bits                                 | 4 bits                                 | 4 bits                                 | 4 bits                                 |
| Hysteresis               | N/A                                    | 0                                      | 0                                      | 0                                      |
| Filter                   | N/A                                    | 0                                      | 0                                      | 0                                      |
| Network Bit Rate         | 1.25 Mbps                              | 78 kbps                                | 625 kbps                               | 1.25 Mbps                              |
| Alternate Rate           | N/A                                    | N/A                                    | N/A                                    | N/A                                    |
| Wakeup Pin Direction     | N/A                                    | N/A                                    | N/A                                    | N/A                                    |
| XCVR Controls Preamble   | N/A                                    | N/A                                    | N/A                                    | N/A                                    |
| General Purpose Data     | N/A                                    | N/A                                    | N/A                                    | N/A                                    |
| Allow Node Override      | N/A                                    | N/A                                    | N/A                                    | N/A                                    |
| Receive Start Delay      | 4.0 bits                               | 1.0 bit                                | 1.0 bit                                | 1.0 bit                                |
| Receive End Delay        | 4.0 bits                               | 0.0 bits                               | 0.0 bits                               | 0.0 bits                               |
| Indeterminate Time       | 4.0 bits                               | 0.0 bits                               | 0.0 bits                               | 0.0 bits                               |
| Min Interpacket Time     | 8.0 bits                               | 0.0 bits                               | 0.0 bits                               | 0.0 bits                               |
| Turnaround Time          | 0 $\mu$ sec                            | 0 $\mu$ sec                            | 0 $\mu$ sec                            | 0 $\mu$ sec                            |

| Parameter       | FO-10    | DC-78    | DC-625   | DC-1250  |
|-----------------|----------|----------|----------|----------|
| Missed Preamble | 4.0 bits | 0.0 bits | 0.0 bits | 0.0 bits |
| Preamble Length | N/A      | N/A      | N/A      | N/A      |
| Use Raw Data    | No       | No       | No       | No       |

# B

## Determining RTR-10 Firmware Version

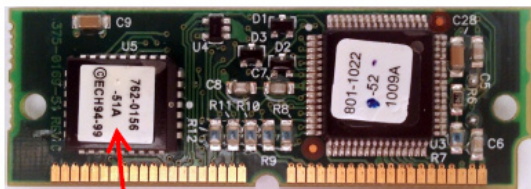
In order to understand the buffer capacity of your RTR-10 router, you need to determine which version of router firmware is included with your product. The photos on the following page will help you make this determination.

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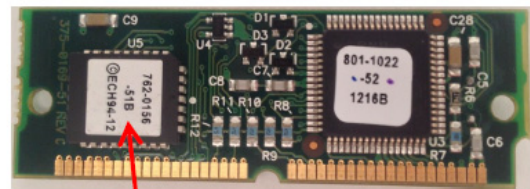
## Router Firmware Version

The router firmware version is printed on the label on the memory containing the router firmware. The label will have a 9-digit part number starting with "726" followed by a version number. The letters "A," "B," or "C" indicate router firmware version A, B, or C.

The following figure illustrates the location of the firmware version label for the **RTR-10 Router Core Module**.



Router Firmware  
Version A



Router Firmware  
Version B

